Urban Simulation 3

The Family of Spatial Interaction Models General Urban Models: LUTI Models

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Outline of Today's Lecture

- Modifying the Generic Model
- The Family of Spatial Interaction Models
- Urban Models: Coupled Spatial Interaction
- Generalising the Model Adding Employment
- Building Spatial Interaction into an Economic Base
- Extending the Retail Model to Deal with Agglomeration and Scale
- Three Basic Applications and a Fourth – the QUANT Model, probably Next Time

Modifying the Generic Model

Inthe last lecture, I have annotated the mass variables in the gravity model – O_i^{α} and D_j^{γ} as coefficients $T_{ij} = KO_i^{\alpha} D_j^{\gamma} \exp(-\beta c_{ij})$. This is because the extended model is used in the practical but because it is more general. We will use $T_{ij} = KO_i D_j \, exp\left(-\beta c_{ij}\right)$ when it comes to the family of gravity models. Thus $\alpha = \gamma = 1$. With different constraints, it is tricky and laborious to notate the origin and destination variables with parameters, so I have dropped these

The Family of Spatial Interaction Models

We already implicitly indicated how we can constrain the model to produce specific types that meet various constraints on the amount of trips attracted to different locations. Our first and most basic model is unconstrained in that we ensure that the only constraints the model meets are on the total trips which are predetermined, that is $T = \sum_{i=1}^{n} \sum_{j=1}^{m} T_{ij}$.

Now if we know the trips attracted to the origin, or to the destination, or to both, then these give rise to different models. Including the basic model there are four in all that meet the following. This is the family……

1. The Unconstrained Model

$T_{ij} = KO_iD_j \exp(-\beta c_{ij})$ subject to $\sum_{i=1}^{n}\sum_{j=1}^{m} T_{ij} = T$

2. The Singly-Constrained Models

2a: The Origin- Constrained Model

$T_{ij} = A_iO_iD_j \exp(-\beta c_{ij})$ subject to $\sum_{j=1}^{m} T_{ij} = O_i$

2b: The Destination-Constrained Model

$T_{ij} = O_iB_jD_j \exp(-\beta c_{ij})$ subject to $\sum_{i=1}^{n} T_{ij} = D_j$

3. The Doubly Constrained Model

$T_{ij} = A_iO_iB_jD_j \exp(-\beta c_{ij})$ st $\sum_{j=1}^{m} T_{ij} = O_i$ & $\sum_{i=1}^{n} T_{ij} = D_j$

We can work out the full models by substituting the model on the left into the constraint on the right

Now these models are **distribution models**: but they not only distribute trips but if we add their flows over origins or destinations or both, the different models give us predictions of what happens at origins and destinations. These are then **location models** and their predictions are in red below

The Models	The Predictions
$T'_{ij} = KO_iD_j \exp(-\beta c_{ij})$;	$\sum_{j=1}^{m} T'_{ij} = O'_i$; $\sum_{i=1}^{n} T'_{ij} = D'_j$
$T'_{ij} = A_iO_iD_j \exp(-\beta c_{ij})$;	$\sum_{j=1}^{m} T'_{ij} = O_i$; $\sum_{i=1}^{n} T'_{ij} = D'_j$
$T'_{ij} = O_iB_jD_j \exp(-\beta c_{ij})$;	$\sum_{j=1}^{m} T'_{ij} = O'_i$; $\sum_{i=1}^{n} T'_{ij} = D_j$
$T'_{ij} = A_iO_iB_jD_j \exp(-\beta c_{ij})$;	$\sum_{j=1}^{m} T'_{ij} = O_i$; $\sum_{i=1}^{n} T'_{ij} = D_j$

It is laborious to work through all the models but we can pick one of these models and show how we can unpack it and work out the scaling constants. We will take the ***origin-constrained model*** and adapt it to a retailing example. Here is the model

$$T'_{ij} = A_i O_i D_j \exp\left(-\beta c_{ij}\right) \; subject\ to \; \sum_{j=1}^{m} T'_{ij} = O_i$$

and this model predicts what we want to know about where people will locate at the destination

$$\sum_{i=1}^{n} T'_{ij} = D'_j$$

Now we must figure out the scaling constant A_i. We substitute the model into $\sum_{j=1}^{m} T'_{ij} = O_i$ and re-arrange as

$$\sum_{j=1}^{m} T'_{ij} = A_i O_i \sum_j D_j \exp\left(-\beta c_{ij}\right) = O_i$$

and then by cancelling O_i from both sides we get

$$A_i = 1/\sum_j D_j \exp\left(-\beta c_{ij}\right)$$

This then lets us predict the amount of activity attracted to destination j

$$D'_j = \sum_{i=1}^{n} T'_{ij}$$

If the origin activity is expenditure on retailing where the population lives and the destination is some measure of the size of activity such as retail floorspace. Then the predicted activity is sales at the destination shopping centres

Now the new computer program on the web site

Spatial-Interaction-Models.ipynb

Lets you choose one of these four models and using the same hypothetical data set shown earlier , it runs the model and presents the input and output data and the correlations of predicted v observed trips, origins and destinations.

```
Enter the Grid Size: The Number of Squares Along One Side of a Square Grid, 5 or 10, say ---  5
The Size of the Hypothetical Spatial System is 5 Zones by 5 Zones, Making 25 in All

Enter the Parameter Value on Distance - beta - that You Think Best Fits the Data: It should be greater than 0 and less than 1 ----  0.5

You now have to input one of the Four Model Variants that you want to run

We will number these as

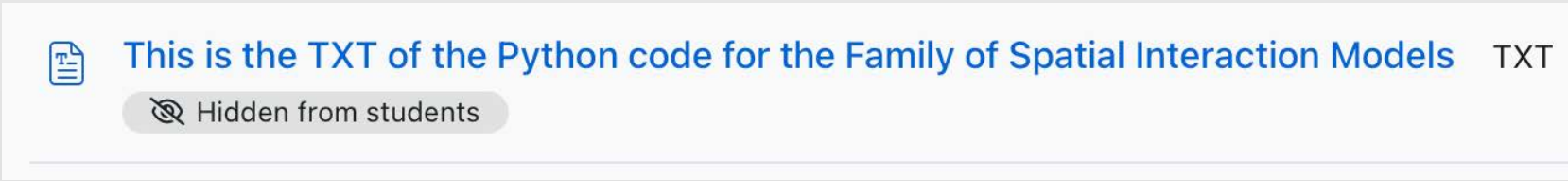
1 The Unconstrained Model as in the Basic Gravity Model Program
2 The Singly-Constrained (Origin-Constrained) Model
3 The Singly-Constrained (Destination-Constrained) Model
4 The Doubly-Constrained (Origin-and-Destination Constrained) Model

Type in the Relevant Number from 1 to 4 3
You have chosen the Destination-Constrained Model
Zone     ObsO    ObsD    PredO   PredD

1        28.93   34.57   22.18   34.57
2        45.28   32.50   41.04   32.50
3        39.63   39.23   37.97   39.23
```

I will now load Anaconda and then go to Moodle and get the text file and copy that to the clipboard and paste it into a new file on the Jupyter Notebook. I will then run the model. Note that we can pick any of the four models from this program and run them and look at the predicted origins and destinations and trips as we did before for the unconstrained model.

You can download the ipynb file – which is a TXT file and paste it into the Jupyter Notebook and run it as I will show here

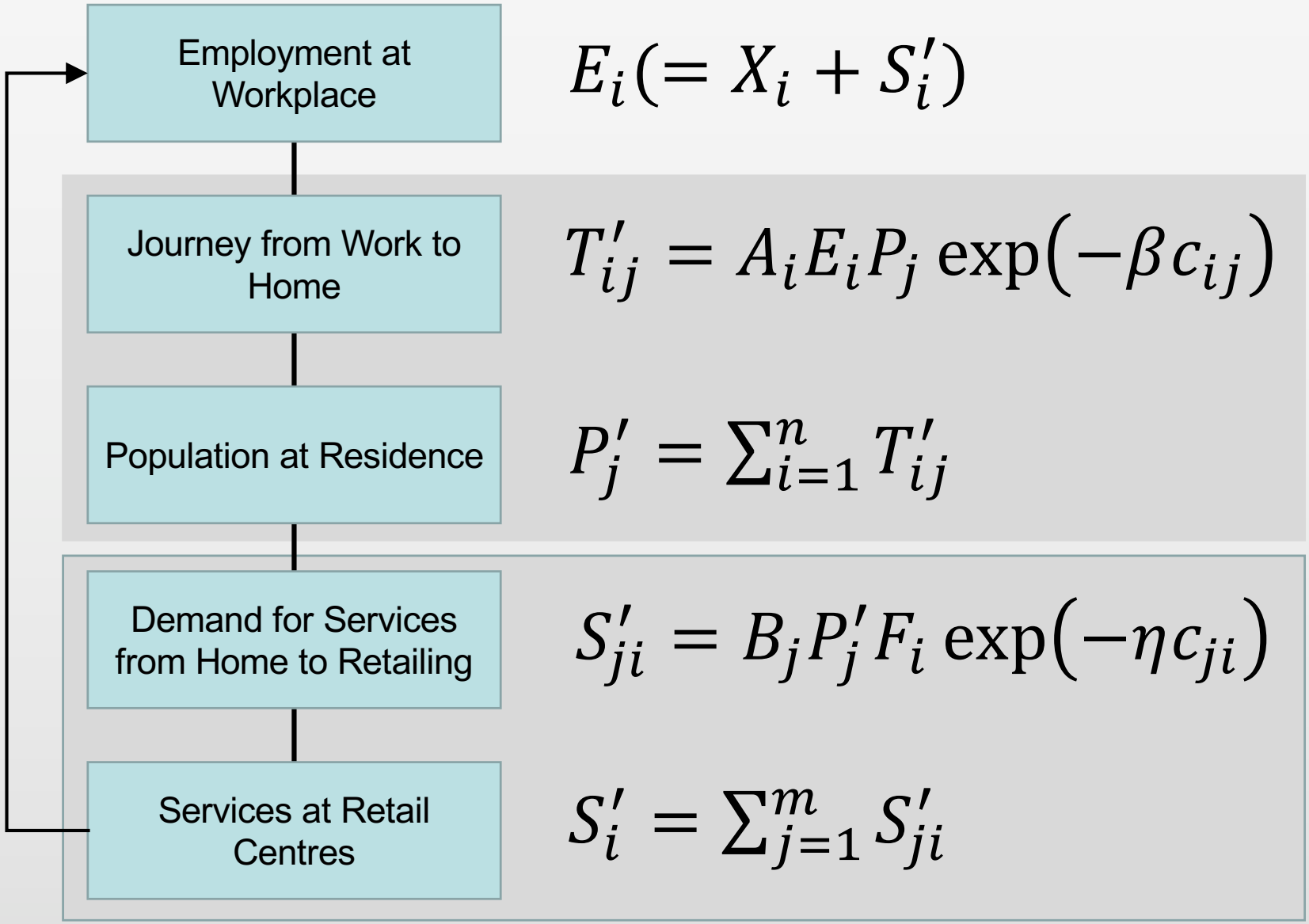


Urban Models: Coupled Spatial Interaction

Ok – we have the ***building blocks*** of more comprehensive urban models now. Imagine we want to build a model of where people work, where they live and where they shop.

We first define where people work as employment at origins E_i and where they live as population at destinations P_j. This can be modelled as a singly constrained model where we predict P_j from E_i and then we predict where these people will shop which relates to the number of workers or size of the shopping centre S_i at the origins.

We thus start with journey to work model defined as



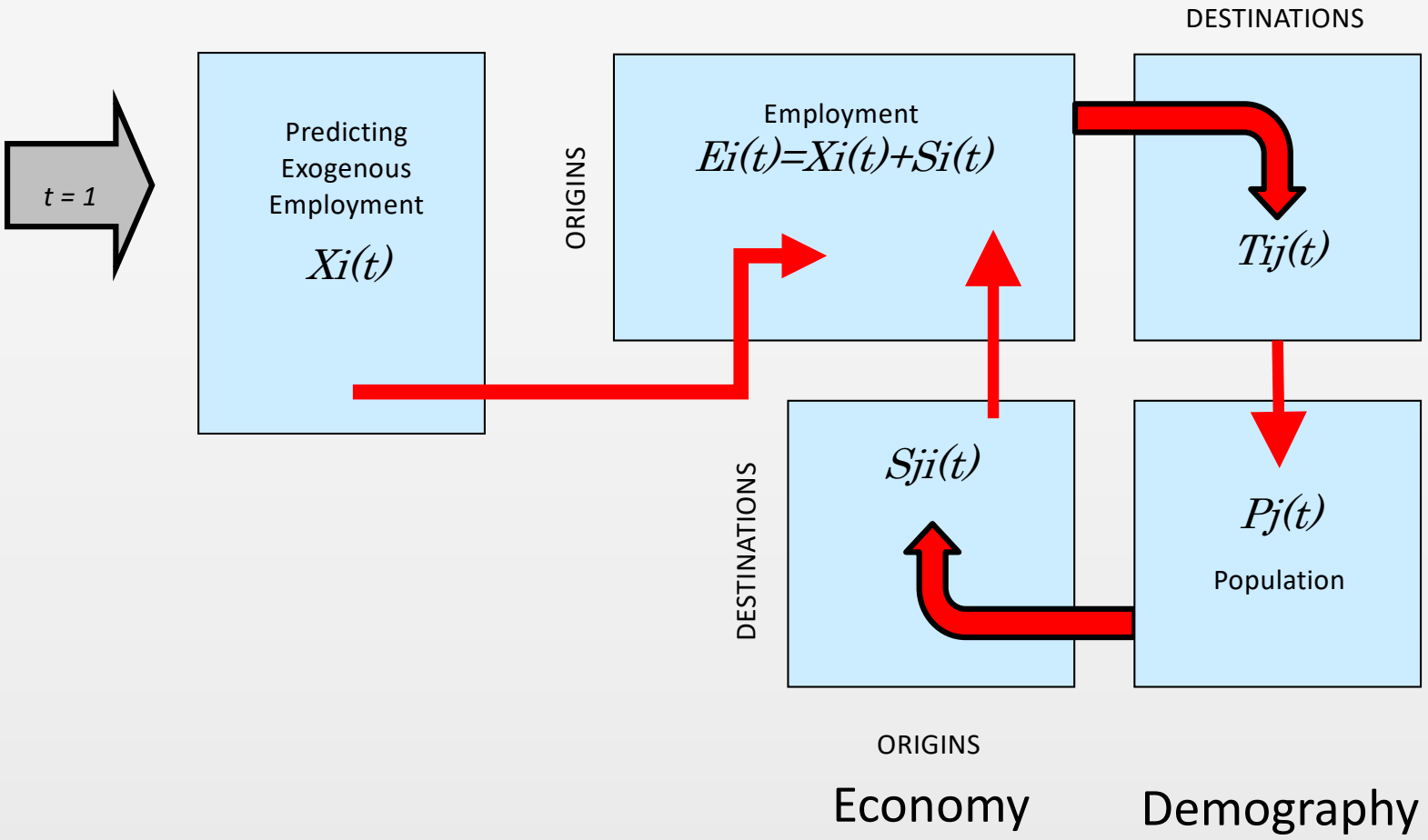
Generalising the Model Adding Employment

I am going to add a third sector – other employment to our model – sometimes it is called 'basic employment' and we are going to model this using a potential function. So we have two kinds of employment –

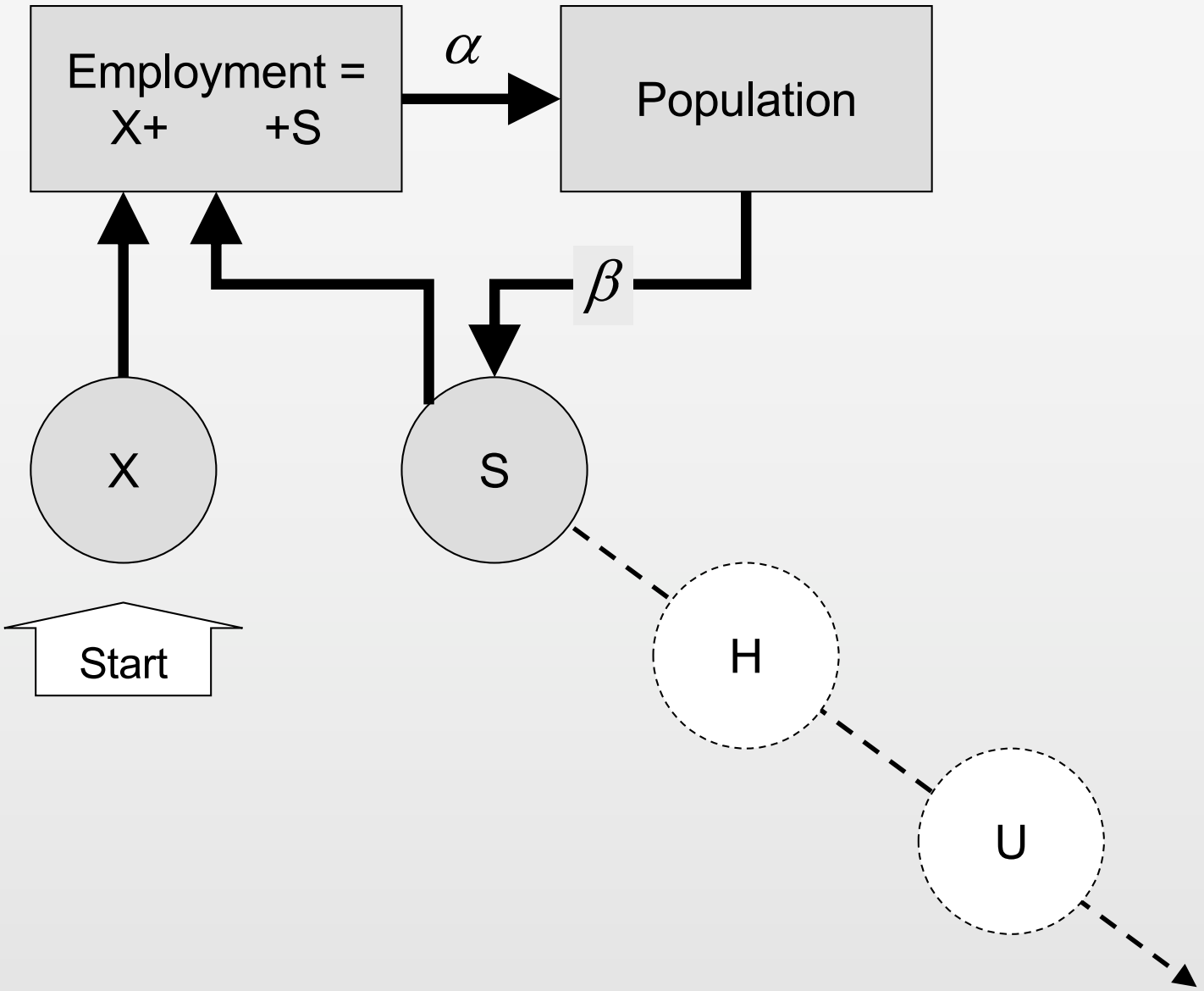
- manufacturing and public service X_i
- retail and commerce S_i

where $E_i = X_i + S_i$ and we have of course population P_j.

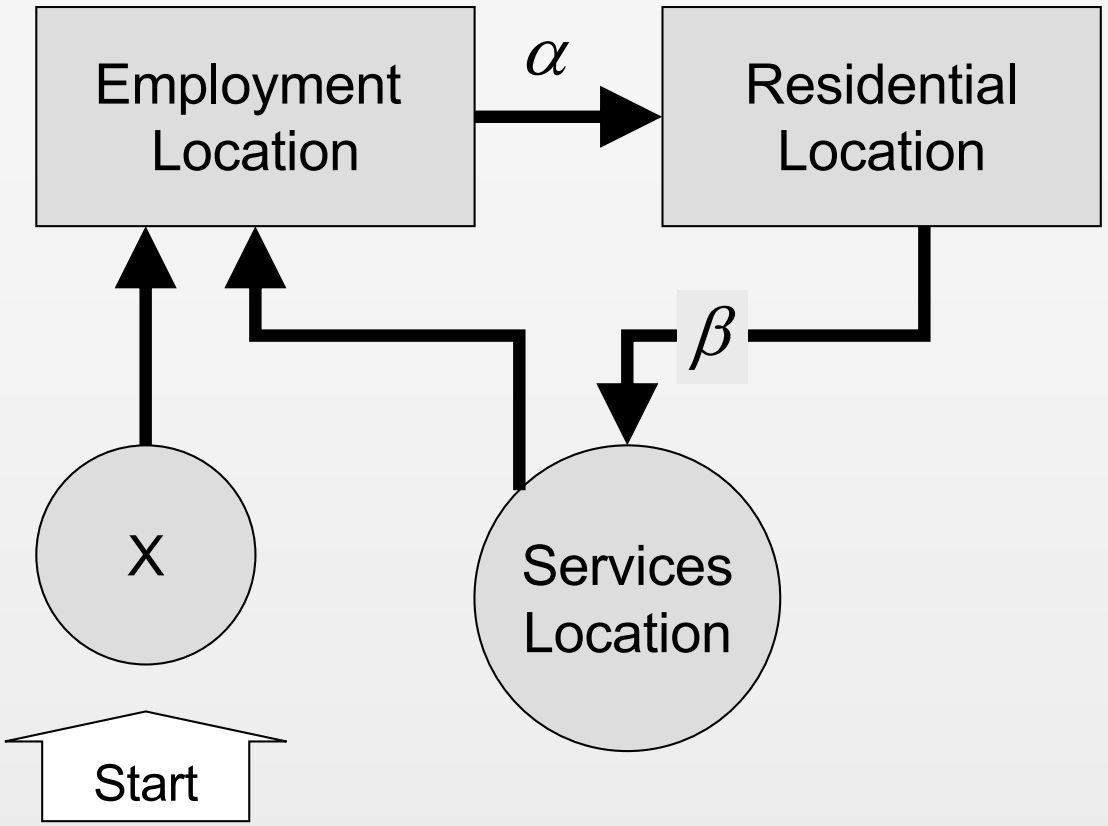
So we can now extend our model as follows – were we have other employment, population and retail employment in that order – our model equations begin with employment



We can keep on extending the model in this way with different sectors but let us look at a simple application for the London region but before this let us demo the extensions



Employment: X=Exogenous; S=Service;



I am not going to detail all these previous models and elaborate the equations but you can see that this is the structure they can fit into – this essentially is the Lowry model framework

Extending the Retail Model to Deal with Agglomeration and Scale

Here is the shopping model that we stated last time:

$$S_{ij} = A_i e P_i F_j \exp(-\beta c_{ij}) = e P_i \frac{F_j \exp(-\beta c_{ij})}{\sum_j F_j \exp(-\beta c_{ij})}$$

Now we can extend it in two ways – first we can add a scaling factor to the attraction on the assumption that as a centre gets bigger, then it has economies of scale. The assumption in economics is that as things get bigger other quantities grow more than proportionately – superlinearly. Thus we can replace

$$F_j \rightarrow\rightarrow\rightarrow F_j^{\alpha}$$

The model thus becomes

$$S_{ij} = eP_i \frac{F_j^{\alpha} \exp(-\beta c_{ij})}{\sum_j F_j^{\alpha} \exp(-\beta c_{ij})}$$

And in fact the example has this kind of scaling factor which we introduced in this and the last lecture. To estimate it, you would need some form of multiple regression because there are two parameters – in fact three including the intercept,

Now we don't calibrate it this way as we will show in a minute but first we need to add an even more appropriate agglomerative effect by taking account of local economies of scales inside a retail centre

We can argue that local shops in a centre exert a positive effect on agglomeration by assuming that the attraction of the centre takes account of these local shops. If we now think of F_j^α as a shop not a set of shops and note that there are other shops F_k^α in the centre j then we can add up the effect of these shops on the attraction using a similar deterrence effect as

$$A_j = F_j^\alpha + \sum_{k \in \Omega_j} F_k^\alpha \exp(-\phi c_{jk})$$

Note we are summing over all the k located shops in the centre j and assuming a deterrent effect – i.e. closer shops exert of a positive effect

Putting this into the model, we get an augmented retail model with three scaling parameters and of course some sort of constraints to ensure normalisation. The model now looks like this

$$S_{ij} = \eta P_i \frac{A_j \exp(-\beta c_{ij})}{\sum_j A_j \exp(-\beta c_{ij})} = \eta P_i \frac{\left[F_j^{\alpha} + \sum_{k \in \Omega_j} F_k^{\alpha} \exp(-\phi c_{jk}) \right] \exp(-\beta c_{ij})}{\sum_j \left[F_j^{\alpha} + \sum_{k \in \Omega_j} F_k^{\alpha} \exp(-\phi c_{jk}) \right] \exp(-\beta c_{ij})}$$

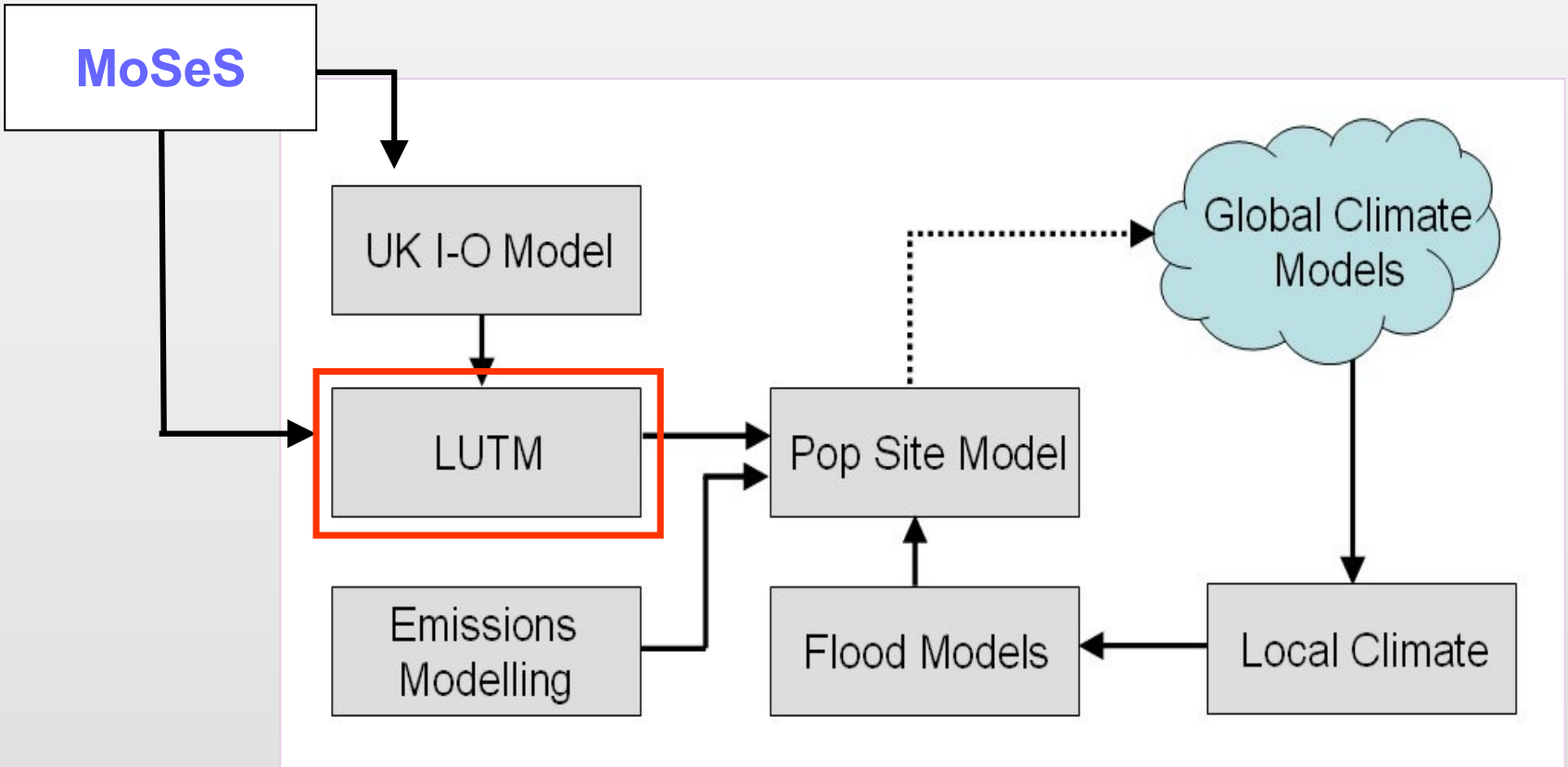
Of course we don't know what sign these parameters can take and it may be that the local effect is truly agglomerative and the parameter is positive

Three Applications and a Fourth Next Time

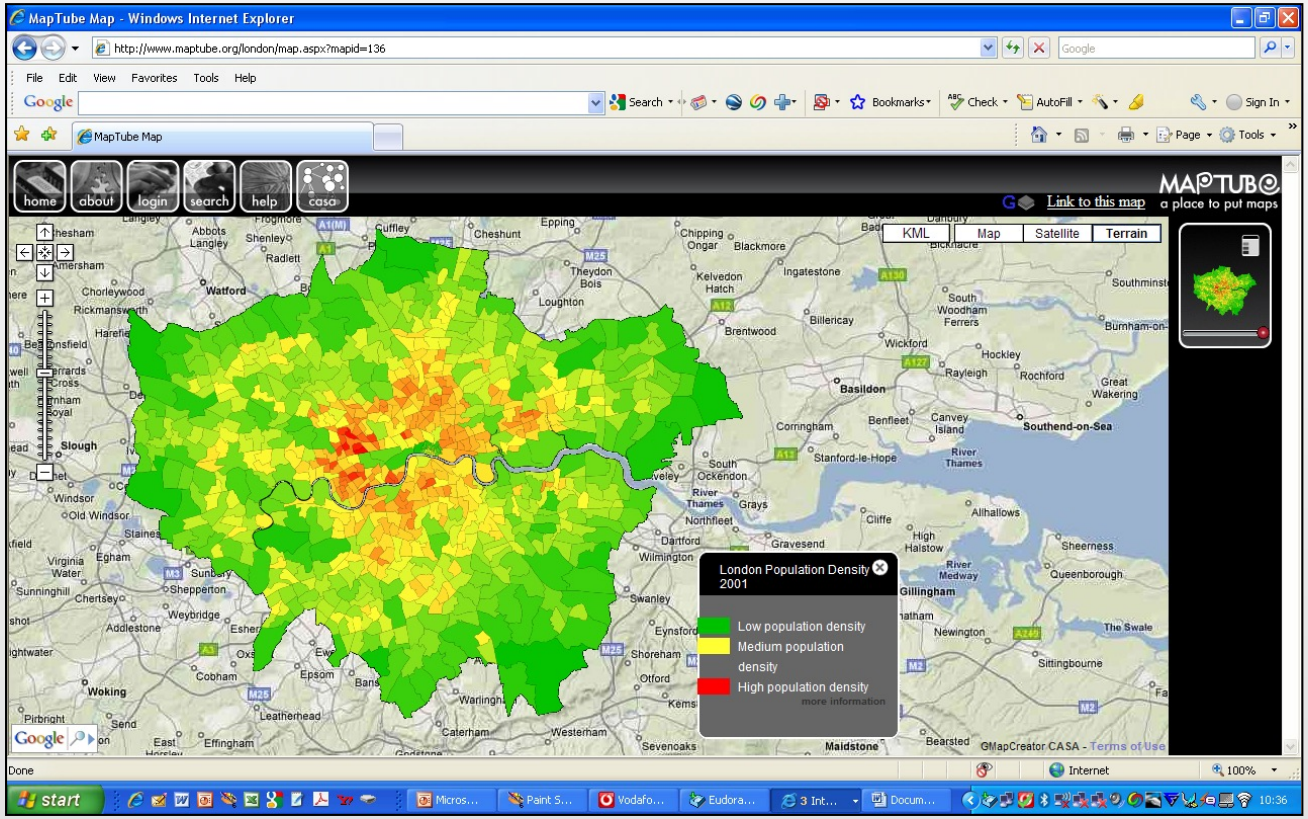
1. A simple gravitational model for Greater London – linked to other models focussed on integrating different infrastructure and economic models
2. The extended model for Greater London and the outer metropolitan area – three sector urban model called SIMULACRA
3. A simple model built rapidly in a two days - a demo for Dubai based on data extracted from the web
4. A large scale LUTI land use transportation interaction model for the UK called QUANT – we will talk about this next time in our fourth lecture

1. The London Tyndall Model Applications

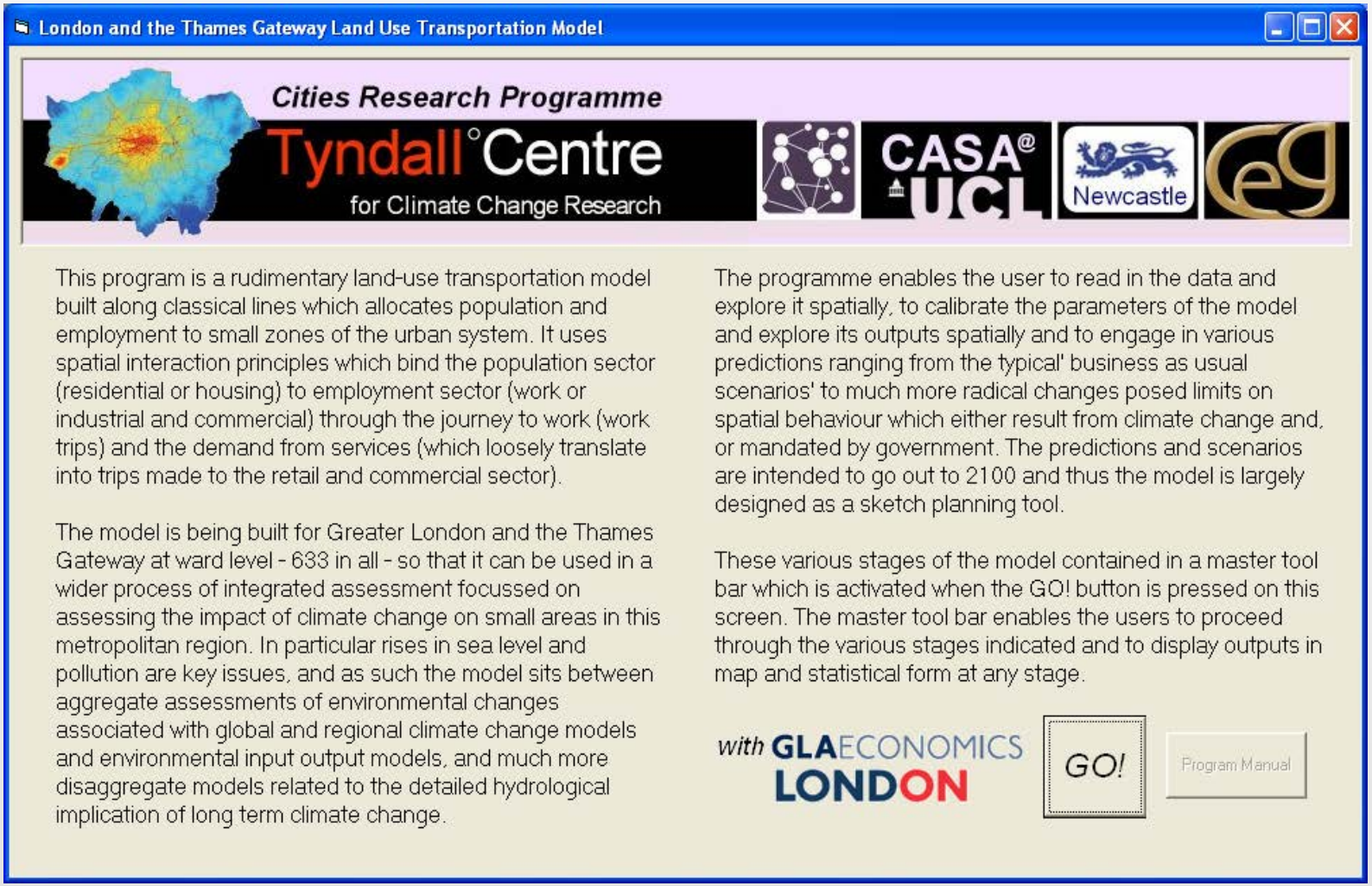
This is an integrated model as shown in the block diagram below. Our focus is on integrating a singly constrained residential local models into the framework.

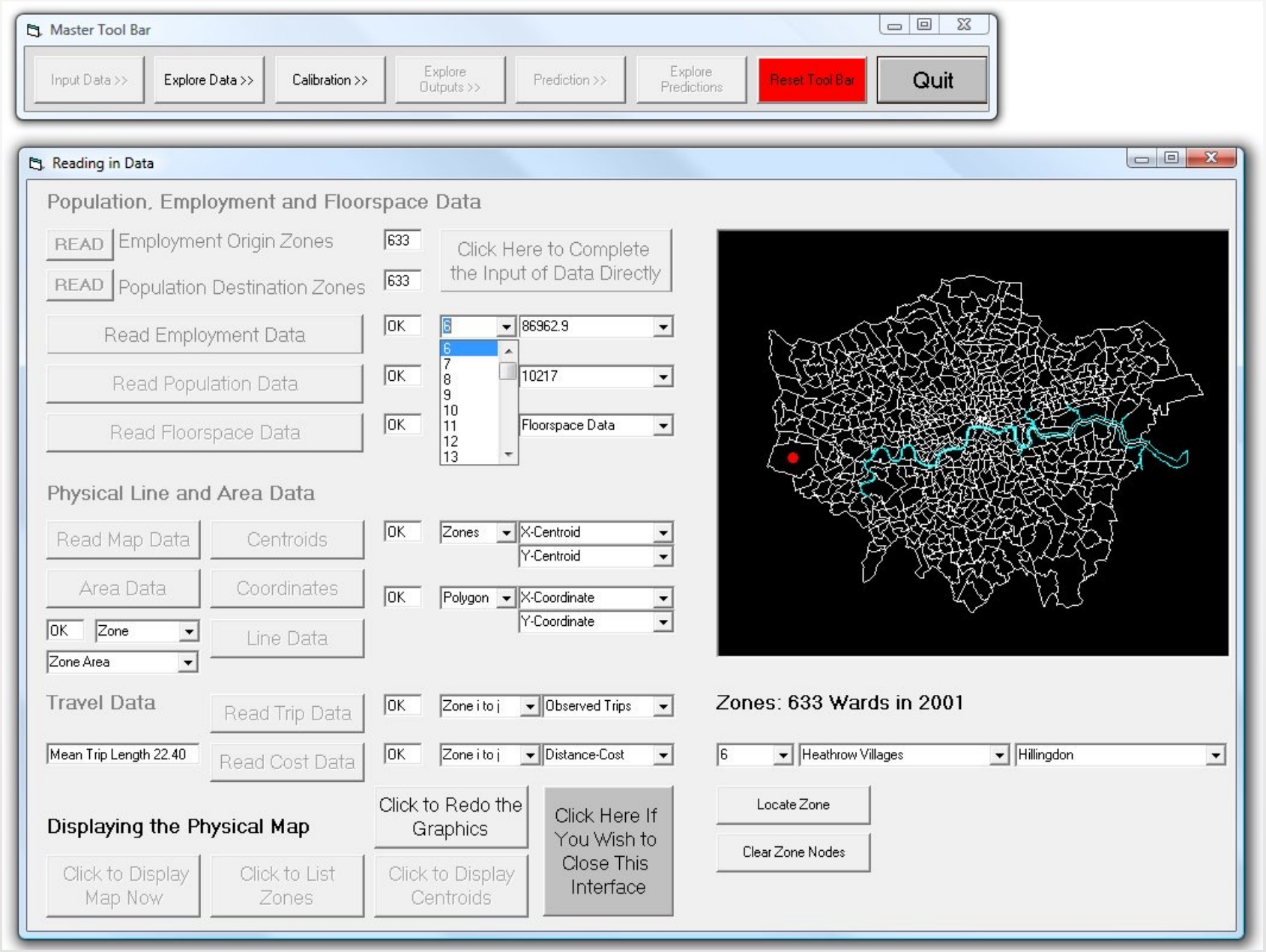


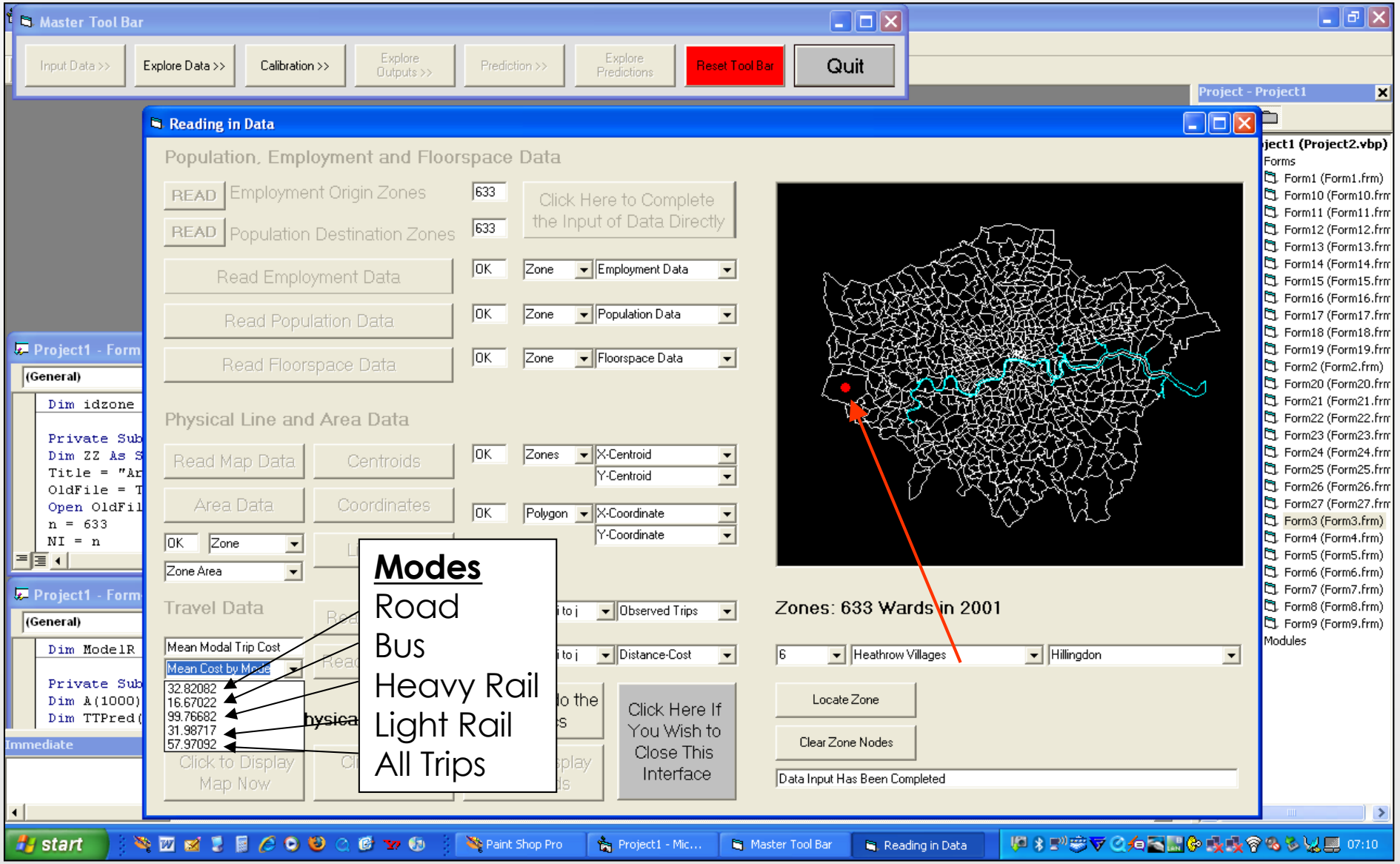
- essentially we have built this model for Greater London which is divided into 633 zones – the area has 7.7m population and about 4.3m jobs –
- we have four modes – road (car), heavy rail, light rail and tube, and bus – walk/bike is a residual mode. First the extent of the area



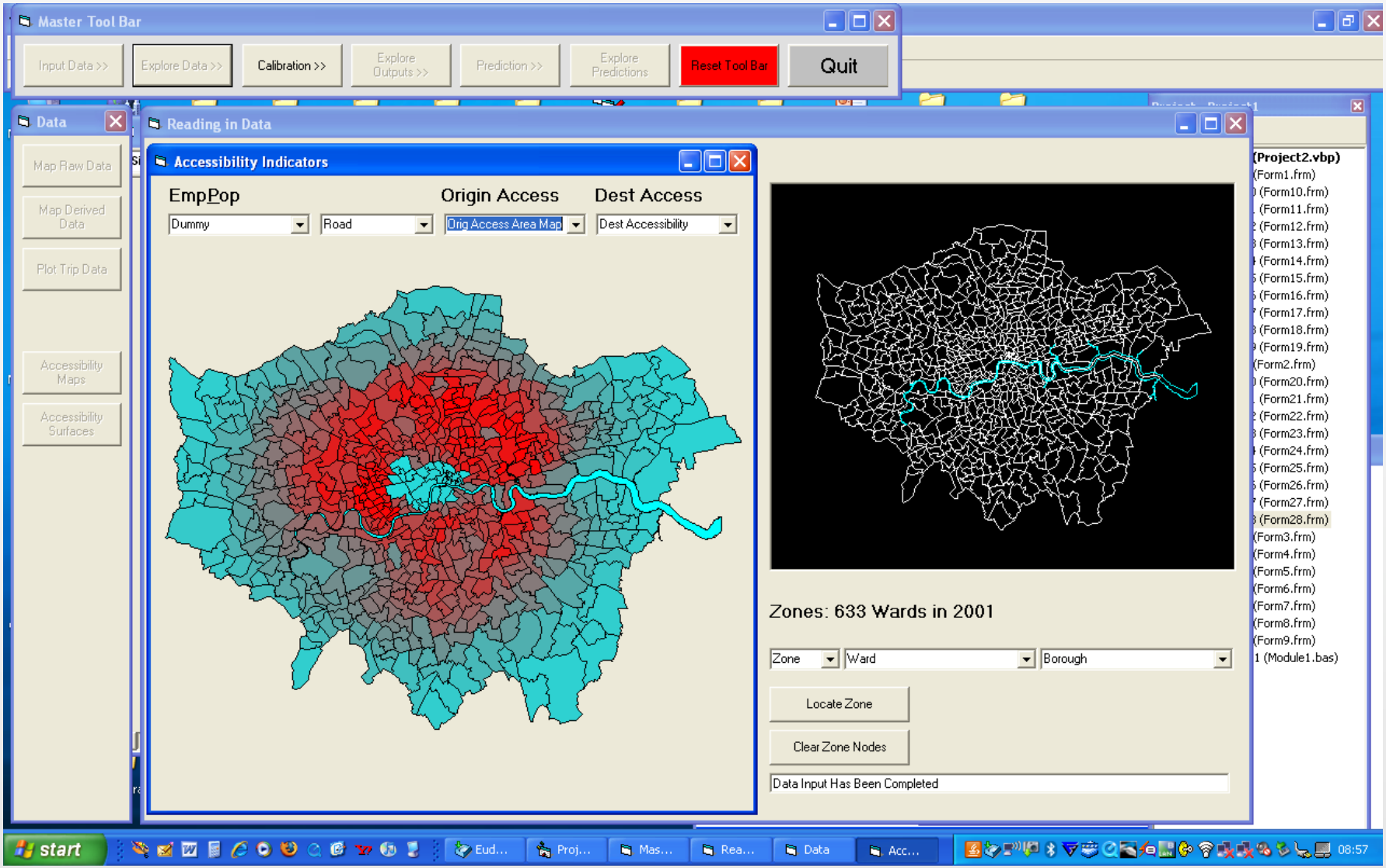
To give a flavour of the LUTI/LUTM model, I will show some screen shots first




Master Tool Bar
Input Data >>
Explore Data >>
Calibration >>
Explore Outputs >>
Prediction >>
Explore Predictions
Reset Tool Bar
Quit
Reading in Data
Population, Employment and Floorspace Data
READ
Employment Origin Zones
633
Click Here to Complete the Input of Data Directly
READ
Population Destination Zones
633
Read Employment Data
OK
6
86962.9
6
7
8
9
10
11
12
13
Read Population Data
OK
10217
Read Floorspace Data
OK
Floorspace Data
Physical Line and Area Data
Read Map Data
Centroids
OK
Zones
X-Centroid
Y-Centroid
Area Data
Coordinates
OK
Polygon
X-Coordinate
Y-Coordinate
OK
Zone
Line Data
Zone Area
Travel Data
Read Trip Data
OK
Zone i to j
Observed Trips
Mean Trip Length 22.40
Read Cost Data
OK
Zone i to j
Distance-Cost
Zones: 633 Wards in 2001
6
Heathrow Villages
Hillingdon
Displaying the Physical Map
Click to Redo the Graphics
Click Here If You Wish to Close This Interface
Locate Zone
Clear Zone Nodes
Click to Display Map Now
Click to List Zones
Click to Display Centroids

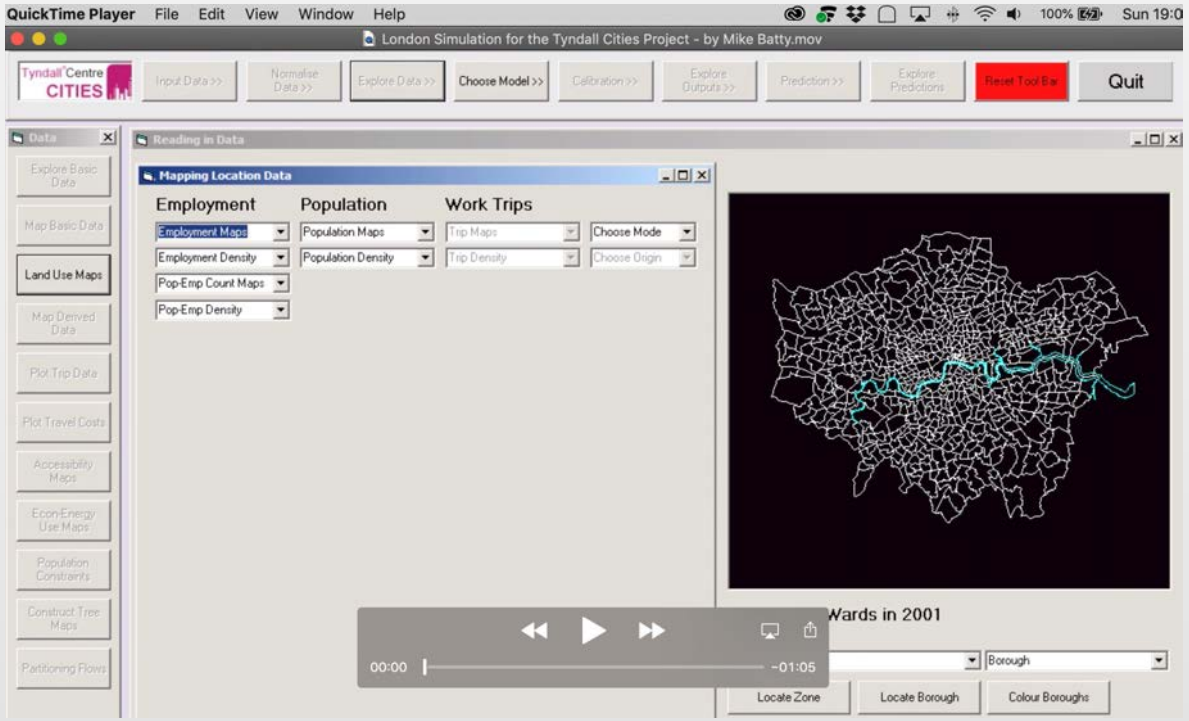


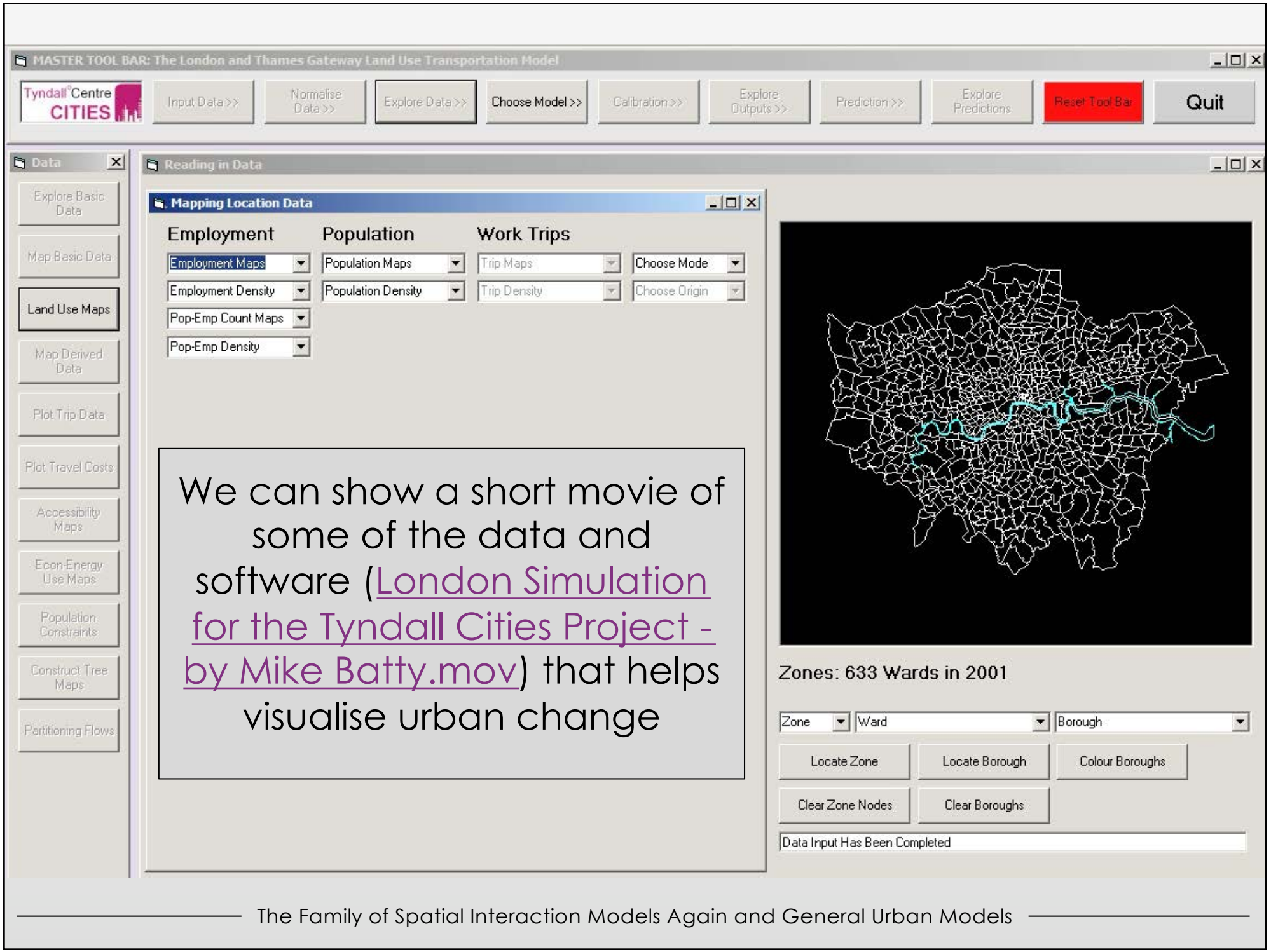
Road: 38%; Bus: 12%: Heavy Rail: 12%: Light Rail 19%; Other (Walk, Bike): 19%


Master Tool Bar
Input Data >>
Explore Data >>
Calibration >>
Explore Outputs >>
Prediction >>
Explore Predictions
Reset Tool Bar
Quit
Data
Map Raw Data
Map Derived Data
Plot Trip Data
Accessibility Maps
Accessibility Surfaces
Reading in Data
Accessibility Indicators
EmpPop
Origin Access
Dest Access
Dummy
Road
Orig Access Area Map
Dest Accessibility
Zones: 633 Wards in 2001
Zone
Ward
Borough
Locate Zone
Clear Zone Nodes
Data Input Has Been Completed
(Project2.vbp)
(Form1.frm)
(Form10.frm)
(Form11.frm)
(Form12.frm)
(Form13.frm)
(Form14.frm)
(Form15.frm)
(Form16.frm)
(Form17.frm)
(Form18.frm)
(Form19.frm)
(Form2.frm)
(Form20.frm)
(Form21.frm)
(Form22.frm)
(Form23.frm)
(Form24.frm)
(Form25.frm)
(Form26.frm)
(Form27.frm)
(Form28.frm)
(Form3.frm)
(Form4.frm)
(Form5.frm)
(Form6.frm)
(Form7.frm)
(Form8.frm)
(Form9.frm)
1 (Module1.bas)
start
08:57

Ok let me load a movie

I will need to leave powerpoint




MASTER TOOL BAR: The London and Thames Gateway Land Use Transportation Model
Tyndall Centre CITIES
Input Data >>
Normalise Data >>
Explore Data >>
Choose Model >>
Calibration >>
Explore Outputs >>
Prediction >>
Explore Predictions
Reset Tool Bar
Quit
Data
Explore Basic Data
Map Basic Data
Land Use Maps
Map Derived Data
Plot Trip Data
Plot Travel Costs
Accessibility Maps
Econ-Energy Use Maps
Population Constraints
Construct Tree Maps
Partitioning Flows
Reading in Data
Mapping Location Data
Employment
Population
Work Trips
Employment Maps
Employment Density
Pop-Emp Count Maps
Pop-Emp Density
Population Maps
Population Density
Trip Maps
Trip Density
Choose Mode
Choose Origin
We can show a short movie of some of the data and software (London Simulation for the Tyndall Cities Project - by Mike Batty.mov) that helps visualise urban change
Zones: 633 Wards in 2001
Zone
Ward
Borough
Locate Zone
Locate Borough
Colour Boroughs
Clear Zone Nodes
Clear Boroughs
Data Input Has Been Completed

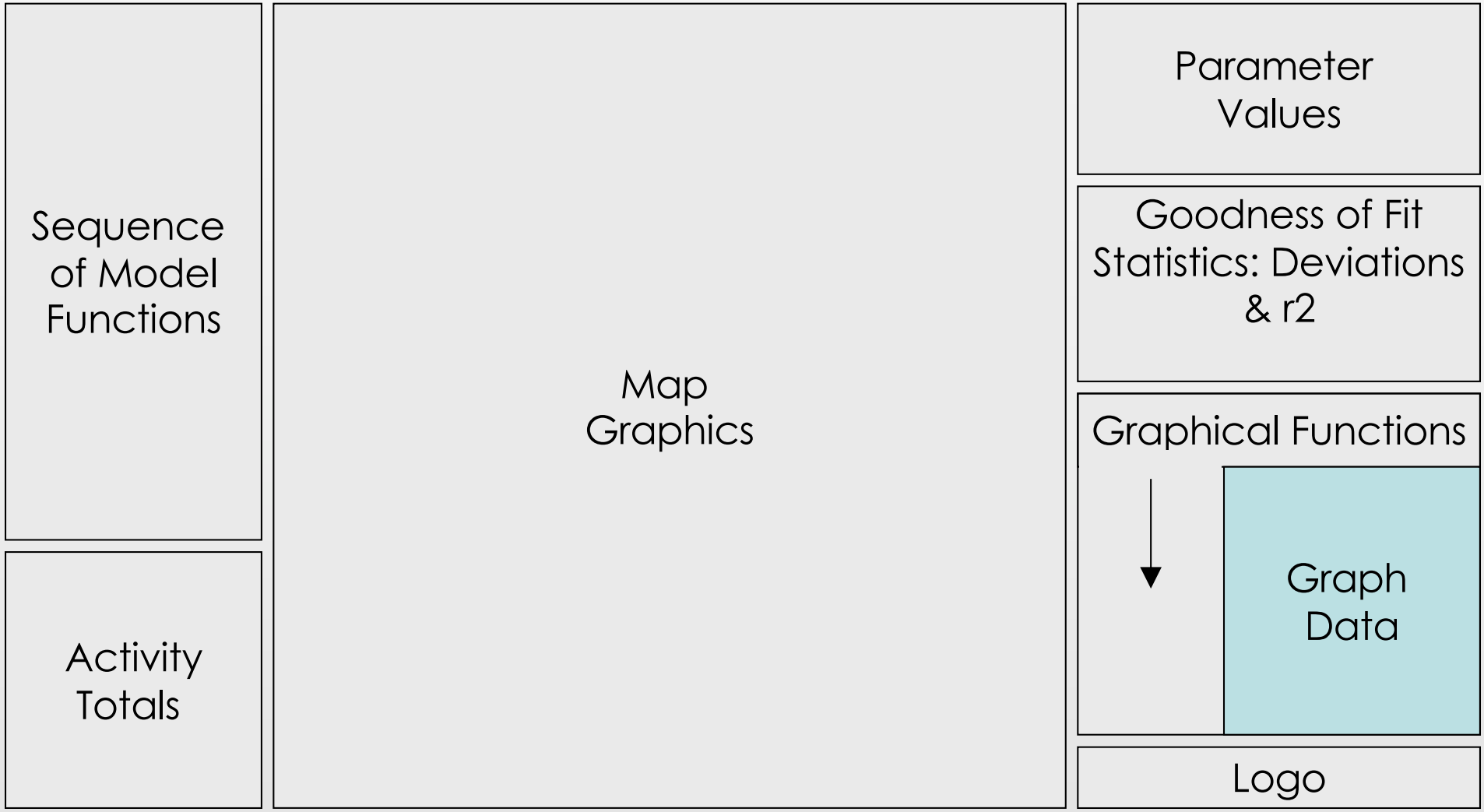
Modular Modelling: Coupled Spatial Interaction

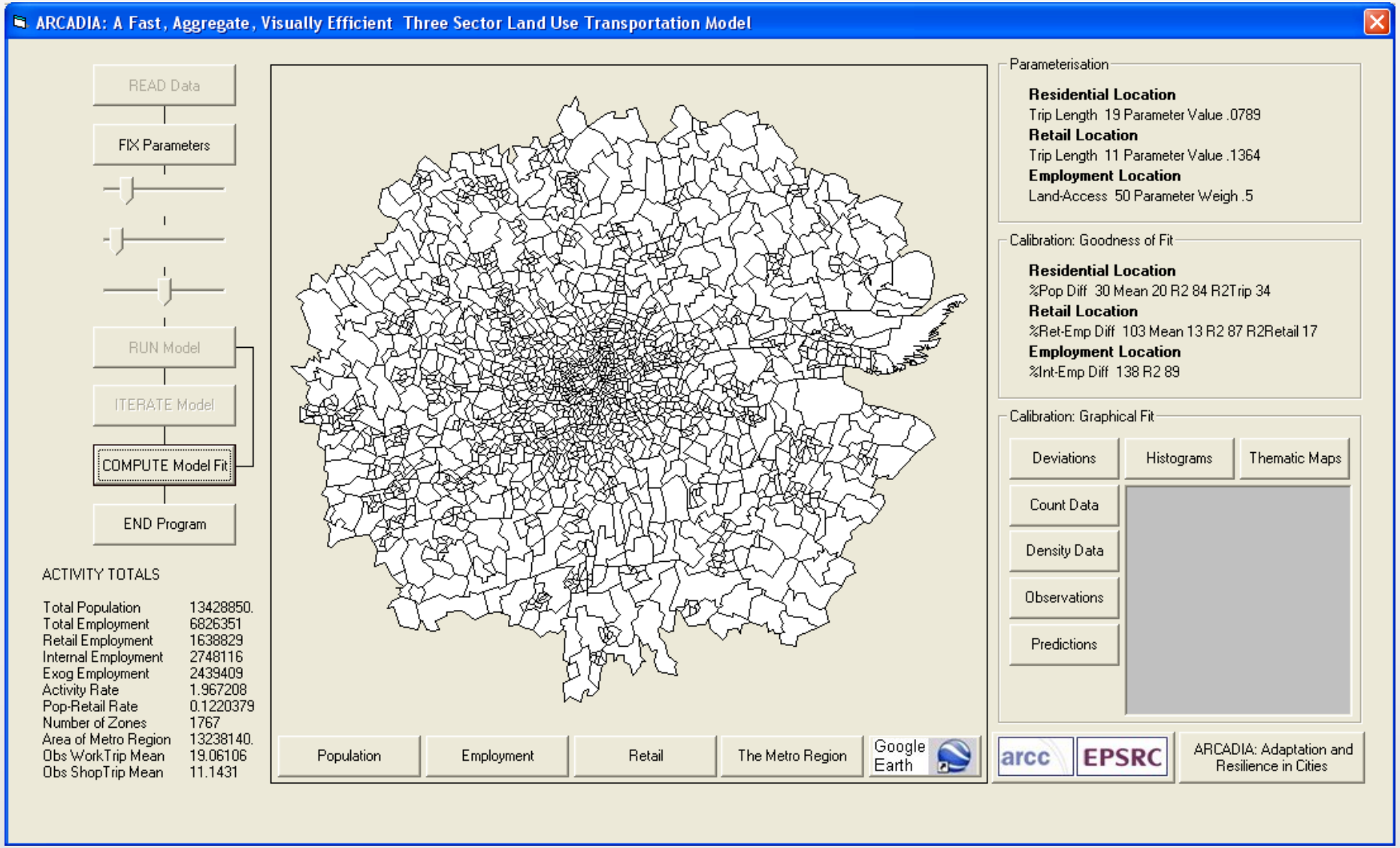
So far we have just singled out a module for one kind of interaction – based on a variant of the gravity model - consider stringing these together as more than one kind of spatial interaction: Model 1 ⇨ Model 2 ⇨ Model 3 ⇨ …..

Classically we might model flows from home to work and home to shop but there are many more and in this sense, we can use these as building blocks for wider models. This is for next time too

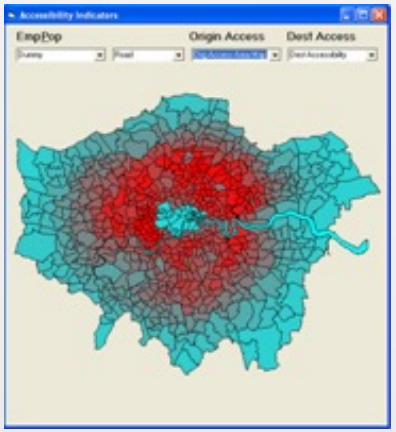
What we will now do is illustrate how we might build such a structure taking a journey to work model from Employment to Population and then to Shopping which we structure as --

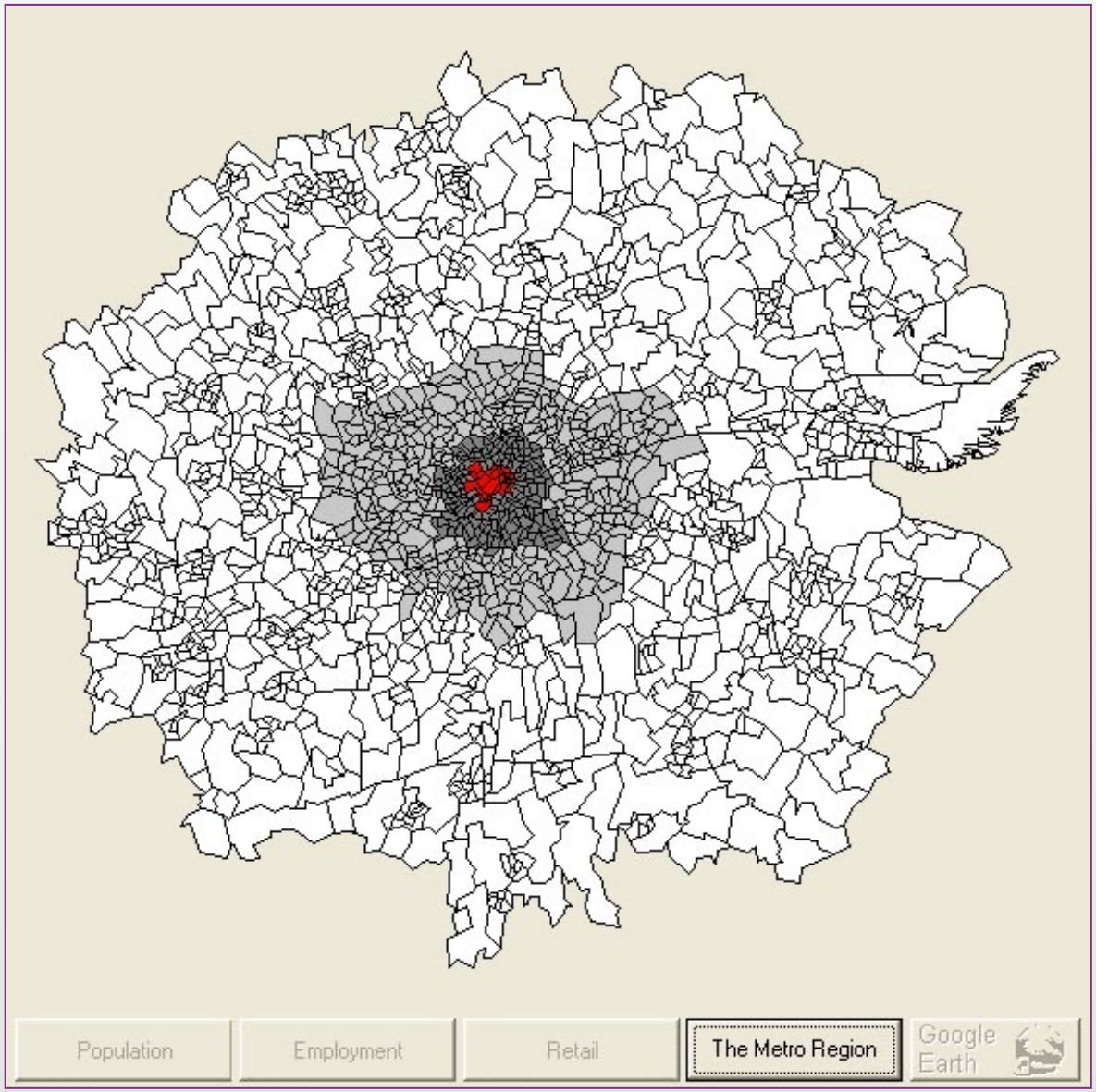
This is the order in which the operations take place

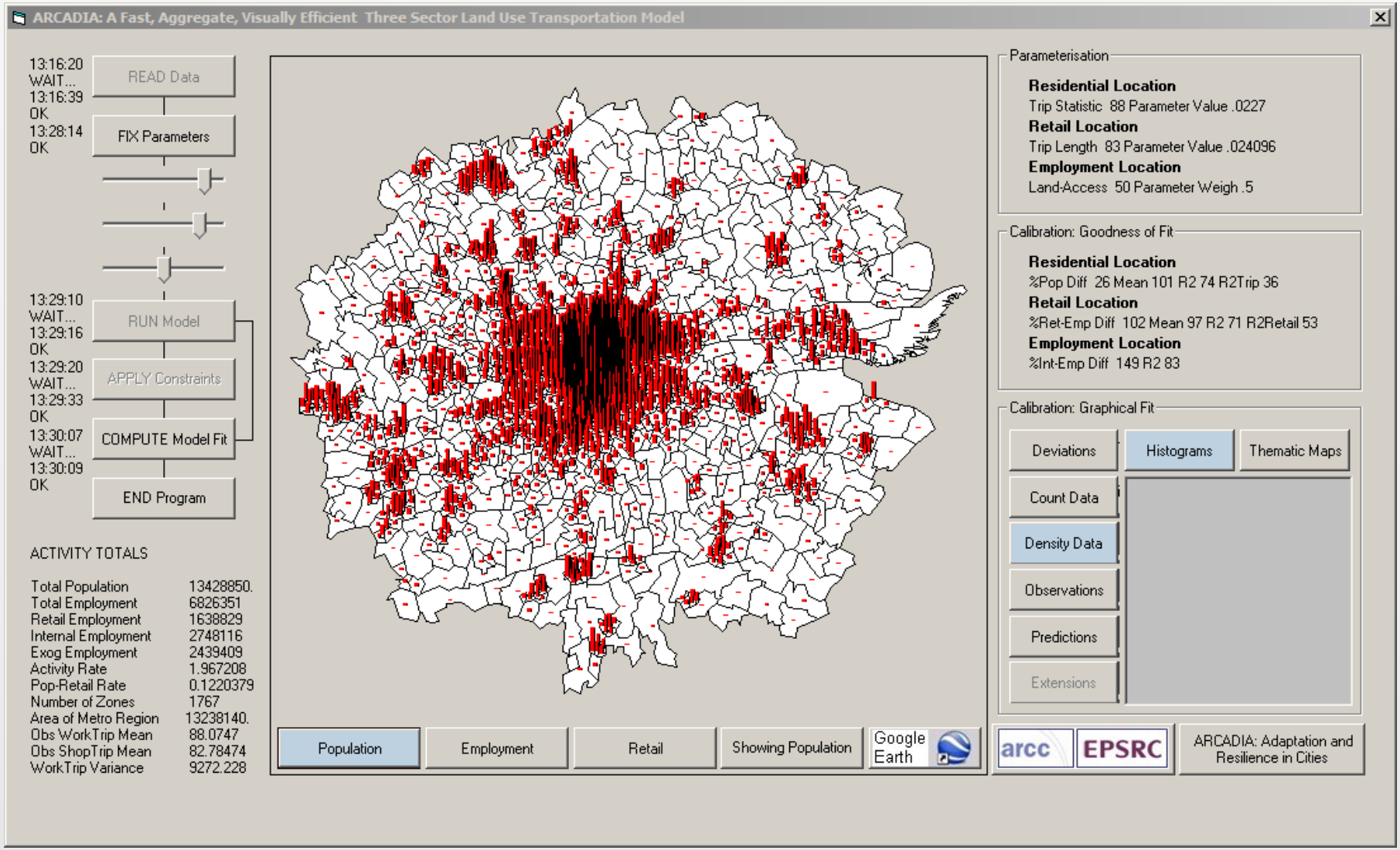


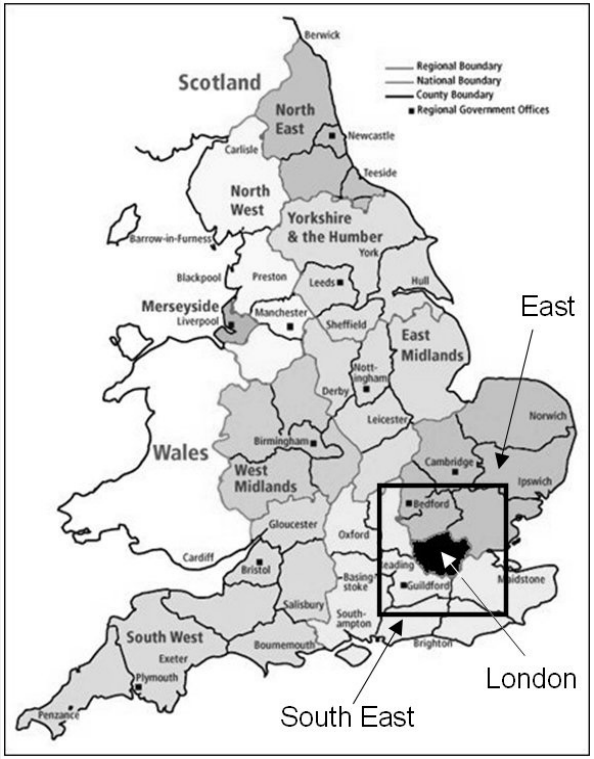

ARCADIA: A Fast, Aggregate, Visually Efficient Three Sector Land Use Transportation Model
READ Data
FIX Parameters
RUN Model
ITERATE Model
COMPUTE Model Fit
END Program
ACTIVITY TOTALS
Total Population 13428850.
Total Employment 6826351
Retail Employment 1638829
Internal Employment 2748116
Exog Employment 2439409
Activity Rate 1.967208
Pop-Retail Rate 0.1220379
Number of Zones 1767
Area of Metro Region 13238140.
Obs WorkTrip Mean 19.06106
Obs ShopTrip Mean 11.1431
Population
Employment
Retail
The Metro Region
Google Earth
Parameterisation
Residential Location
Trip Length 19 Parameter Value .0789
Retail Location
Trip Length 11 Parameter Value .1364
Employment Location
Land-Access 50 Parameter Weigh .5
Calibration: Goodness of Fit
Residential Location
%Pop Diff 30 Mean 20 R2 84 R2Trip 34
Retail Location
%Ret-Emp Diff 103 Mean 13 R2 87 R2Retail 17
Employment Location
%Int-Emp Diff 138 R2 89
Calibration: Graphical Fit
Deviations
Histograms
Thematic Maps
Count Data
Density Data
Observations
Predictions
arcc
EPSRC
ARCADIA: Adaptation and Resilience in Cities


Employment
Population
Work Trips

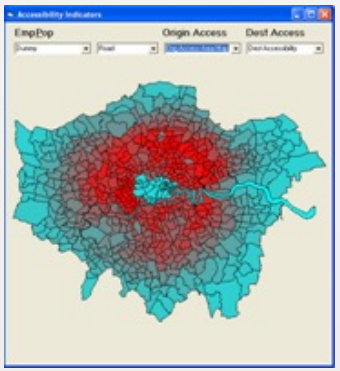

EmpPop
Origin Access
Dest Access

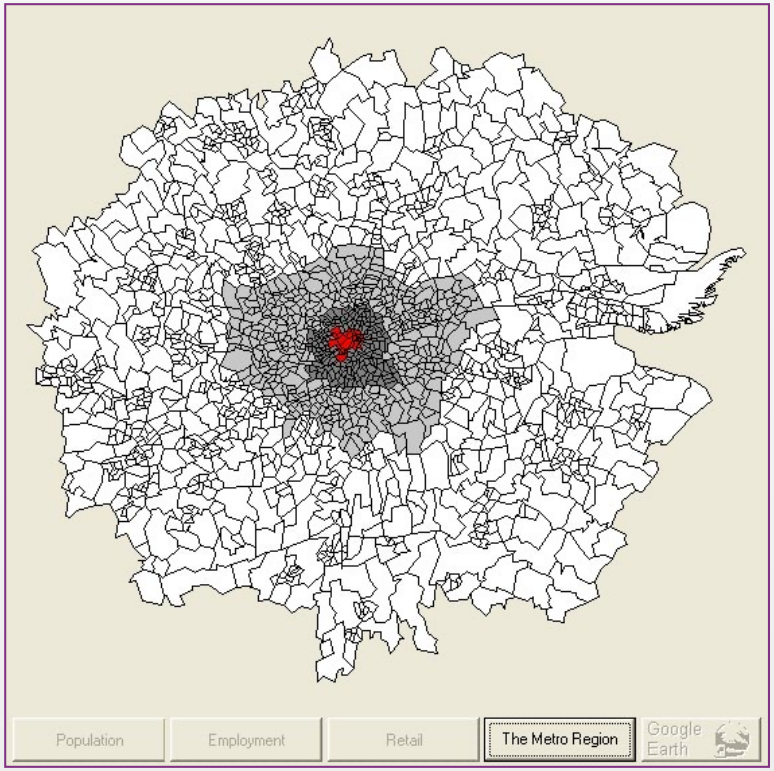

Population
Employment
Retail
The Metro Region
Google Earth

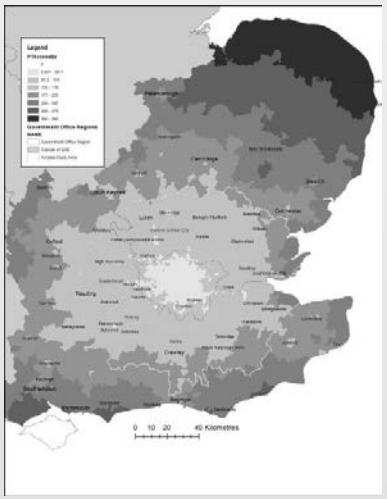

ARCADIA: A Fast, Aggregate, Visually Efficient Three Sector Land Use Transportation Model
13:16:20 WAIT...
13:16:39 OK
READ Data
13:28:14 OK
FIX Parameters
13:29:10 WAIT...
13:29:16 OK
RUN Model
13:29:20 WAIT...
13:29:33 OK
APPLY Constraints
13:30:07 WAIT...
13:30:09 OK
COMPUTE Model Fit
END Program
ACTIVITY TOTALS
Total Population 13428850.
Total Employment 6826351
Retail Employment 1638829
Internal Employment 2748116
Exog Employment 2439409
Activity Rate 1.967208
Pop-Retail Rate 0.1220379
Number of Zones 1767
Area of Metro Region 13238140.
Obs WorkTrip Mean 88.0747
Obs ShopTrip Mean 82.78474
WorkTrip Variance 9272.228
Population
Employment
Retail
Showing Population
Google Earth
Parameterisation
Residential Location
Trip Statistic 88 Parameter Value .0227
Retail Location
Trip Length 83 Parameter Value .024096
Employment Location
Land-Access 50 Parameter Weigh .5
Calibration: Goodness of Fit
Residential Location
%Pop Diff 26 Mean 101 R2 74 R2Trip 36
Retail Location
%Ret-Emp Diff 102 Mean 97 R2 71 R2Retail 53
Employment Location
%Int-Emp Diff 149 R2 83
Calibration: Graphical Fit
Deviations
Histograms
Thematic Maps
Count Data
Density Data
Observations
Predictions
Extensions
arcc
EPSRC
ARCADIA: Adaptation and Resilience in Cities

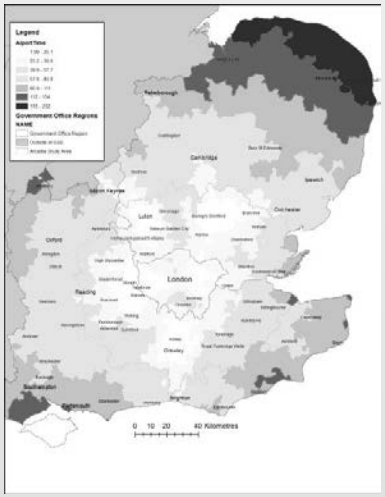

Scotland
North East
North West
Yorkshire & the Humber
East Midlands
East
Wales
West Midlands
South West
South East
London
Regional Boundary
National Boundary
County Boundary
Regional Government Offices

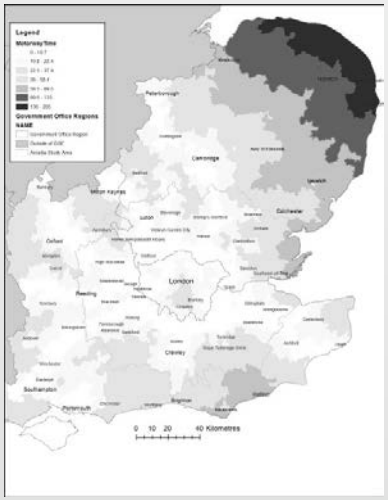

Employment
Population
Work Trips

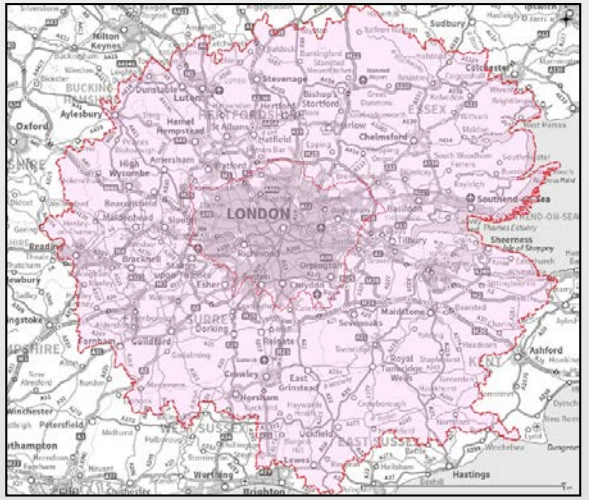

Emp/Pop
Origin Access
Dest Access


Population
Employment
Retail
The Metro Region
Google Earth



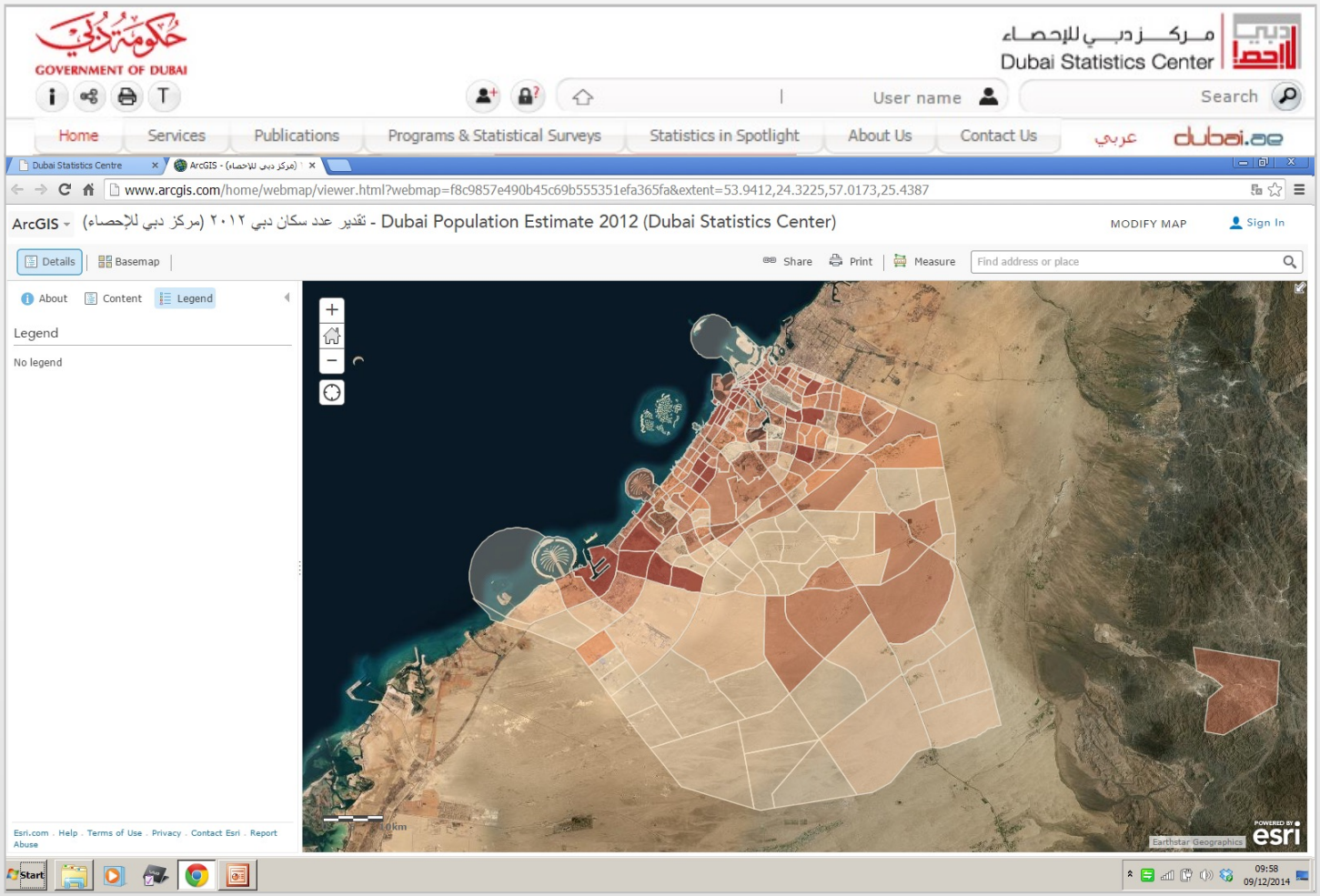




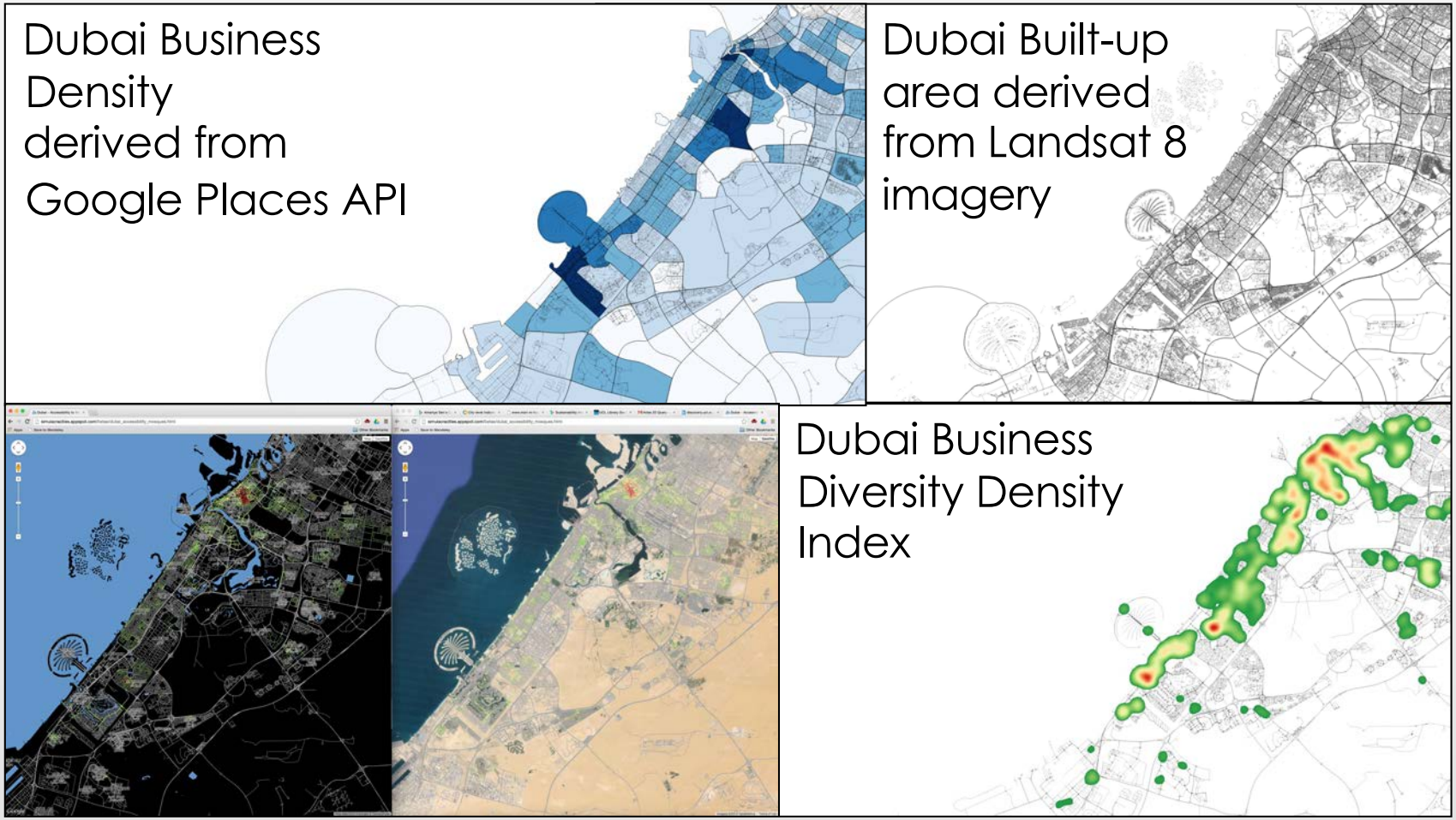

LONDON

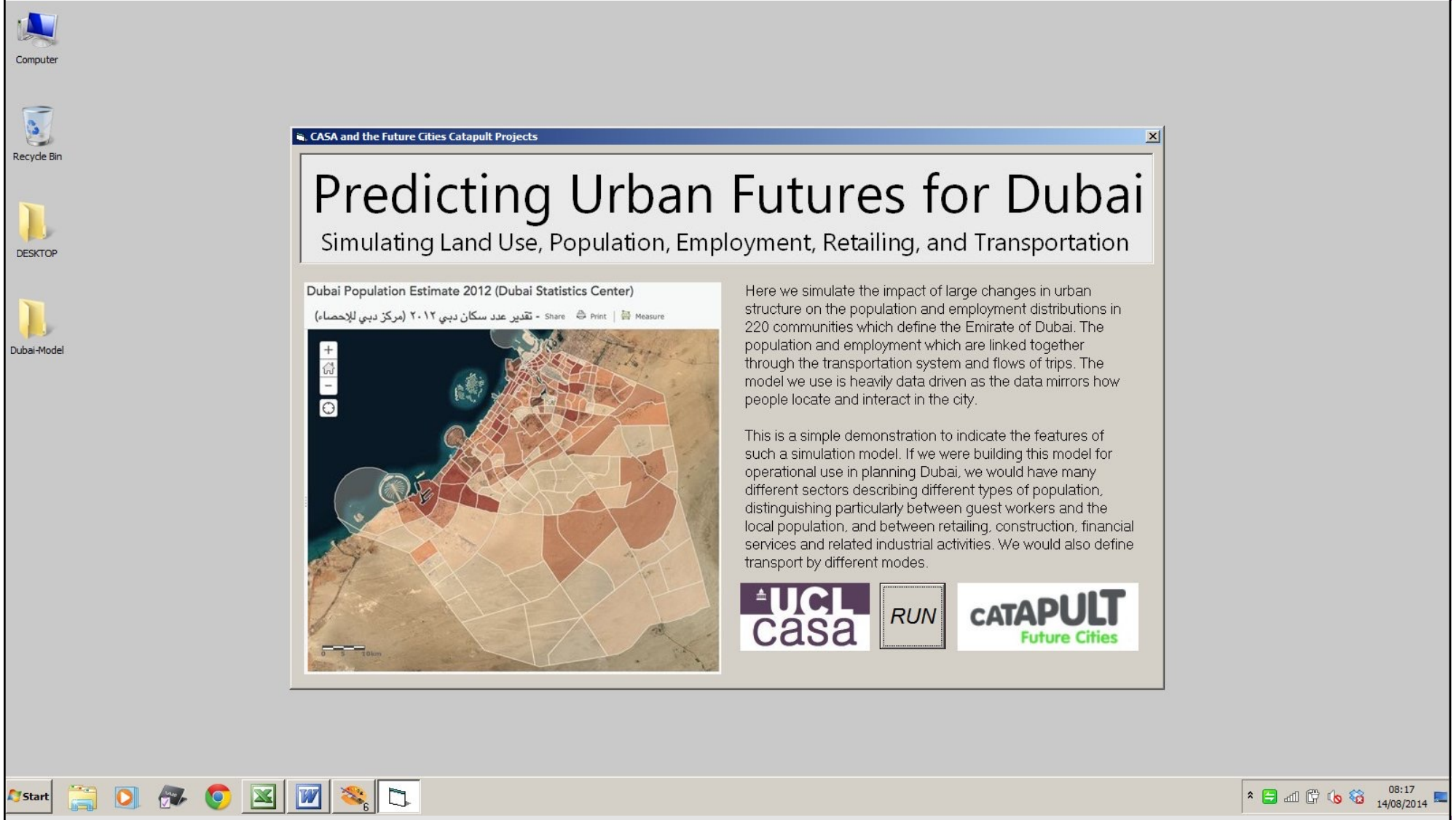
Quick and Dirty Models

A New Retail Centre in Dubai

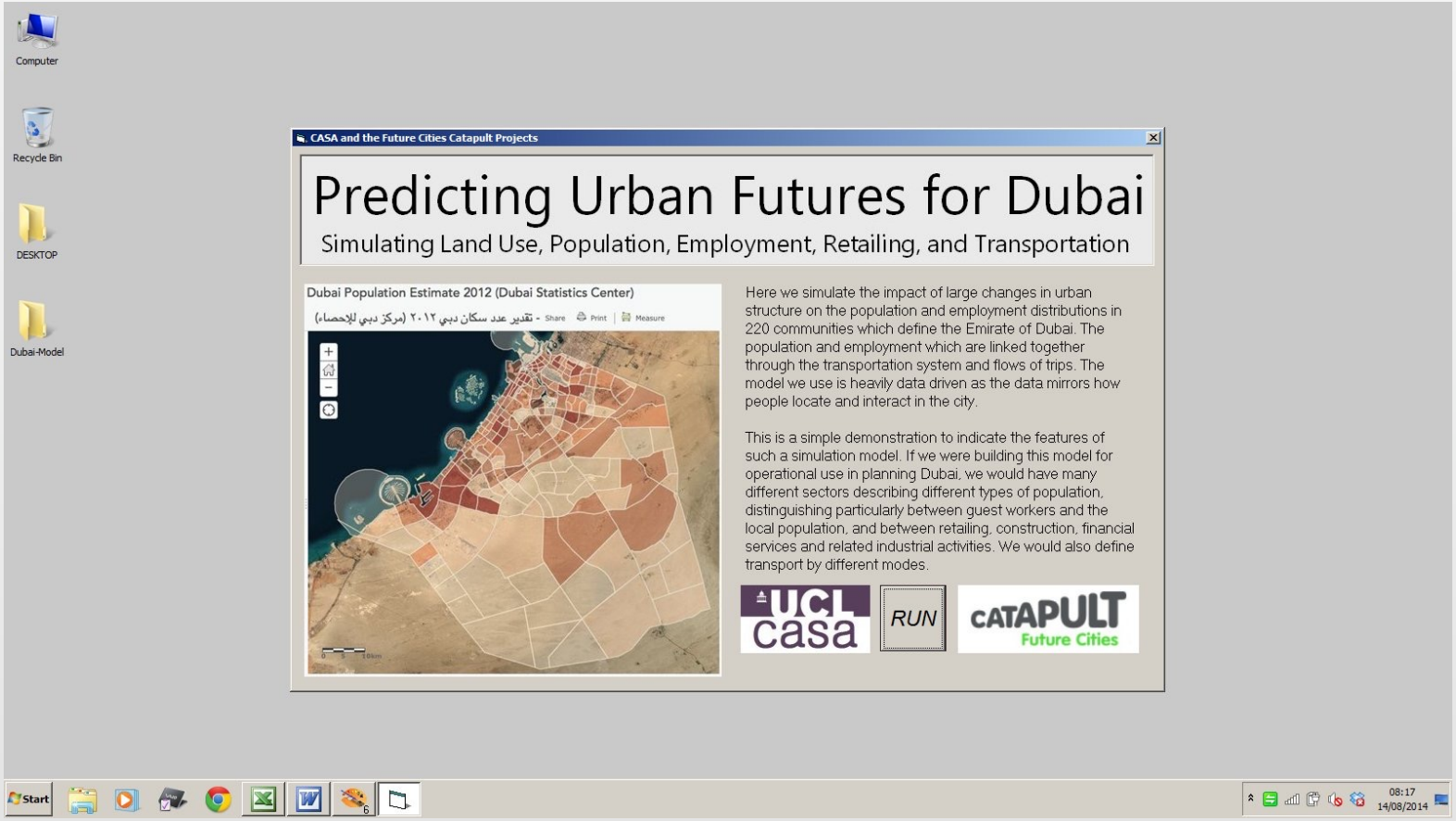


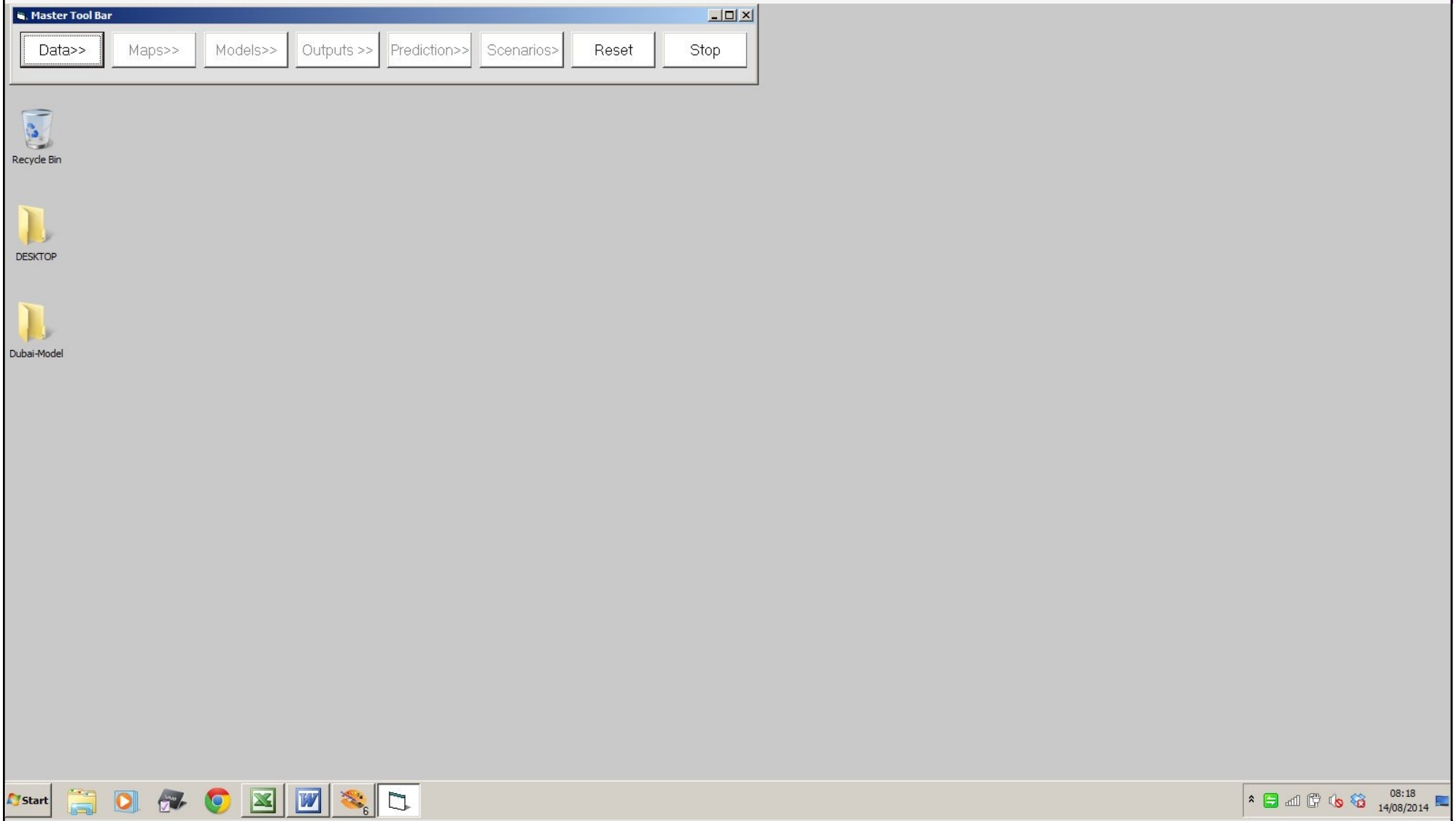
Where did we get the data – in a data poor environment?

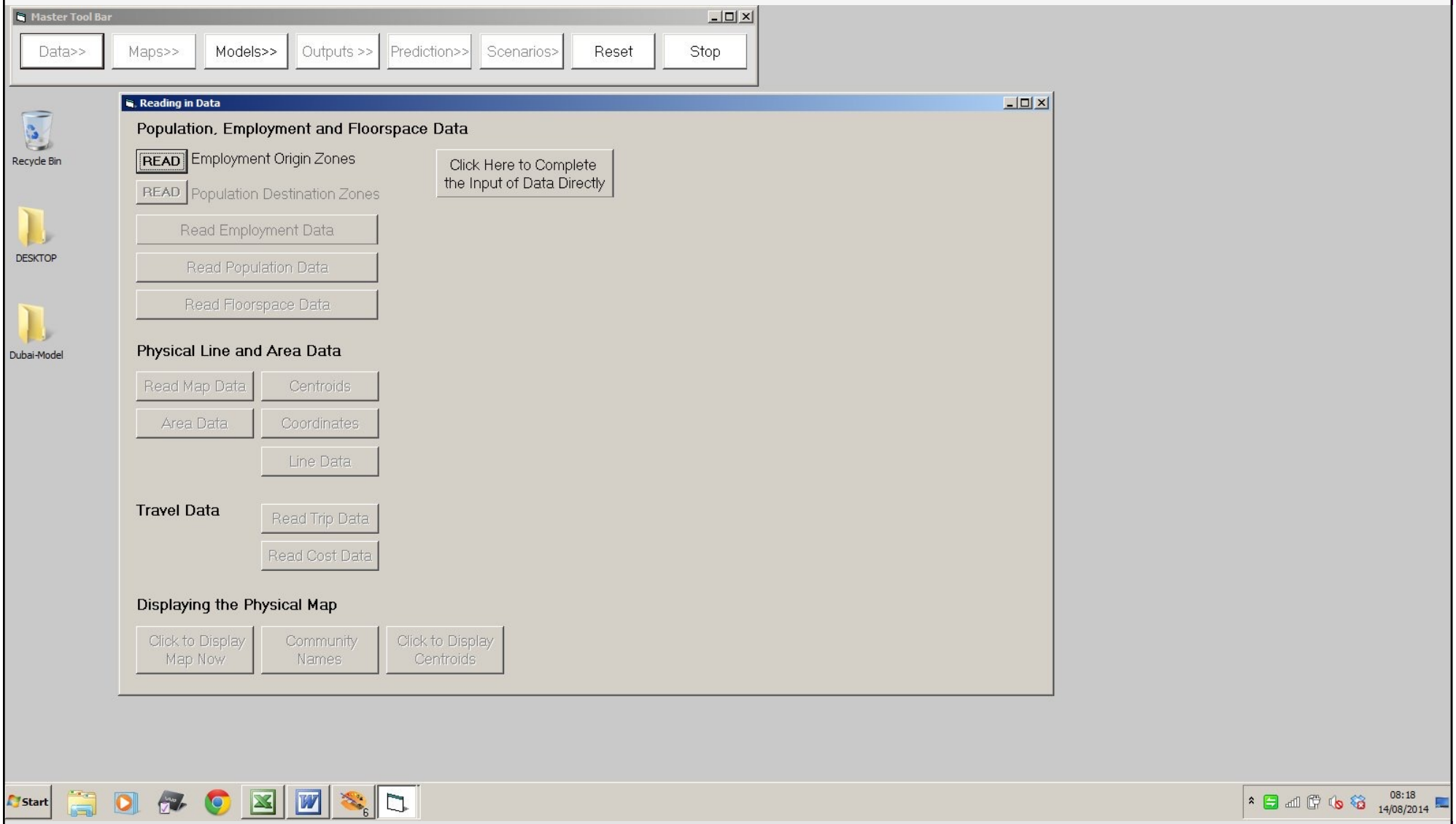


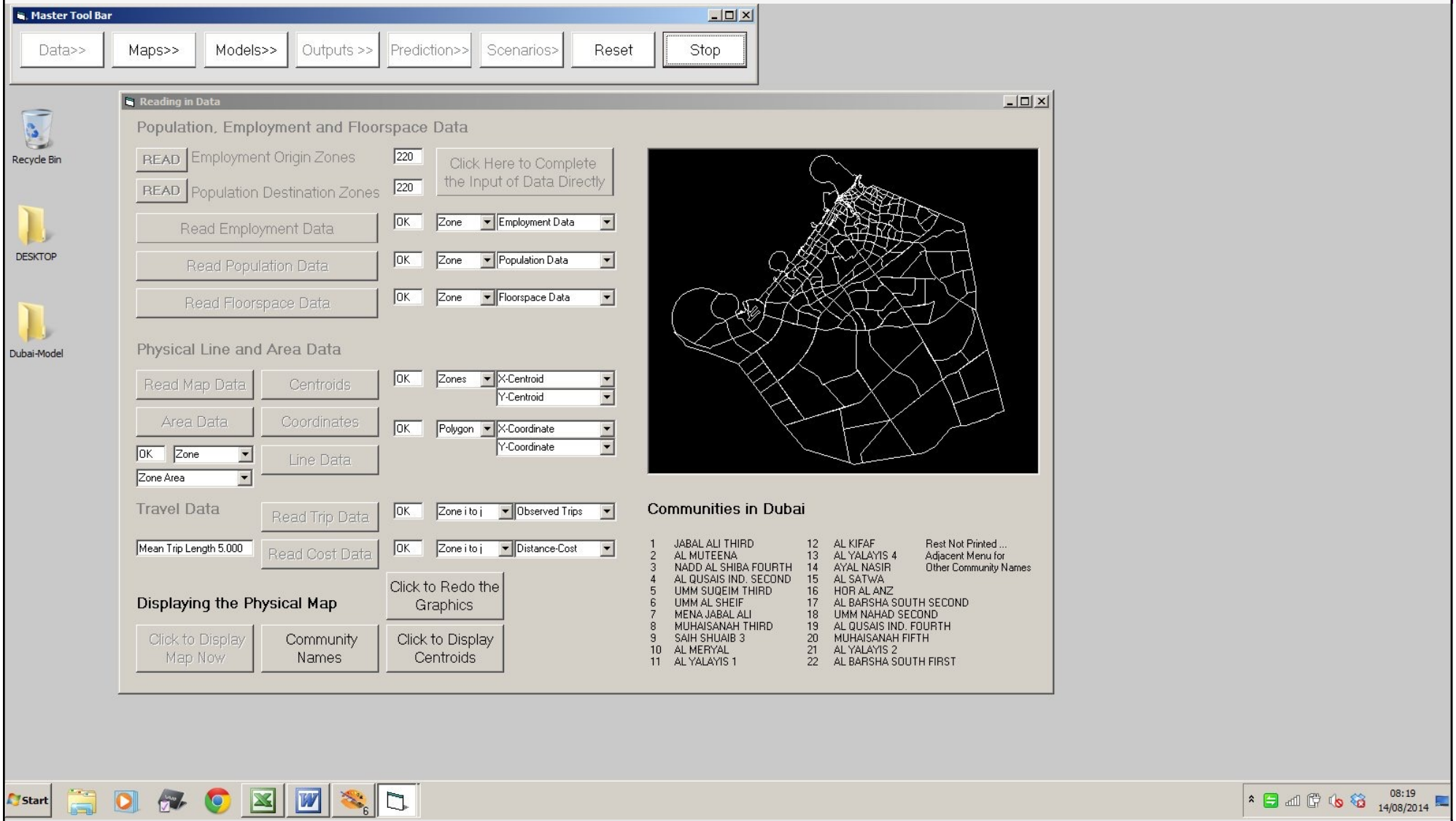

Computer
Recycle Bin
DESKTOP
Dubai-Model
CASA and the Future Cities Catapult Projects
Predicting Urban Futures for Dubai
Simulating Land Use, Population, Employment, Retailing, and Transportation
Dubai Population Estimate 2012 (Dubai Statistics Center)
تقدير عدد سكان دبي ٢٠١٢ (مركز دبي للإحصاء) -
Share
Print
Measure
Here we simulate the impact of large changes in urban structure on the population and employment distributions in 220 communities which define the Emirate of Dubai. The population and employment which are linked together through the transportation system and flows of trips. The model we use is heavily data driven as the data mirrors how people locate and interact in the city.
This is a simple demonstration to indicate the features of such a simulation model. If we were building this model for operational use in planning Dubai, we would have many different sectors describing different types of population, distinguishing particularly between guest workers and the local population, and between retailing, construction, financial services and related industrial activities. We would also define transport by different modes.
UCL casa
RUN
CATAPULT Future Cities
Start
08:17
14/08/2014

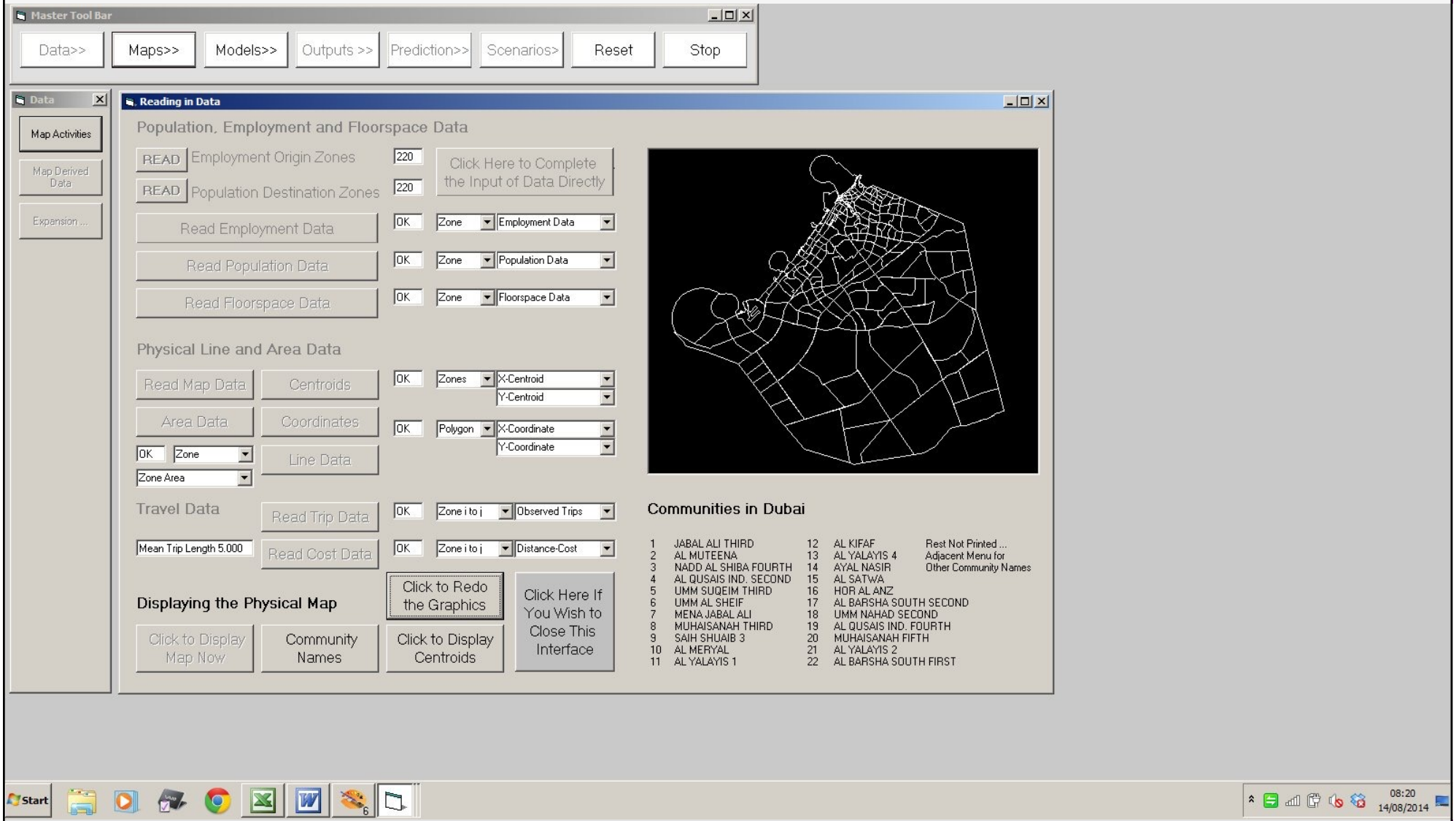
I can show you a quick movie that we demoed to the guys from Dubai who came to see us about 8 years ago

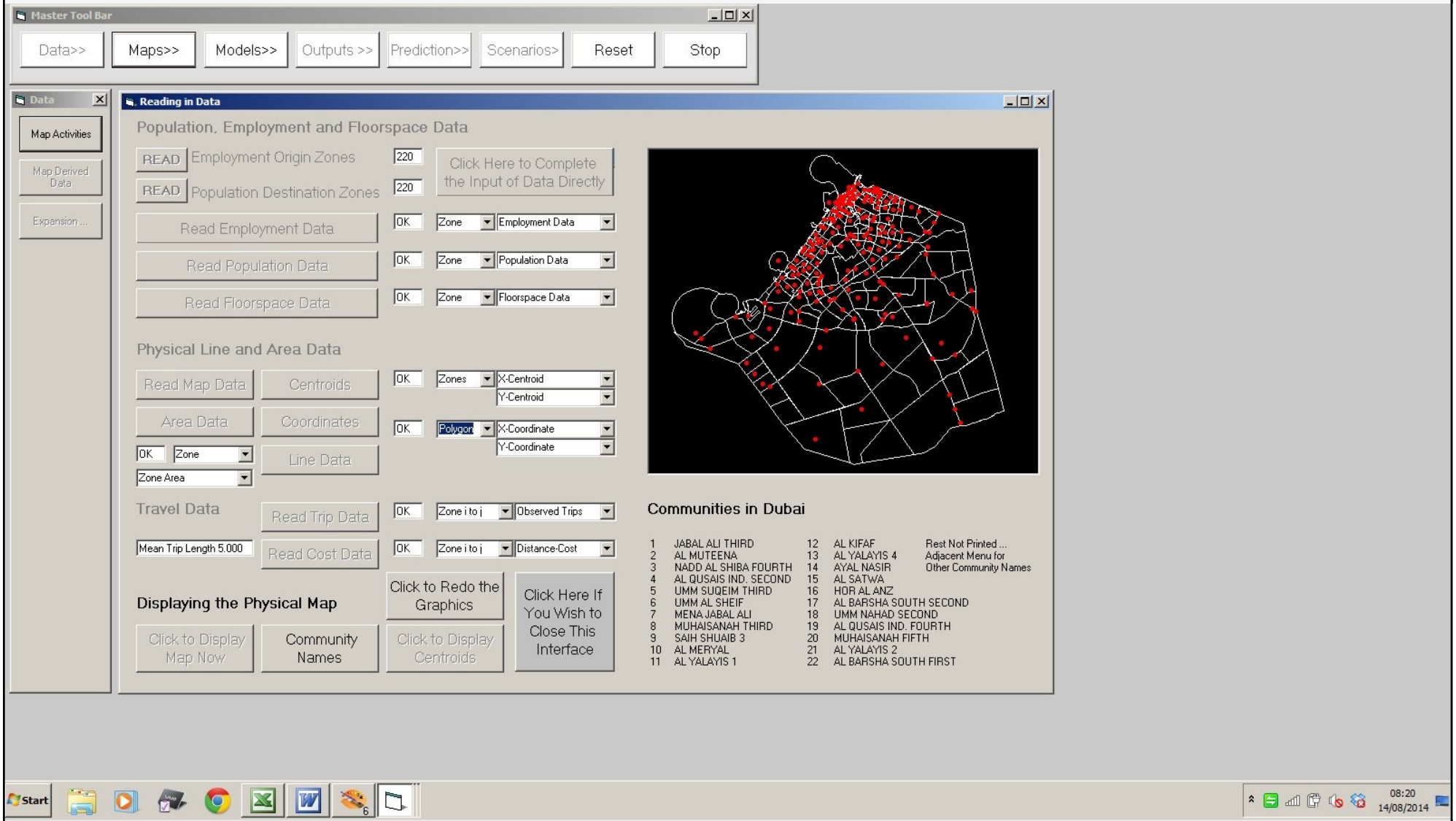


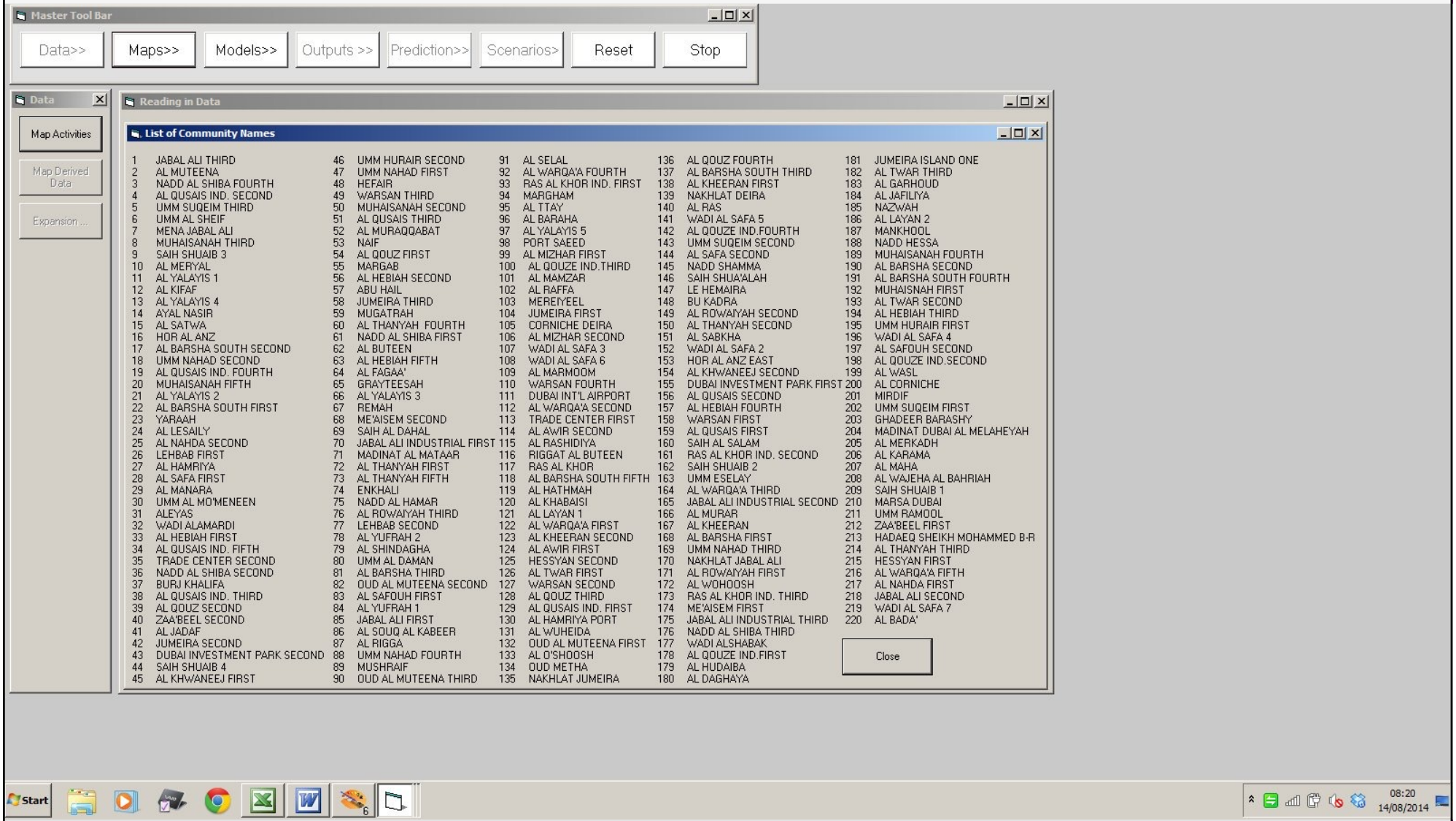

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Maps>>
Models>>
Outputs >>
Prediction>>
Scenarios>
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Stop
Recycle Bin
DESKTOP
Dubai-Model
Start
08:18
14/08/2014

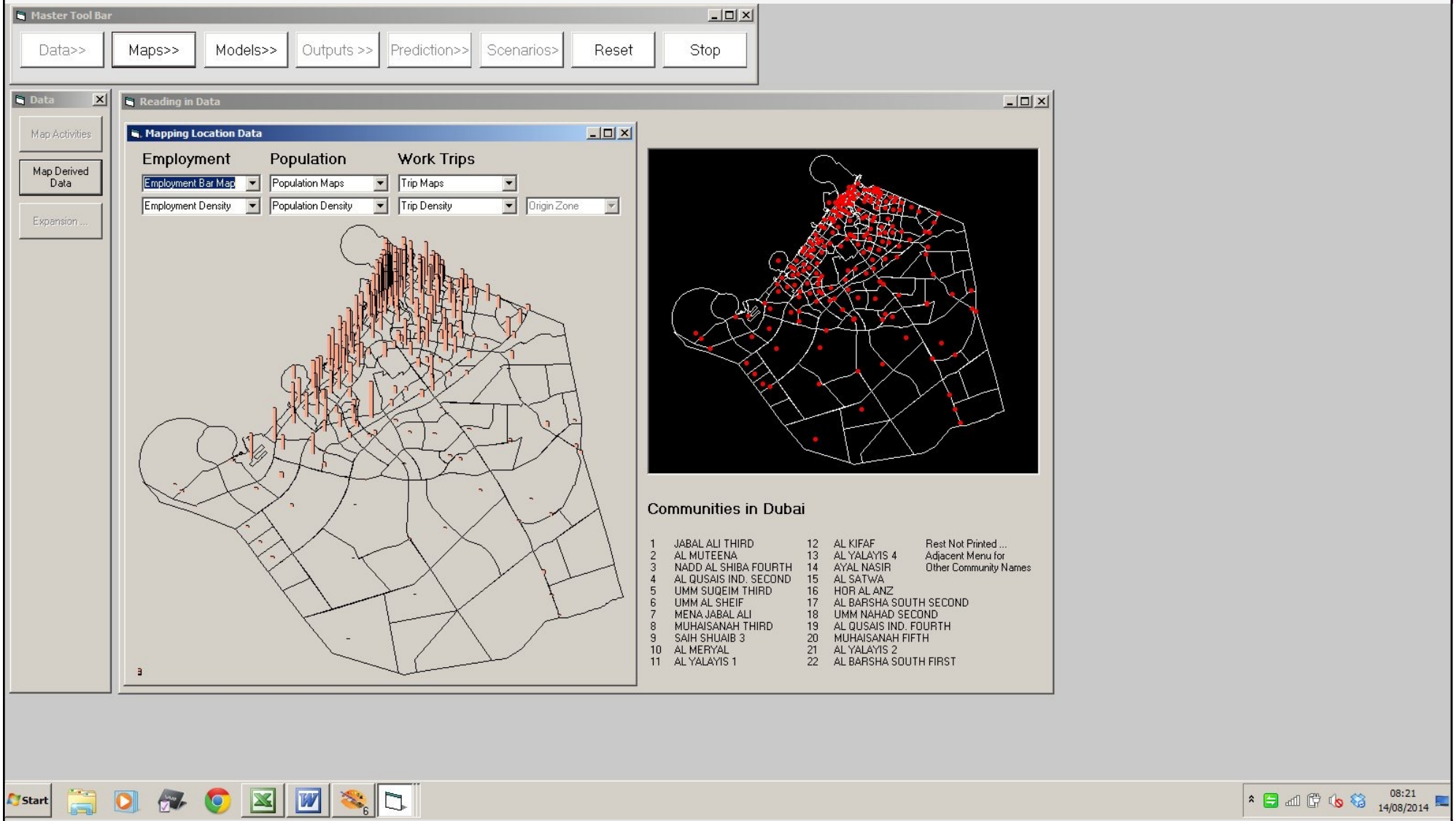

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Models>>
Outputs >>
Prediction>>
Scenarios>
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DESKTOP
Dubai-Model
Reading in Data
Population, Employment and Floorspace Data
READ
Employment Origin Zones
READ
Population Destination Zones
Click Here to Complete the Input of Data Directly
Read Employment Data
Read Population Data
Read Floorspace Data
Physical Line and Area Data
Read Map Data
Centroids
Area Data
Coordinates
Line Data
Travel Data
Read Trip Data
Read Cost Data
Displaying the Physical Map
Click to Display Map Now
Community Names
Click to Display Centroids
Start
08:18
14/08/2014

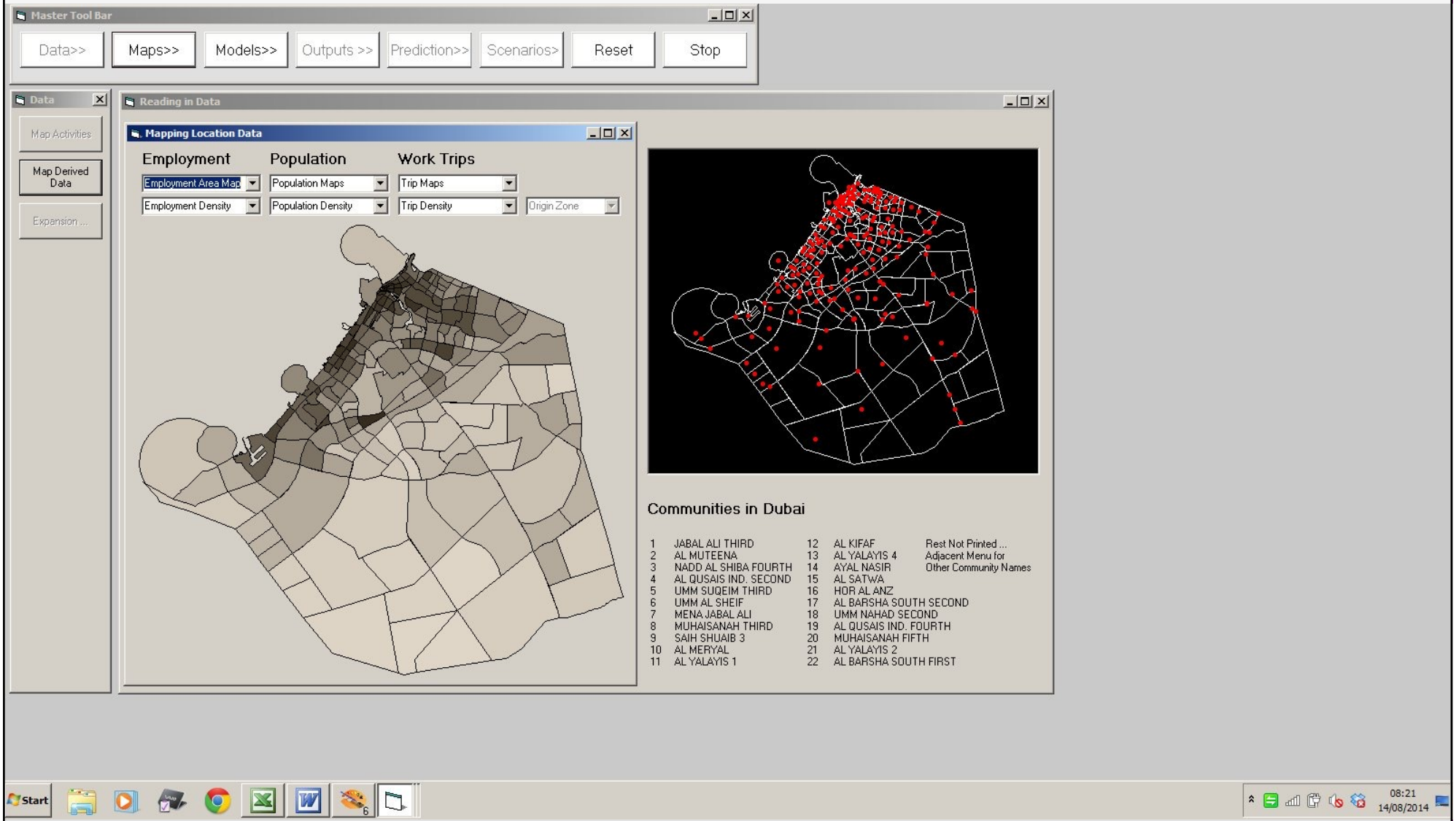

Master Tool Bar
Data>>
Maps>>
Models>>
Outputs >>
Prediction>>
Scenarios>
Reset
Stop
Recycle Bin
DESKTOP
Dubai-Model
Reading in Data
Population, Employment and Floorspace Data
READ
Employment Origin Zones
220
READ
Population Destination Zones
220
Click Here to Complete the Input of Data Directly
Read Employment Data
OK
Zone
Employment Data
Read Population Data
OK
Zone
Population Data
Read Floorspace Data
OK
Zone
Floorspace Data
Physical Line and Area Data
Read Map Data
Centroids
OK
Zones
X-Centroid
Y-Centroid
Area Data
Coordinates
OK
Polygon
X-Coordinate
Y-Coordinate
OK
Zone
Line Data
Zone Area
Travel Data
Read Trip Data
OK
Zone i to j
Observed Trips
Mean Trip Length 5.000
Read Cost Data
OK
Zone i to j
Distance-Cost
Displaying the Physical Map
Click to Redo the Graphics
Click to Display Map Now
Community Names
Click to Display Centroids
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
5 UMM SUQEIM THIRD
6 UMM AL SHEIF
7 MENA JABAL ALI
8 MUHAISANAH THIRD
9 SAIH SHUAIB 3
10 AL MERYAL
11 AL YALAYIS 1
12 AL KIFAF
13 AL YALAYIS 4
14 AYAL NASIR
15 AL SATWA
16 HOR AL ANZ
17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
19 AL QUSAIS IND. FOURTH
20 MUHAISANAH FIFTH
21 AL YALAYIS 2
22 AL BARSHA SOUTH FIRST
Rest Not Printed ... Adjacent Menu for Other Community Names
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14/08/2014

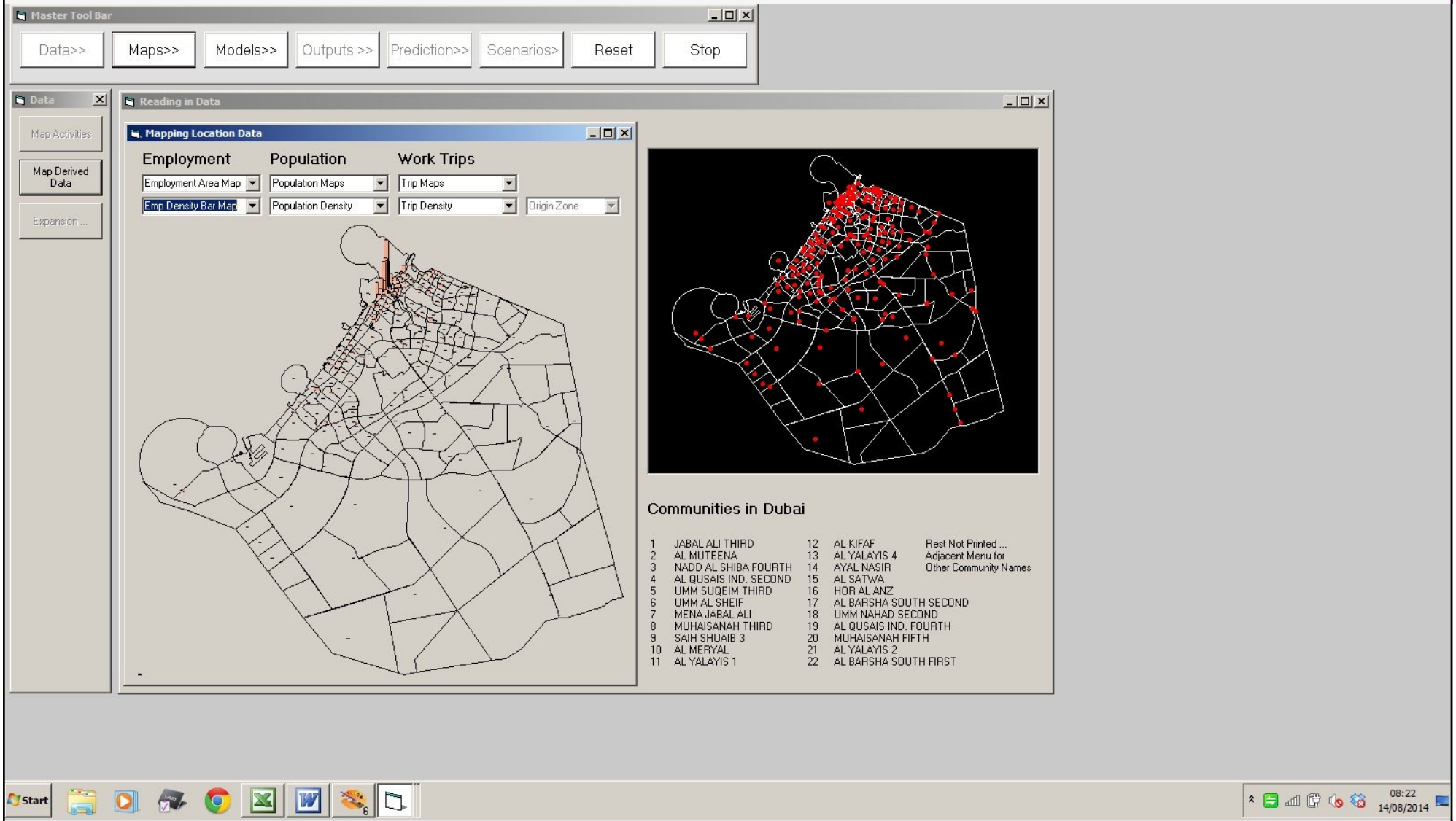

Master Tool Bar
Data>>
Maps>>
Models>>
Outputs >>
Prediction>>
Scenarios>
Reset
Stop
Data
Map Activities
Map Derived Data
Expansion ...
Reading in Data
Population, Employment and Floorspace Data
READ
Employment Origin Zones
220
READ
Population Destination Zones
220
Click Here to Complete the Input of Data Directly
Read Employment Data
Read Population Data
Read Floorspace Data
OK
Zone
Employment Data
Population Data
Floorspace Data
Physical Line and Area Data
Read Map Data
Centroids
Area Data
Coordinates
Line Data
Zones
X-Centroid
Y-Centroid
Polygon
X-Coordinate
Y-Coordinate
Zone Area
Travel Data
Read Trip Data
Read Cost Data
Zone i to j
Observed Trips
Distance-Cost
Mean Trip Length 5.000
Displaying the Physical Map
Click to Redo the Graphics
Click Here If You Wish to Close This Interface
Click to Display Map Now
Community Names
Click to Display Centroids
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
5 UMM SUQEIM THIRD
6 UMM AL SHEIF
7 MENA JABAL ALI
8 MUHAISANAH THIRD
9 SAIH SHUAIB 3
10 AL MERYAL
11 AL YALAYIS 1
12 AL KIFAF
13 AL YALAYIS 4
14 AYAL NASIR
15 AL SATWA
16 HOR AL ANZ
17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
19 AL QUSAIS IND. FOURTH
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Rest Not Printed ... Adjacent Menu for Other Community Names
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14/08/2014

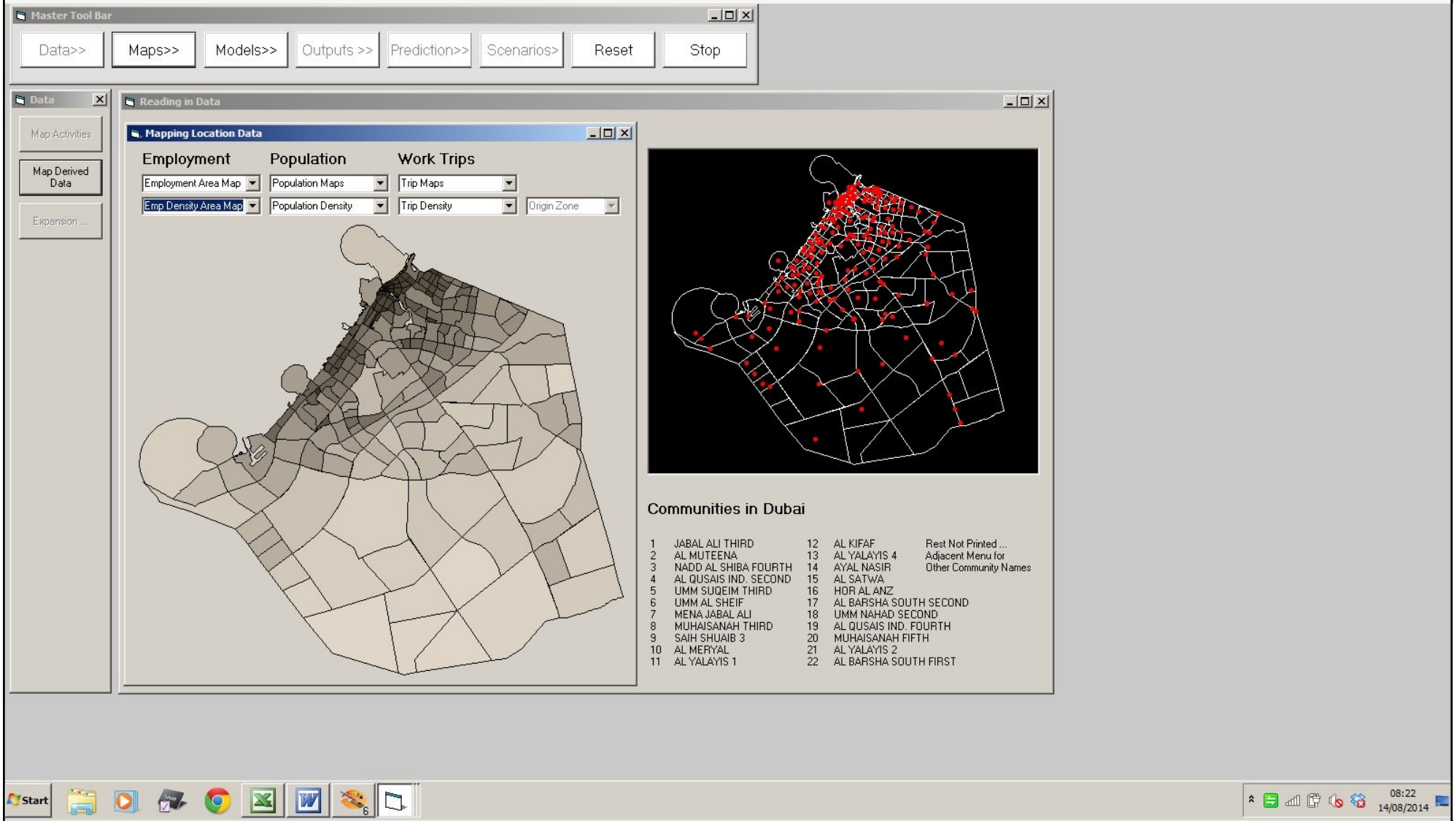

Master Tool Bar
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Maps>>
Models>>
Outputs >>
Prediction>>
Scenarios>
Reset
Stop
Data
Map Activities
Map Derived Data
Expansion ...
Reading in Data
Population, Employment and Floorspace Data
READ
Employment Origin Zones
220
READ
Population Destination Zones
220
Click Here to Complete the Input of Data Directly
Read Employment Data
OK
Zone
Employment Data
Read Population Data
OK
Zone
Population Data
Read Floorspace Data
OK
Zone
Floorspace Data
Physical Line and Area Data
Read Map Data
Centroids
OK
Zones
X-Centroid
Y-Centroid
Area Data
Coordinates
OK
Polygon
X-Coordinate
Y-Coordinate
OK
Zone
Line Data
Zone Area
Travel Data
Read Trip Data
OK
Zone i to j
Observed Trips
Mean Trip Length 5.000
Read Cost Data
OK
Zone i to j
Distance-Cost
Displaying the Physical Map
Click to Redo the Graphics
Click Here If You Wish to Close This Interface
Click to Display Map Now
Community Names
Click to Display Centroids
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
5 UMM SUQEIM THIRD
6 UMM AL SHEIF
7 MENA JABAL ALI
8 MUHAISANAH THIRD
9 SAIH SHUAIB 3
10 AL MERYAL
11 AL YALAYIS 1
12 AL KIFAF
13 AL YALAYIS 4
14 AYAL NASIR
15 AL SATWA
16 HOR AL ANZ
17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
19 AL QUSAIS IND. FOURTH
20 MUHAISANAH FIFTH
21 AL YALAYIS 2
22 AL BARSHA SOUTH FIRST
Rest Not Printed ... Adjacent Menu for Other Community Names
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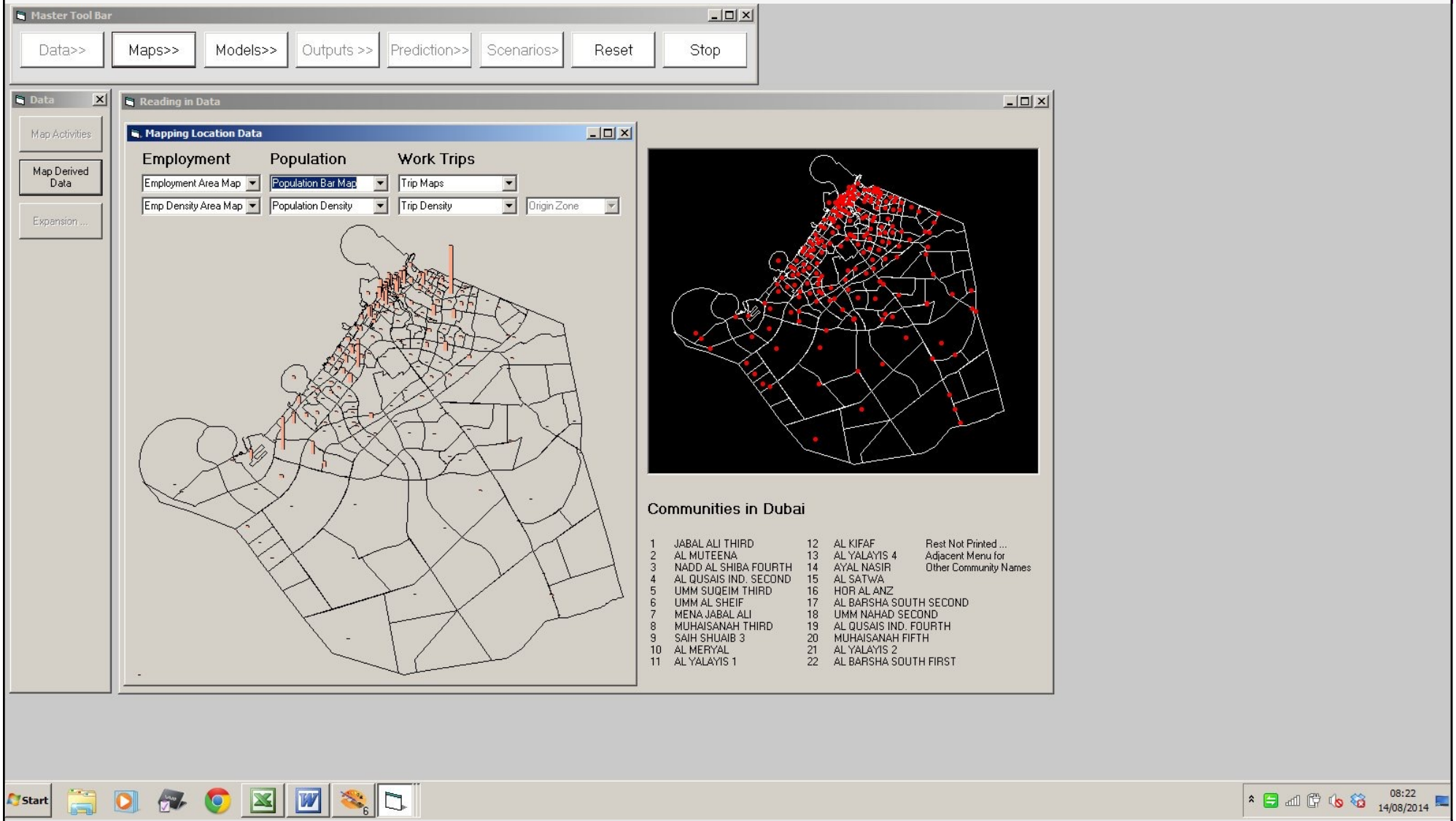

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Maps>>
Models>>
Outputs >>
Prediction>>
Scenarios>
Reset
Stop
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Map Activities
Map Derived Data
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1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
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6 UMM AL SHEIF
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8 MUHAISANAH THIRD
9 SAIH SHUAIB 3
10 AL MERYAL
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12 AL KIFAF
13 AL YALAYIS 4
14 AYAL NASIR
15 AL SATWA
16 HOR AL ANZ
17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
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20 MUHAISANAH FIFTH
21 AL YALAYIS 2
22 AL BARSHA SOUTH FIRST
23 YARAAH
24 AL LESAILY
25 AL NAHDA SECOND
26 LEHBAB FIRST
27 AL HAMRIYA
28 AL SAFA FIRST
29 AL MANARA
30 UMM AL MO'MENEEN
31 ALEYAS
32 WADI ALAMARDI
33 AL HEBIAH FIRST
34 AL QUSAIS IND. FIFTH
35 TRADE CENTER SECOND
36 NADD AL SHIBA SECOND
37 BURJ KHALIFA
38 AL QUSAIS IND. THIRD
39 AL QOUZ SECOND
40 ZAA'BEEL SECOND
41 AL JADAF
42 JUMEIRA SECOND
43 DUBAI INVESTMENT PARK SECOND
44 SAIH SHUAIB 4
45 AL KHWANEEJ FIRST
46 UMM HURAIR SECOND
47 UMM NAHAD FIRST
48 HEFAIR
49 WARSAN THIRD
50 MUHAISANAH SECOND
51 AL QUSAIS THIRD
52 AL MURAQQABAT
53 NAIF
54 AL QOUZ FIRST
55 MARGAB
56 AL HEBIAH SECOND
57 ABU HAIL
58 JUMEIRA THIRD
59 MUGATRAH
60 AL THANYAH FOURTH
61 NADD AL SHIBA FIRST
62 AL BUTEEN
63 AL HEBIAH FIFTH
64 AL FAGAA'
65 GRAYTEESAH
66 AL YALAYIS 3
67 REMAH
68 ME'AISEM SECOND
69 SAIH AL DAHAL
70 JABAL ALI INDUSTRIAL FIRST
71 MADINAT AL MATAAR
72 AL THANYAH FIRST
73 AL THANYAH FIFTH
74 ENKHALI
75 NADD AL HAMAR
76 AL ROWAIYAH THIRD
77 LEHBAB SECOND
78 AL YUFRAH 2
79 AL SHINDAGHA
80 UMM AL DAMAN
81 AL BARSHA THIRD
82 OUD AL MUTEENA SECOND
83 AL SAFOUH FIRST
84 AL YUFRAH 1
85 JABAL ALI FIRST
86 AL SOUQ AL KABEER
87 AL RIGGA
88 UMM NAHAD FOURTH
89 MUSHRAIF
90 OUD AL MUTEENA THIRD
91 AL SELAL
92 AL WARQA'A FOURTH
93 RAS AL KHOR IND. FIRST
94 MARGHAM
95 AL TTAY
96 AL BARAHA
97 AL YALAYIS 5
98 PORT SAEED
99 AL MIZHAR FIRST
100 AL QOUZE IND.THIRD
101 AL MAMZAR
102 AL RAFFA
103 MEREIYEEL
104 JUMEIRA FIRST
105 CORNICHE DEIRA
106 AL MIZHAR SECOND
107 WADI AL SAFA 3
108 WADI AL SAFA 6
109 AL MARMOOM
110 WARSAN FOURTH
111 DUBAI INT'L AIRPORT
112 AL WARQA'A SECOND
113 TRADE CENTER FIRST
114 AL AWIR SECOND
115 AL RASHIDIYA
116 RIGGAT AL BUTEEN
117 RAS AL KHOR
118 AL BARSHA SOUTH FIFTH
119 AL HATHMAH
120 AL KHABAISI
121 AL LAYAN 1
122 AL WARQA'A FIRST
123 AL KHEERAN SECOND
124 AL AWIR FIRST
125 HESSYAN SECOND
126 AL TWAR FIRST
127 WARSAN SECOND
128 AL QOUZ THIRD
129 AL QUSAIS IND. FIRST
130 AL HAMRIYA PORT
131 AL WUHEIDA
132 OUD AL MUTEENA FIRST
133 AL O'SHOOSH
134 OUD METHA
135 NAKHLAT JUMEIRA
136 AL QOUZ FOURTH
137 AL BARSHA SOUTH THIRD
138 AL KHEERAN FIRST
139 NAKHLAT DEIRA
140 AL RAS
141 WADI AL SAFA 5
142 AL QOUZE IND.FOURTH
143 UMM SUQEIM SECOND
144 AL SAFA SECOND
145 NADD SHAMMA
146 SAIH SHUA'ALAH
147 LE HEMAIRA
148 BU KADRA
149 AL ROWAIYAH SECOND
150 AL THANYAH SECOND
151 AL SABKHA
152 WADI AL SAFA 2
153 HOR AL ANZ EAST
154 AL KHWANEEJ SECOND
155 DUBAI INVESTMENT PARK FIRST
156 AL QUSAIS SECOND
157 AL HEBIAH FOURTH
158 WARSAN FIRST
159 AL QUSAIS FIRST
160 SAIH AL SALAM
161 RAS AL KHOR IND. SECOND
162 SAIH SHUAIB 2
163 UMM ESELAY
164 AL WARQA'A THIRD
165 JABAL ALI INDUSTRIAL SECOND
166 AL MURAR
167 AL KHEERAN
168 AL BARSHA FIRST
169 UMM NAHAD THIRD
170 NAKHLAT JABAL ALI
171 AL ROWAIYAH FIRST
172 AL WOHOOSH
173 RAS AL KHOR IND. THIRD
174 ME'AISEM FIRST
175 JABAL ALI INDUSTRIAL THIRD
176 NADD AL SHIBA THIRD
177 WADI ALSHABAK
178 AL QOUZE IND.FIRST
179 AL HUDAIBA
180 AL DAGHAYA
181 JUMEIRA ISLAND ONE
182 AL TWAR THIRD
183 AL GARHOUD
184 AL JAFILIYA
185 NAZWAH
186 AL LAYAN 2
187 MANKHOOL
188 NADD HESSA
189 MUHAISANAH FOURTH
190 AL BARSHA SECOND
191 AL BARSHA SOUTH FOURTH
192 MUHAISNAH FIRST
193 AL TWAR SECOND
194 AL HEBIAH THIRD
195 UMM HURAIR FIRST
196 WADI AL SAFA 4
197 AL SAFOUH SECOND
198 AL QOUZE IND.SECOND
199 AL WASL
200 AL CORNICHE
201 MIRDIF
202 UMM SUQEIM FIRST
203 GHADEER BARASHY
204 MADINAT DUBAI AL MELAHEYAH
205 AL MERKADH
206 AL KARAMA
207 AL MAHA
208 AL WAJEHA AL BAHRIAH
209 SAIH SHUAIB 1
210 MARSA DUBAI
211 UMM RAMOOL
212 ZAA'BEEL FIRST
213 HADAEQ SHEIKH MOHAMMED B-R
214 AL THANYAH THIRD
215 HESSYAN FIRST
216 AL WARQA'A FIFTH
217 AL NAHDA FIRST
218 JABAL ALI SECOND
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220 AL BADA'
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14/08/2014

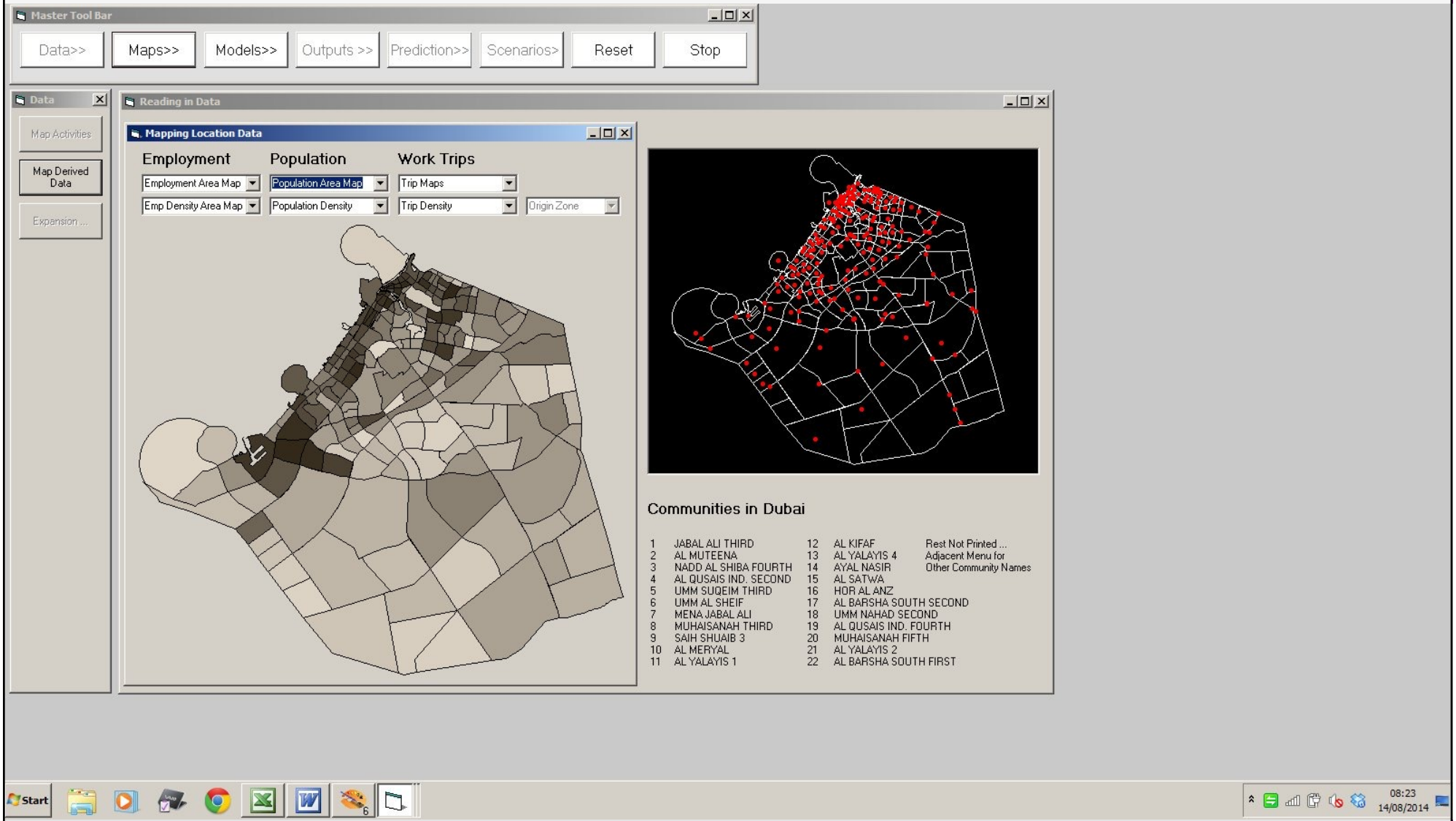

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Models>>
Outputs >>
Prediction>>
Scenarios>
Reset
Stop
Data
Map Activities
Map Derived Data
Expansion ...
Reading in Data
Mapping Location Data
Employment
Population
Work Trips
Employment Bar Map
Population Maps
Trip Maps
Employment Density
Population Density
Trip Density
Origin Zone
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
5 UMM SUQEIM THIRD
6 UMM AL SHEIF
7 MENA JABAL ALI
8 MUHAISANAH THIRD
9 SAIH SHUAIB 3
10 AL MERYAL
11 AL YALAYIS 1
12 AL KIFAF
13 AL YALAYIS 4
14 AYAL NASIR
15 AL SATWA
16 HOR AL ANZ
17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
19 AL QUSAIS IND. FOURTH
20 MUHAISANAH FIFTH
21 AL YALAYIS 2
22 AL BARSHA SOUTH FIRST
Rest Not Printed ...
Adjacent Menu for
Other Community Names
Start
08:21
14/08/2014

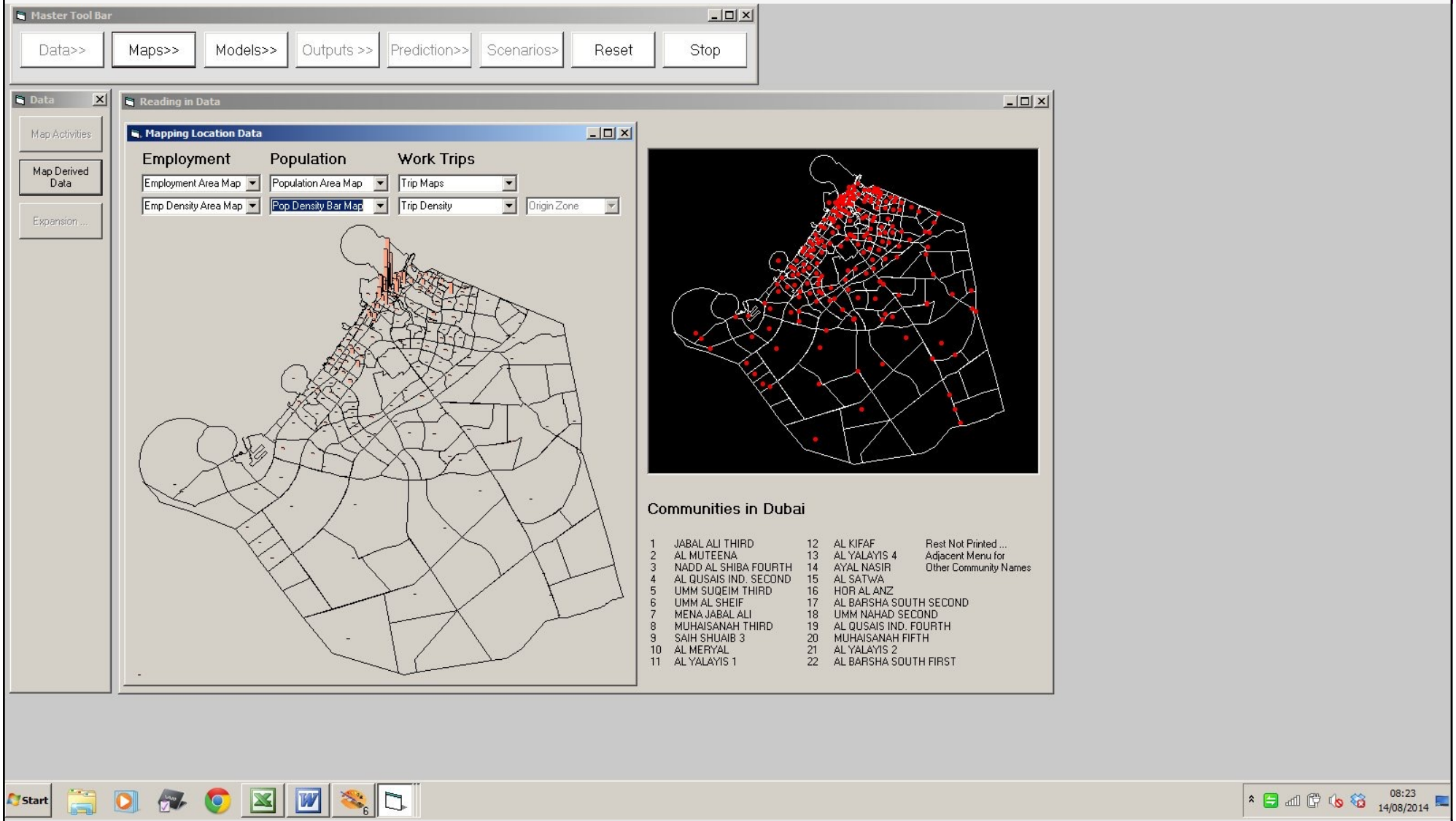

Master Tool Bar
Data>>
Maps>>
Models>>
Outputs >>
Prediction>>
Scenarios>
Reset
Stop
Data
Map Activities
Map Derived Data
Expansion ...
Reading in Data
Mapping Location Data
Employment
Population
Work Trips
Employment Area Map
Employment Density
Population Maps
Population Density
Trip Maps
Trip Density
Origin Zone
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
5 UMM SUQEIM THIRD
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8 MUHAISANAH THIRD
9 SAIH SHUAIB 3
10 AL MERYAL
11 AL YALAYIS 1
12 AL KIFAF
13 AL YALAYIS 4
14 AYAL NASIR
15 AL SATWA
16 HOR AL ANZ
17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
19 AL QUSAIS IND. FOURTH
20 MUHAISANAH FIFTH
21 AL YALAYIS 2
22 AL BARSHA SOUTH FIRST
Rest Not Printed ...
Adjacent Menu for
Other Community Names
Start
08:21
14/08/2014

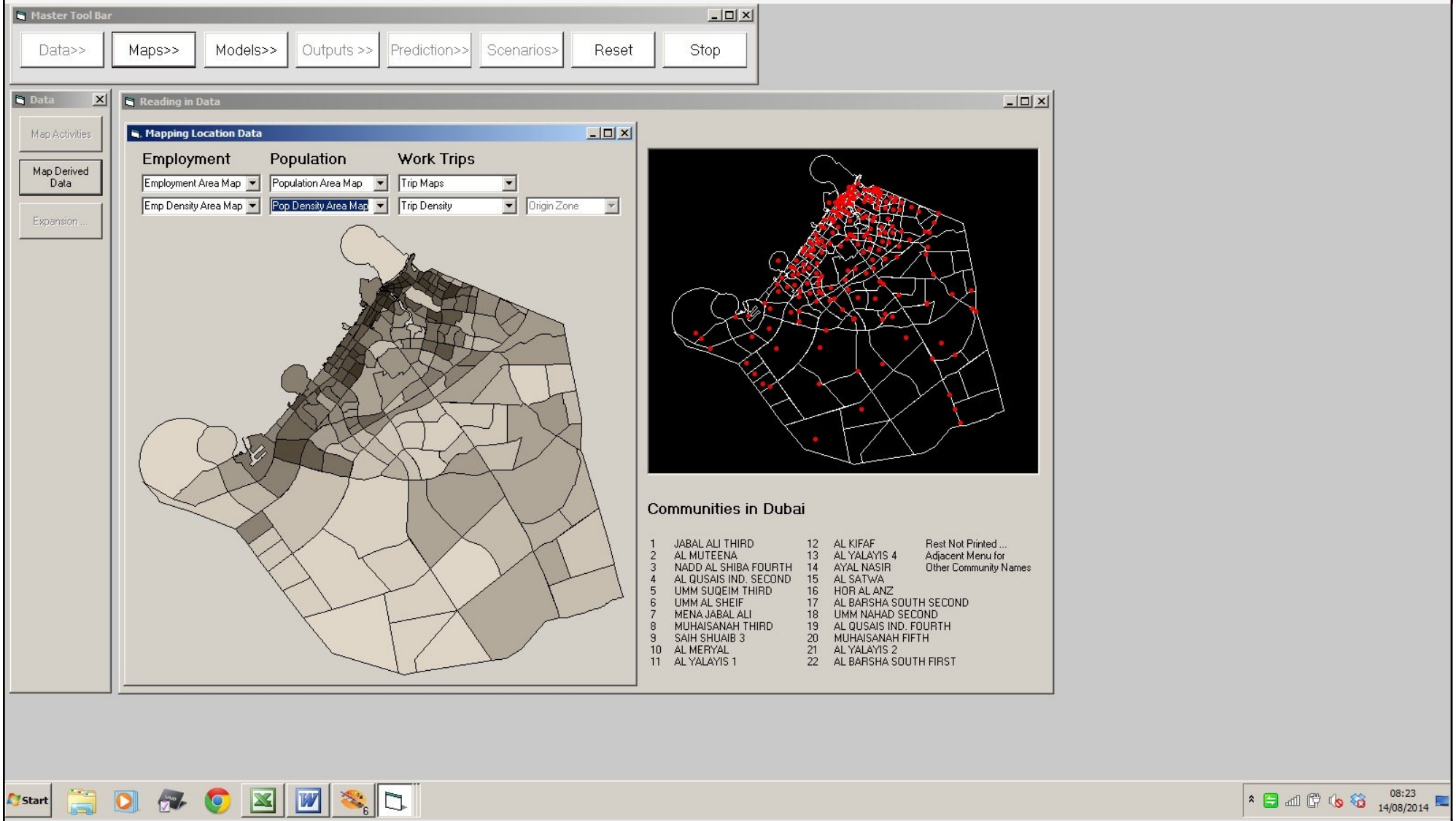

Master Tool Bar
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Scenarios>
Reset
Stop
Data
Map Activities
Map Derived Data
Expansion ...
Reading in Data
Mapping Location Data
Employment
Population
Work Trips
Employment Area Map
Population Maps
Trip Maps
Emp Density Bar Map
Population Density
Trip Density
Origin Zone
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
5 UMM SUQEIM THIRD
6 UMM AL SHEIF
7 MENA JABAL ALI
8 MUHAISANAH THIRD
9 SAIH SHUAIB 3
10 AL MERYAL
11 AL YALAYIS 1
12 AL KIFAF
13 AL YALAYIS 4
14 AYAL NASIR
15 AL SATWA
16 HOR AL ANZ
17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
19 AL QUSAIS IND. FOURTH
20 MUHAISANAH FIFTH
21 AL YALAYIS 2
22 AL BARSHA SOUTH FIRST
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Adjacent Menu for
Other Community Names
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14/08/2014

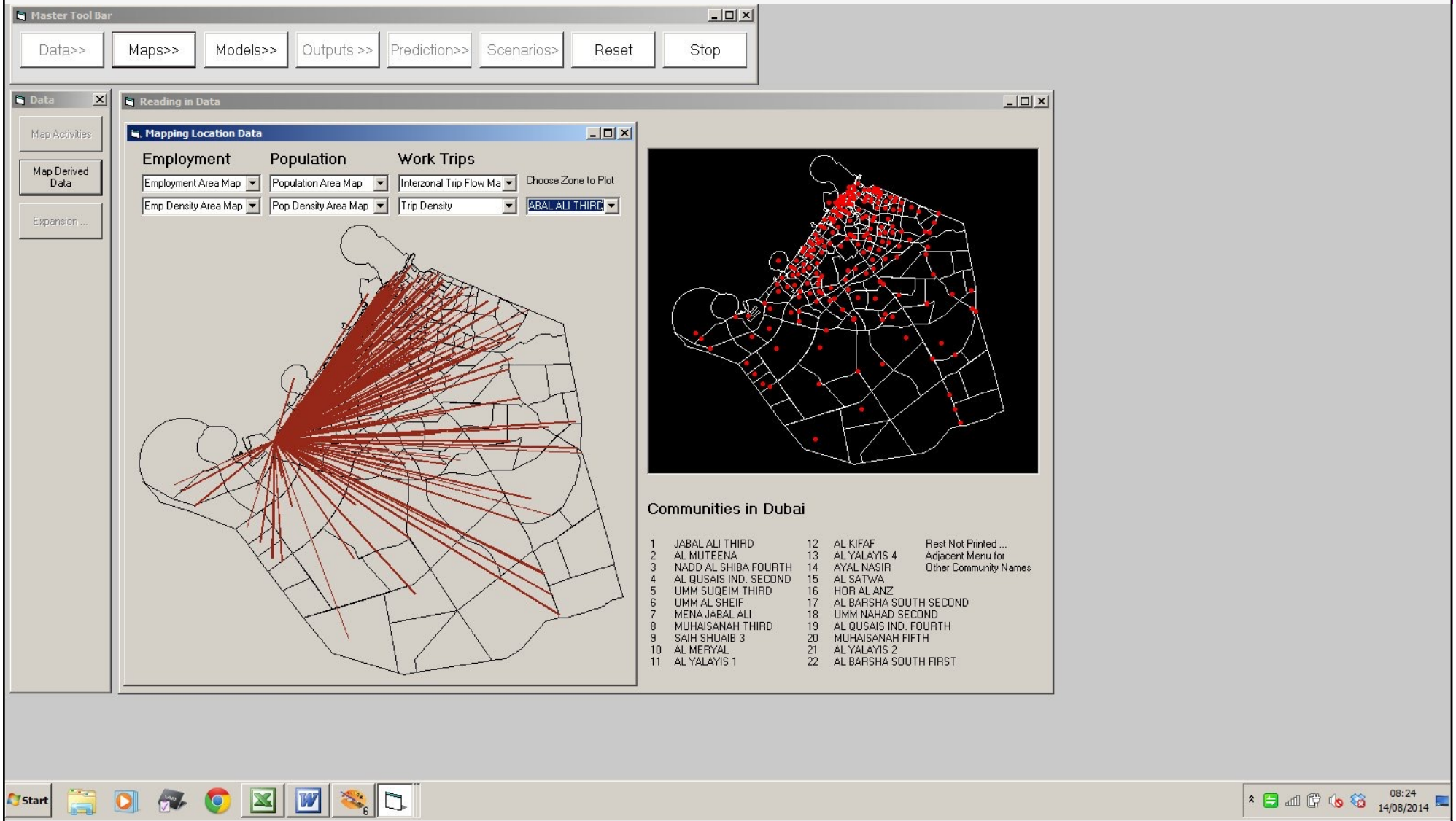

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Stop
Data
Map Activities
Map Derived Data
Expansion ...
Reading in Data
Mapping Location Data
Employment
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Work Trips
Employment Area Map
Emp Density Area Map
Population Maps
Population Density
Trip Maps
Trip Density
Origin Zone
Communities in Dubai
1 JABAL ALI THIRD
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18 UMM NAHAD SECOND
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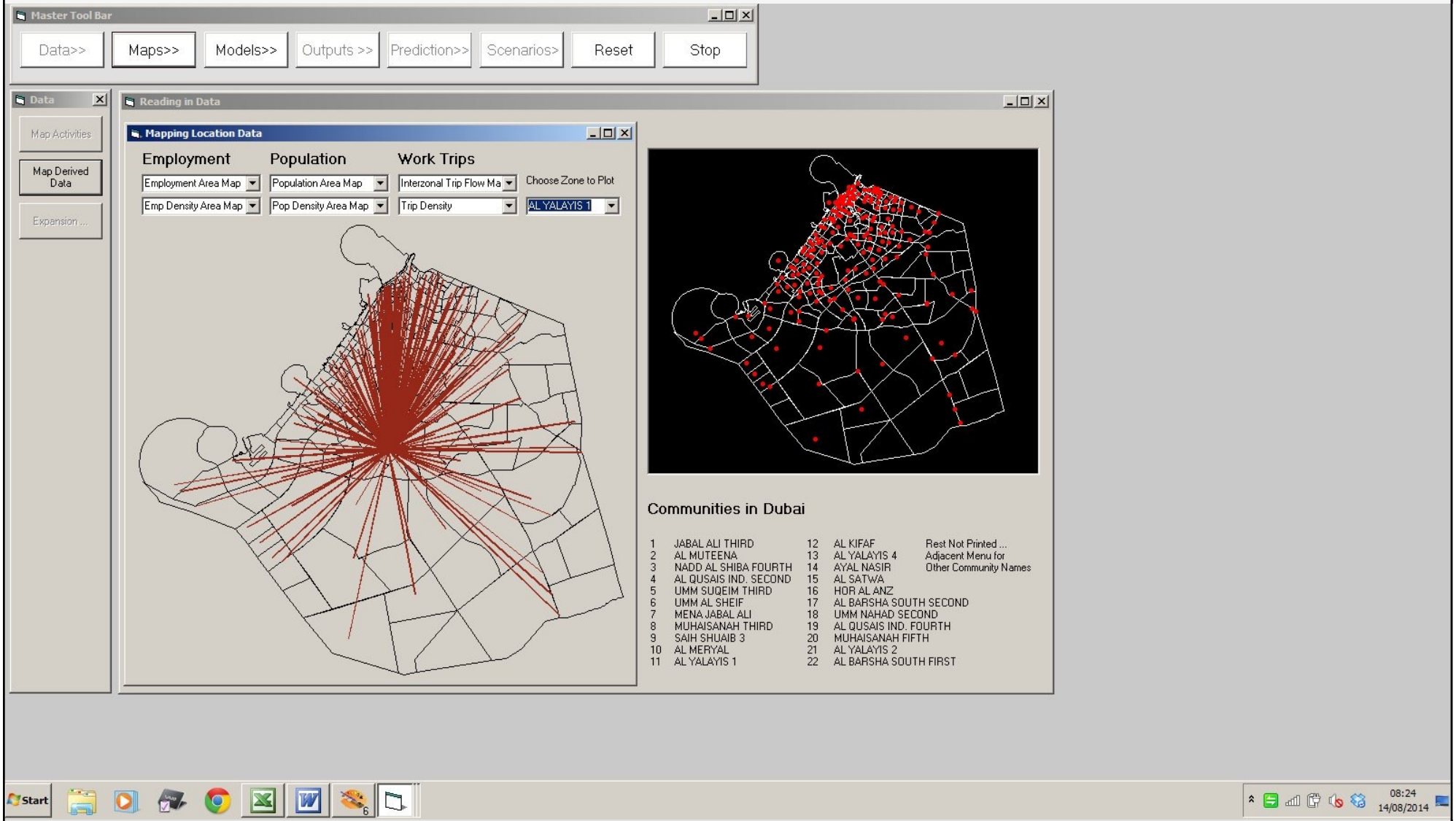

Master Tool Bar
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Reset
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Map Activities
Map Derived Data
Expansion ...
Reading in Data
Mapping Location Data
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Employment Area Map
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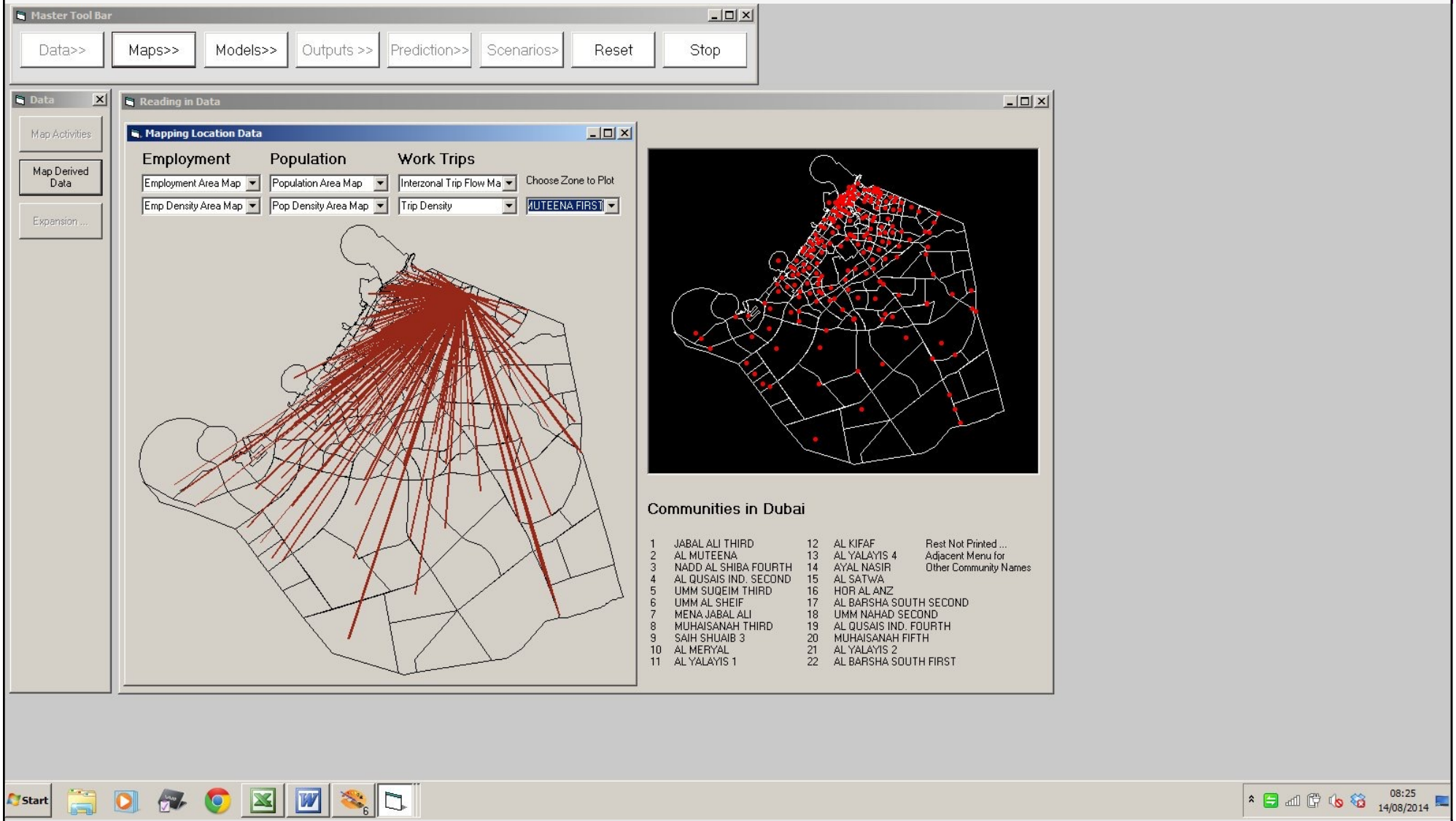

Master Tool Bar
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Map Activities
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Reading in Data
Mapping Location Data
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Work Trips
Employment Area Map
Population Area Map
Trip Maps
Emp Density Area Map
Population Density
Trip Density
Origin Zone
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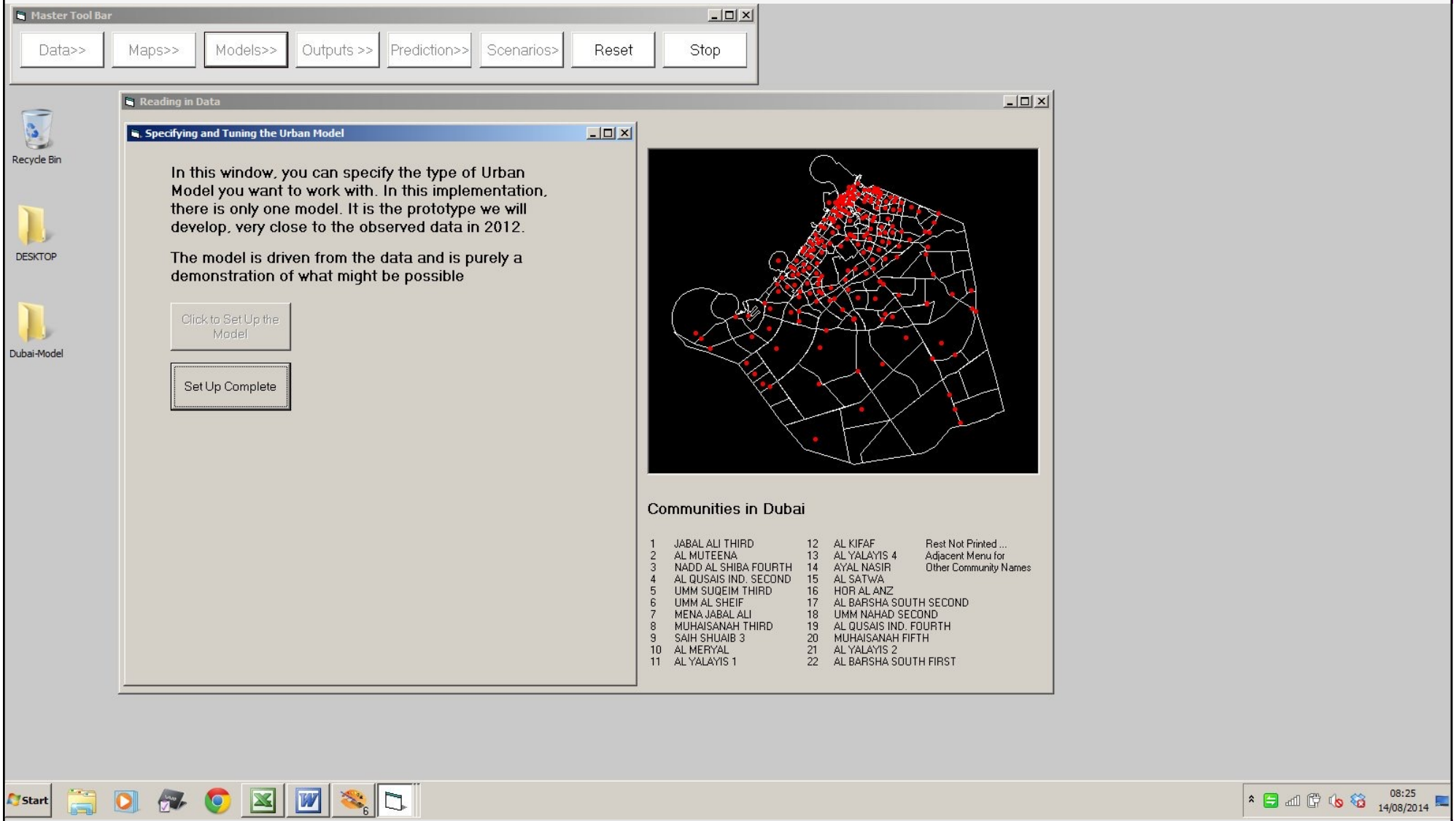

Master Tool Bar
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Scenarios>
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Data
Map Activities
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Reading in Data
Mapping Location Data
Employment
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Work Trips
Employment Area Map
Population Area Map
Trip Maps
Emp Density Area Map
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Trip Density
Origin Zone
Communities in Dubai
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11 AL YALAYIS 1
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13 AL YALAYIS 4
14 AYAL NASIR
15 AL SATWA
16 HOR AL ANZ
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18 UMM NAHAD SECOND
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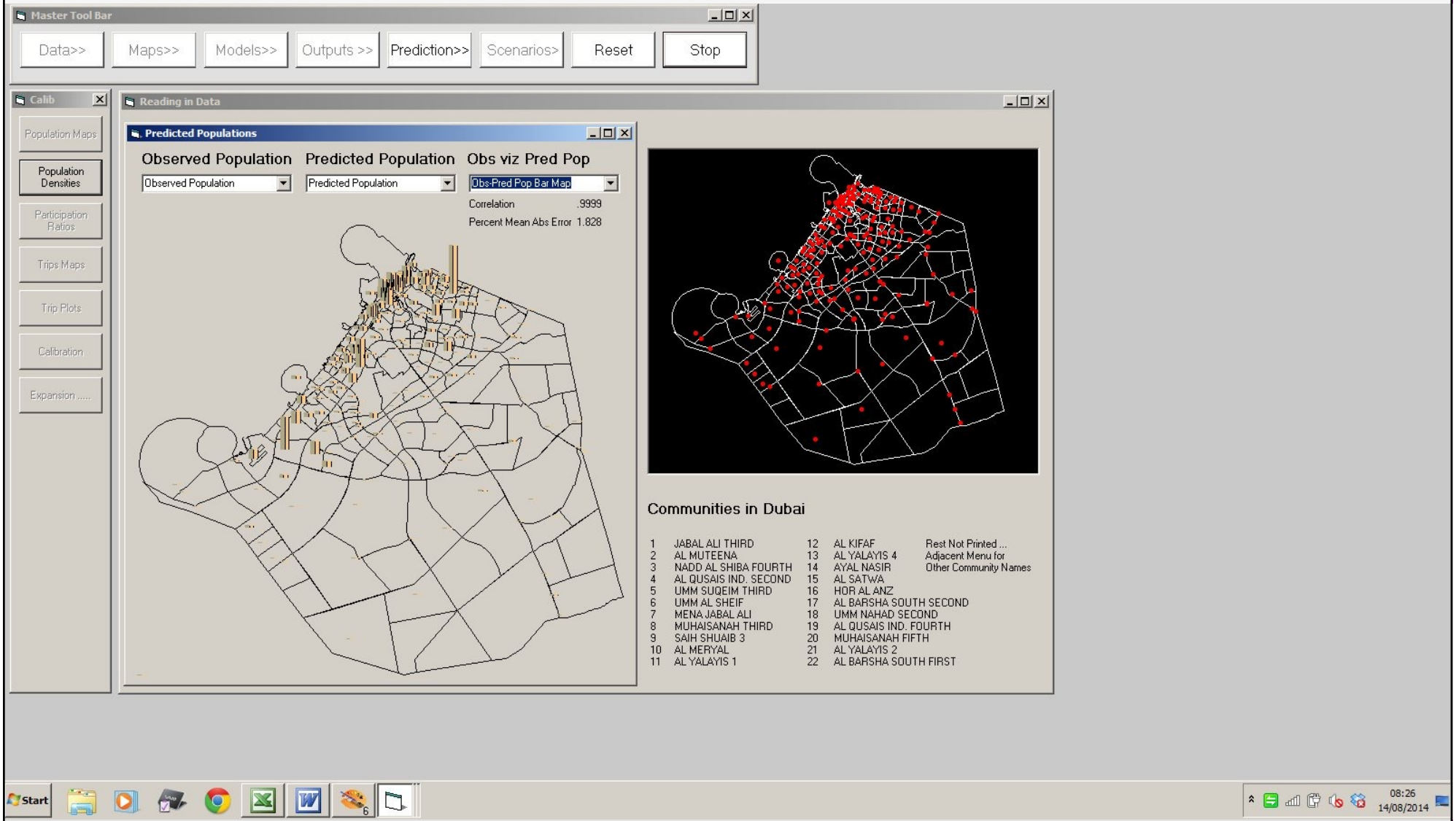

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Map Activities
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Reading in Data
Mapping Location Data
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Population
Work Trips
Employment Area Map
Population Area Map
Trip Maps
Emp Density Area Map
Pop Density Area Map
Trip Density
Origin Zone
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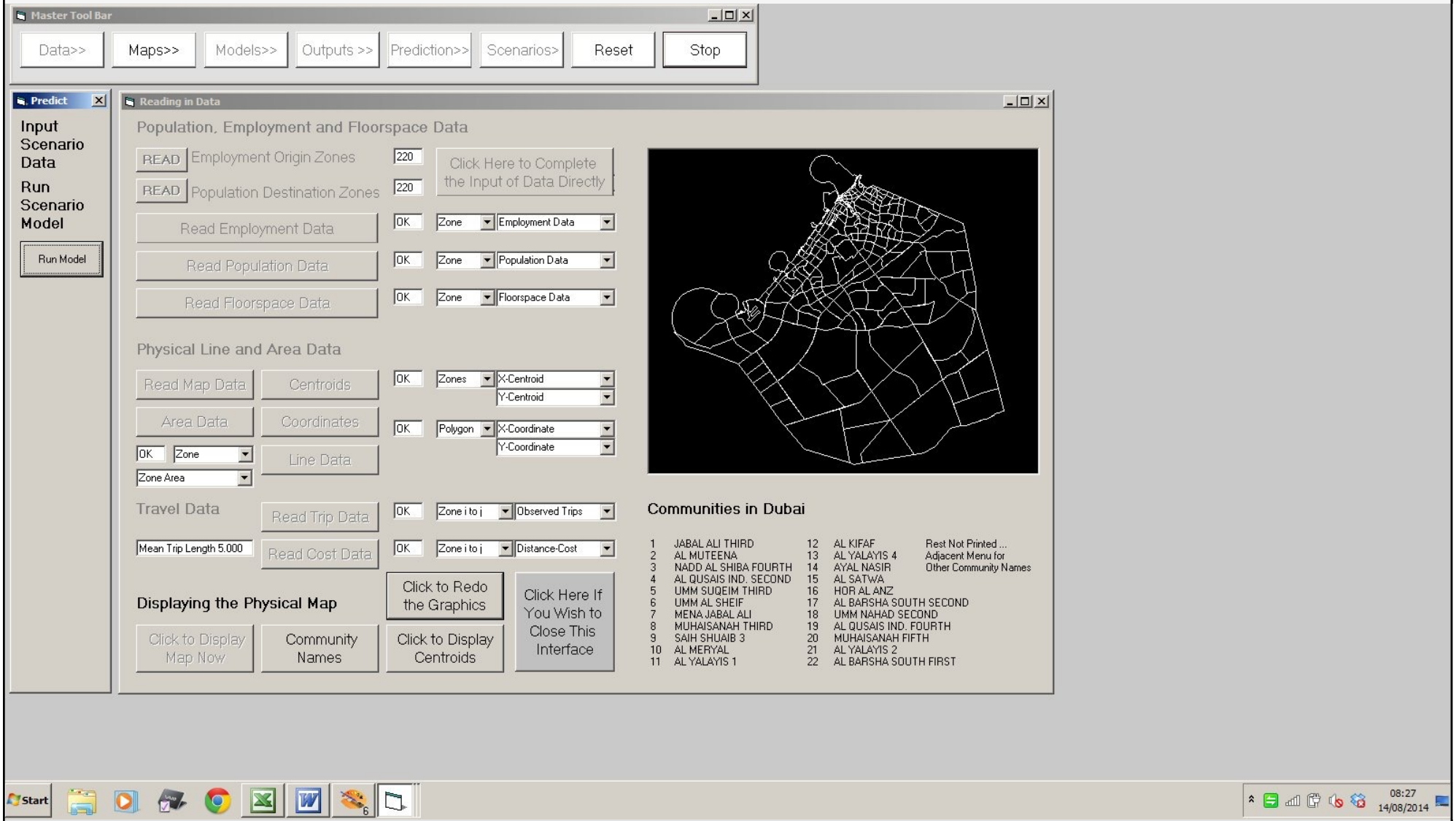

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Map Derived Data
Expansion ...
Reading in Data
Mapping Location Data
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Population
Work Trips
Employment Area Map
Population Area Map
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Choose Zone to Plot
Emp Density Area Map
Pop Density Area Map
Trip Density
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Communities in Dubai
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2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
5 UMM SUQEIM THIRD
6 UMM AL SHEIF
7 MENA JABAL ALI
8 MUHAISANAH THIRD
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11 AL YALAYIS 1
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14 AYAL NASIR
15 AL SATWA
16 HOR AL ANZ
17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
19 AL QUSAIS IND. FOURTH
20 MUHAISANAH FIFTH
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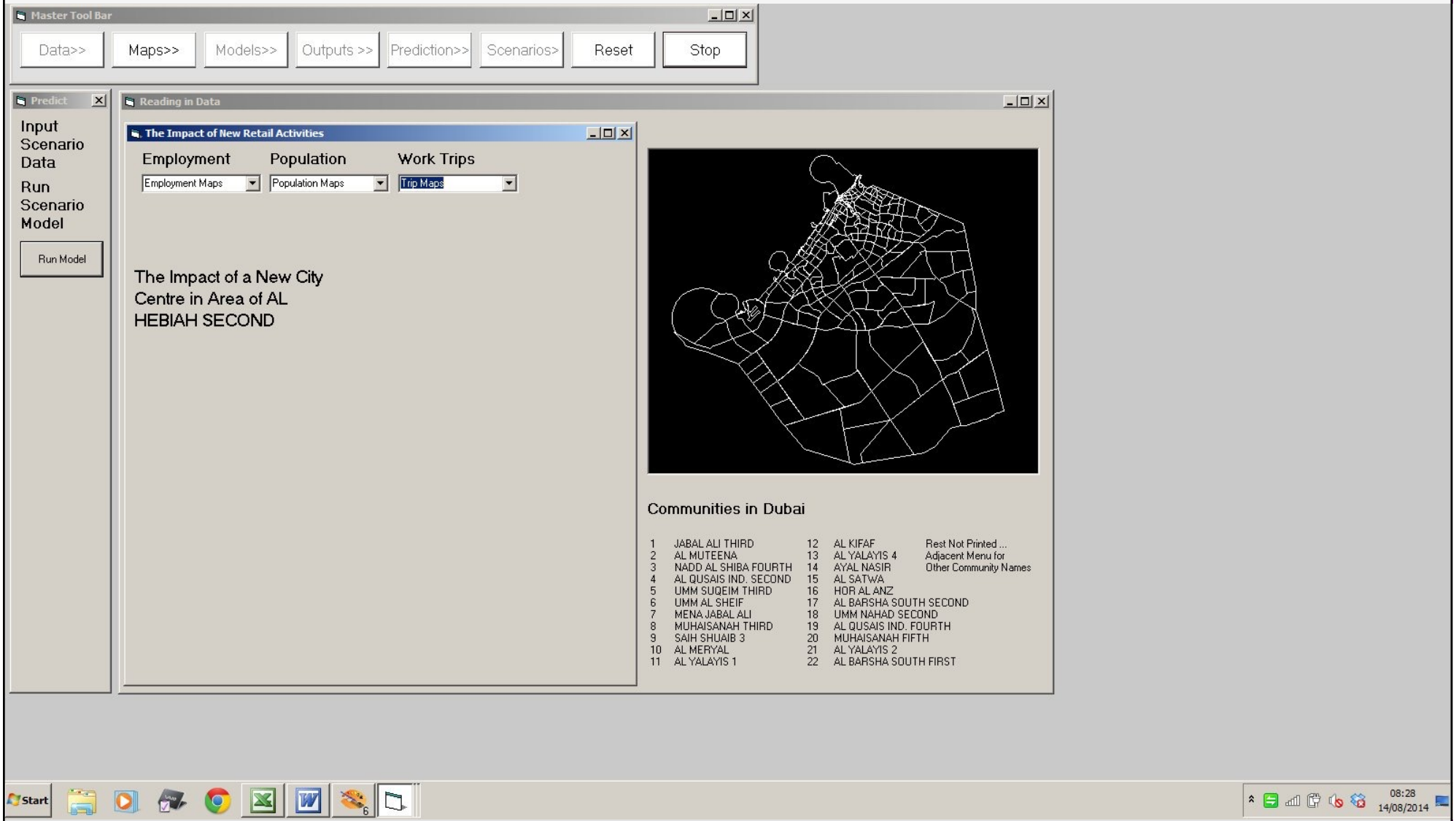

Master Tool Bar
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Stop
Data
Map Activities
Map Derived Data
Expansion ...
Reading in Data
Mapping Location Data
Employment
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Work Trips
Employment Area Map
Population Area Map
Interzonal Trip Flow Ma
Choose Zone to Plot
Emp Density Area Map
Pop Density Area Map
Trip Density
AL YALAYIS 1
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
5 UMM SUQEIM THIRD
6 UMM AL SHEIF
7 MENA JABAL ALI
8 MUHAISANAH THIRD
9 SAIH SHUAIB 3
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11 AL YALAYIS 1
12 AL KIFAF
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14 AYAL NASIR
15 AL SATWA
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17 AL BARSHA SOUTH SECOND
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21 AL YALAYIS 2
22 AL BARSHA SOUTH FIRST
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Other Community Names
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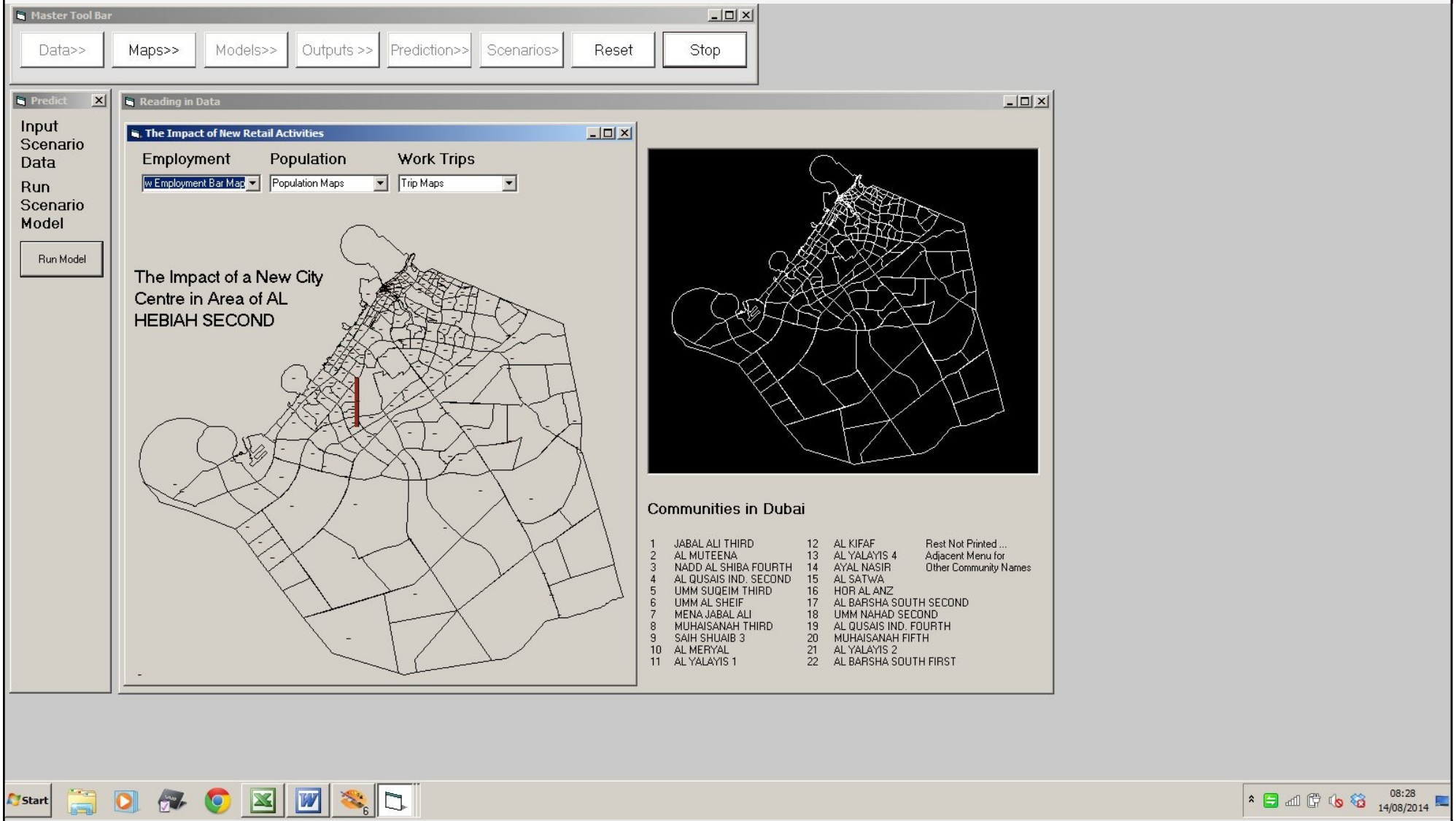

Master Tool Bar
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Stop
Data
Map Activities
Map Derived Data
Expansion ...
Reading in Data
Mapping Location Data
Employment
Population
Work Trips
Employment Area Map
Population Area Map
Interzonal Trip Flow Ma
Choose Zone to Plot
Emp Density Area Map
Pop Density Area Map
Trip Density
AUTEENA FIRST
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
5 UMM SUQEIM THIRD
6 UMM AL SHEIF
7 MENA JABAL ALI
8 MUHAISANAH THIRD
9 SAIH SHUAIB 3
10 AL MERYAL
11 AL YALAYIS 1
12 AL KIFAF
13 AL YALAYIS 4
14 AYAL NASIR
15 AL SATWA
16 HOR AL ANZ
17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
19 AL QUSAIS IND. FOURTH
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21 AL YALAYIS 2
22 AL BARSHA SOUTH FIRST
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Other Community Names
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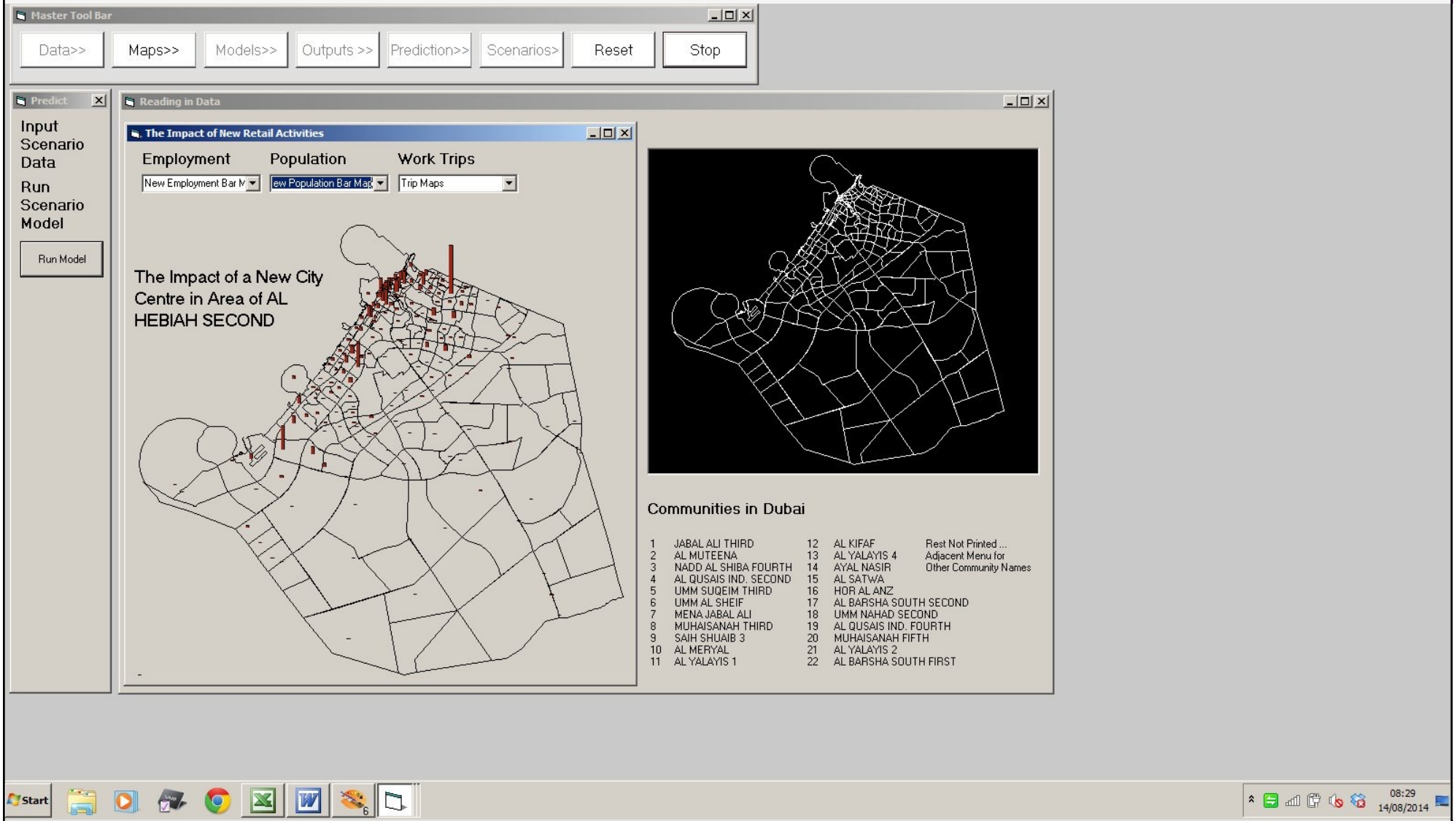

Master Tool Bar
Data>>
Maps>>
Models>>
Outputs >>
Prediction>>
Scenarios>
Reset
Stop
Reading in Data
Specifying and Tuning the Urban Model
In this window, you can specify the type of Urban Model you want to work with. In this implementation, there is only one model. It is the prototype we will develop, very close to the observed data in 2012.
The model is driven from the data and is purely a demonstration of what might be possible
Click to Set Up the Model
Set Up Complete
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
5 UMM SUQEIM THIRD
6 UMM AL SHEIF
7 MENA JABAL ALI
8 MUHAISANAH THIRD
9 SAIH SHUAIB 3
10 AL MERYAL
11 AL YALAYIS 1
12 AL KIFAF
13 AL YALAYIS 4
14 AYAL NASIR
15 AL SATWA
16 HOR AL ANZ
17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
19 AL QUSAIS IND. FOURTH
20 MUHAISANAH FIFTH
21 AL YALAYIS 2
22 AL BARSHA SOUTH FIRST
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Adjacent Menu for
Other Community Names
Recycle Bin
DESKTOP
Dubai-Model
Start
08:25
14/08/2014

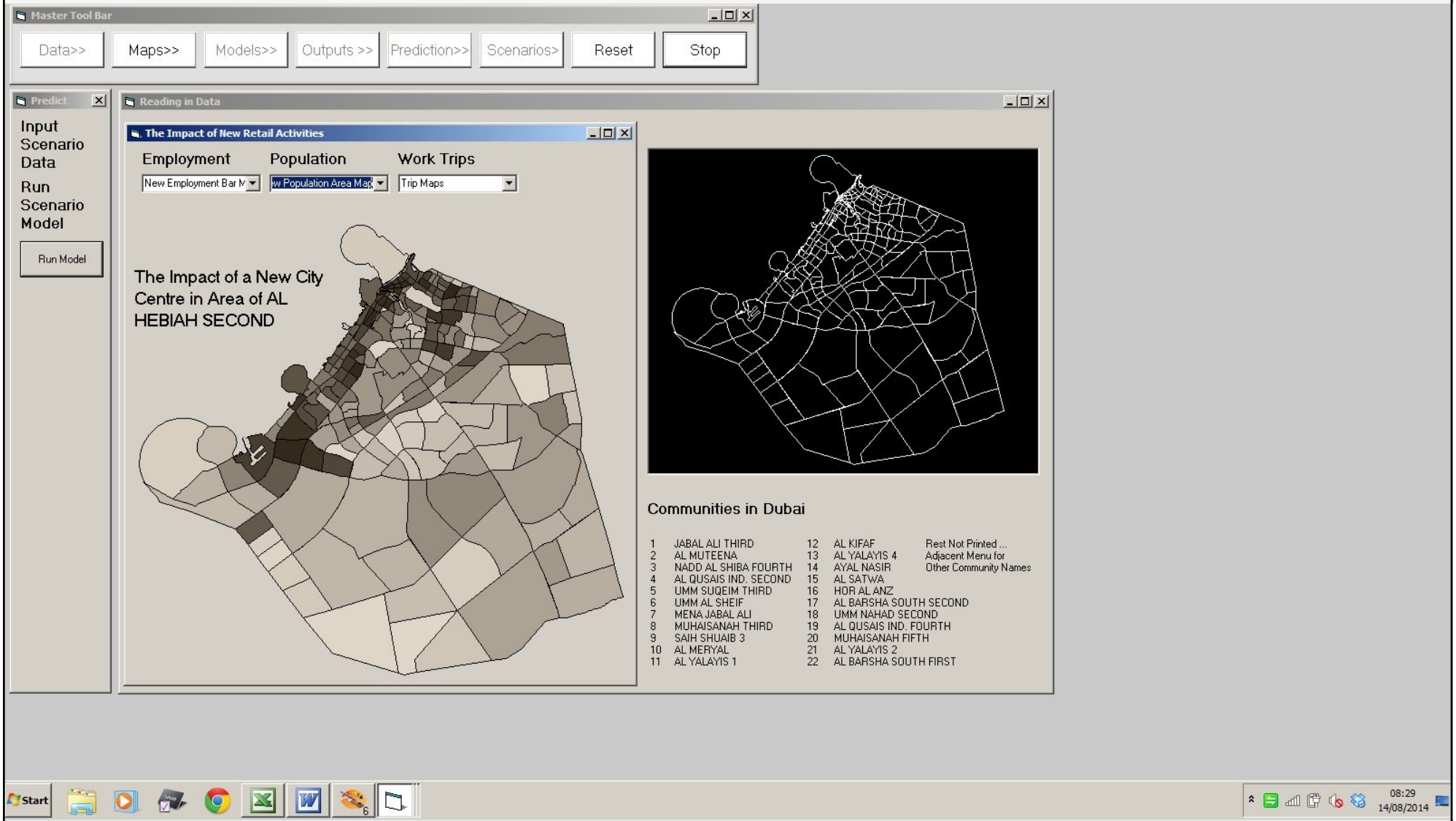

Master Tool Bar
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Outputs >>
Prediction>>
Scenarios>
Reset
Stop
Calib
Population Maps
Population Densities
Participation Ratios
Trips Maps
Trip Plots
Calibration
Expansion .....
Reading in Data
Predicted Populations
Observed Population
Predicted Population
Obs viz Pred Pop
Observed Population
Predicted Population
Obs-Pred Pop Bar Map
Correlation .9999
Percent Mean Abs Error 1.828
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
5 UMM SUQEIM THIRD
6 UMM AL SHEIF
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17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
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Other Community Names
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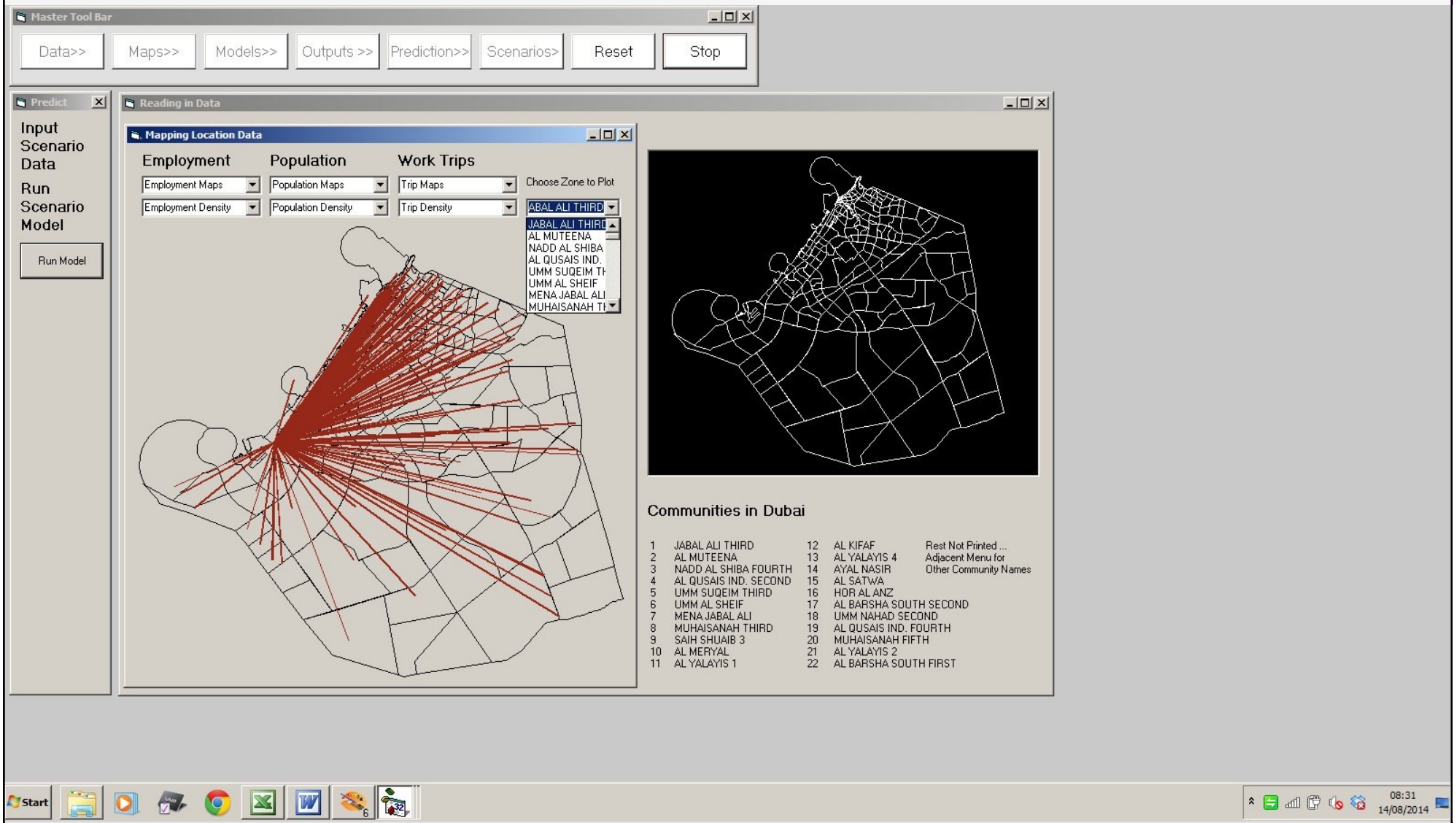

Master Tool Bar
Data>>
Maps>>
Models>>
Outputs >>
Prediction>>
Scenarios>
Reset
Stop
Predict
Input Scenario Data
Run Scenario Model
Run Model
Reading in Data
Population, Employment and Floorspace Data
READ
Employment Origin Zones
220
READ
Population Destination Zones
220
Click Here to Complete the Input of Data Directly
Read Employment Data
OK
Zone
Employment Data
Read Population Data
OK
Zone
Population Data
Read Floorspace Data
OK
Zone
Floorspace Data
Physical Line and Area Data
Read Map Data
Centroids
OK
Zones
X-Centroid
Y-Centroid
Area Data
Coordinates
OK
Polygon
X-Coordinate
Y-Coordinate
OK
Zone
Line Data
Zone Area
Travel Data
Read Trip Data
OK
Zone i to j
Observed Trips
Mean Trip Length 5.000
Read Cost Data
OK
Zone i to j
Distance-Cost
Displaying the Physical Map
Click to Redo the Graphics
Click Here If You Wish to Close This Interface
Click to Display Map Now
Community Names
Click to Display Centroids
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
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16 HOR AL ANZ
17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
19 AL QUSAIS IND. FOURTH
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22 AL BARSHA SOUTH FIRST
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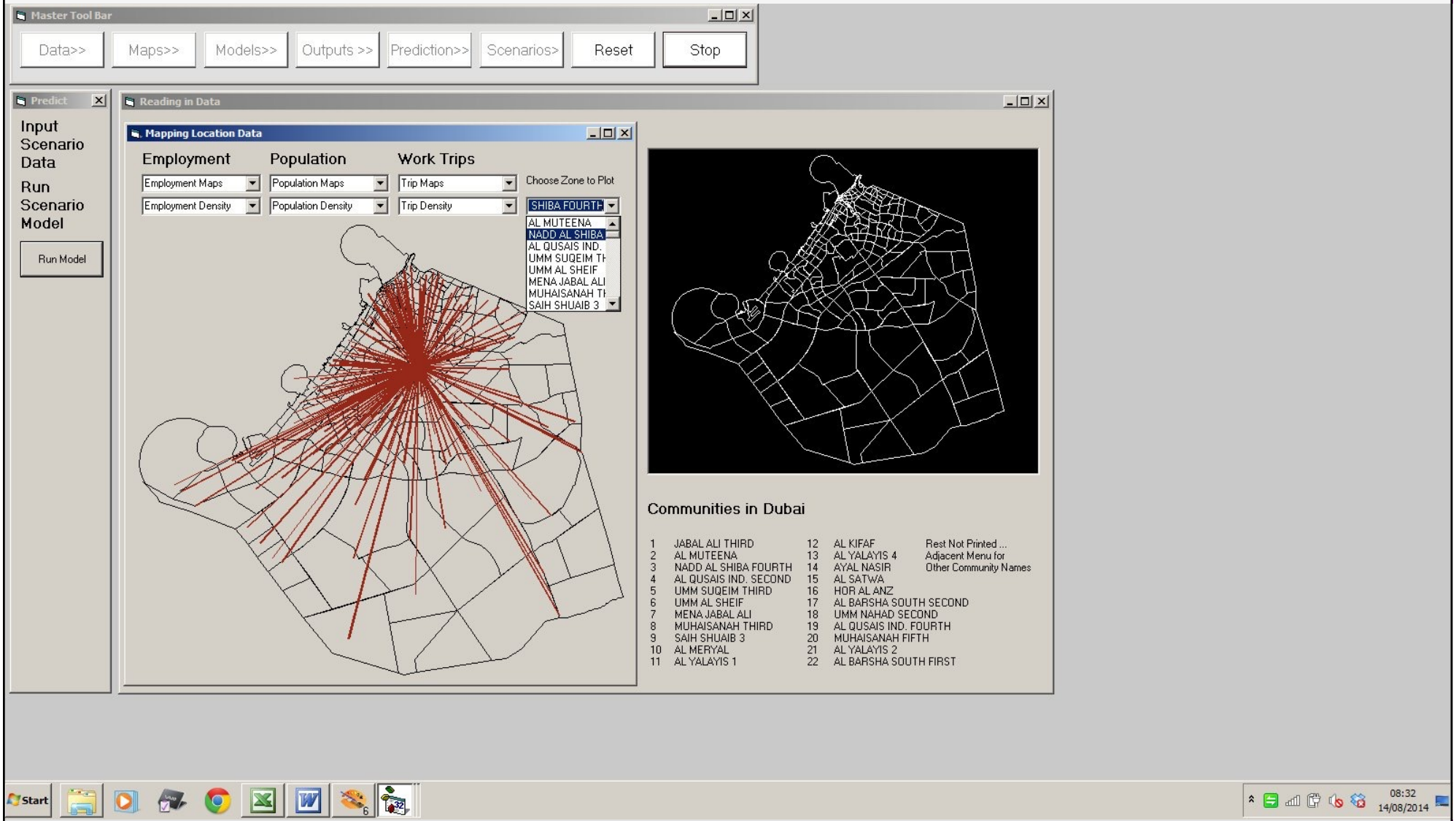

Master Tool Bar
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Maps>>
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Outputs >>
Prediction>>
Scenarios>
Reset
Stop
Predict
Input Scenario Data
Run Scenario Model
Run Model
Reading in Data
The Impact of New Retail Activities
Employment
Population
Work Trips
Employment Maps
Population Maps
Trip Maps
The Impact of a New City Centre in Area of AL HEBIAH SECOND
Communities in Dubai
1 JABAL ALI THIRD
2 AL MUTEENA
3 NADD AL SHIBA FOURTH
4 AL QUSAIS IND. SECOND
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17 AL BARSHA SOUTH SECOND
18 UMM NAHAD SECOND
19 AL QUSAIS IND. FOURTH
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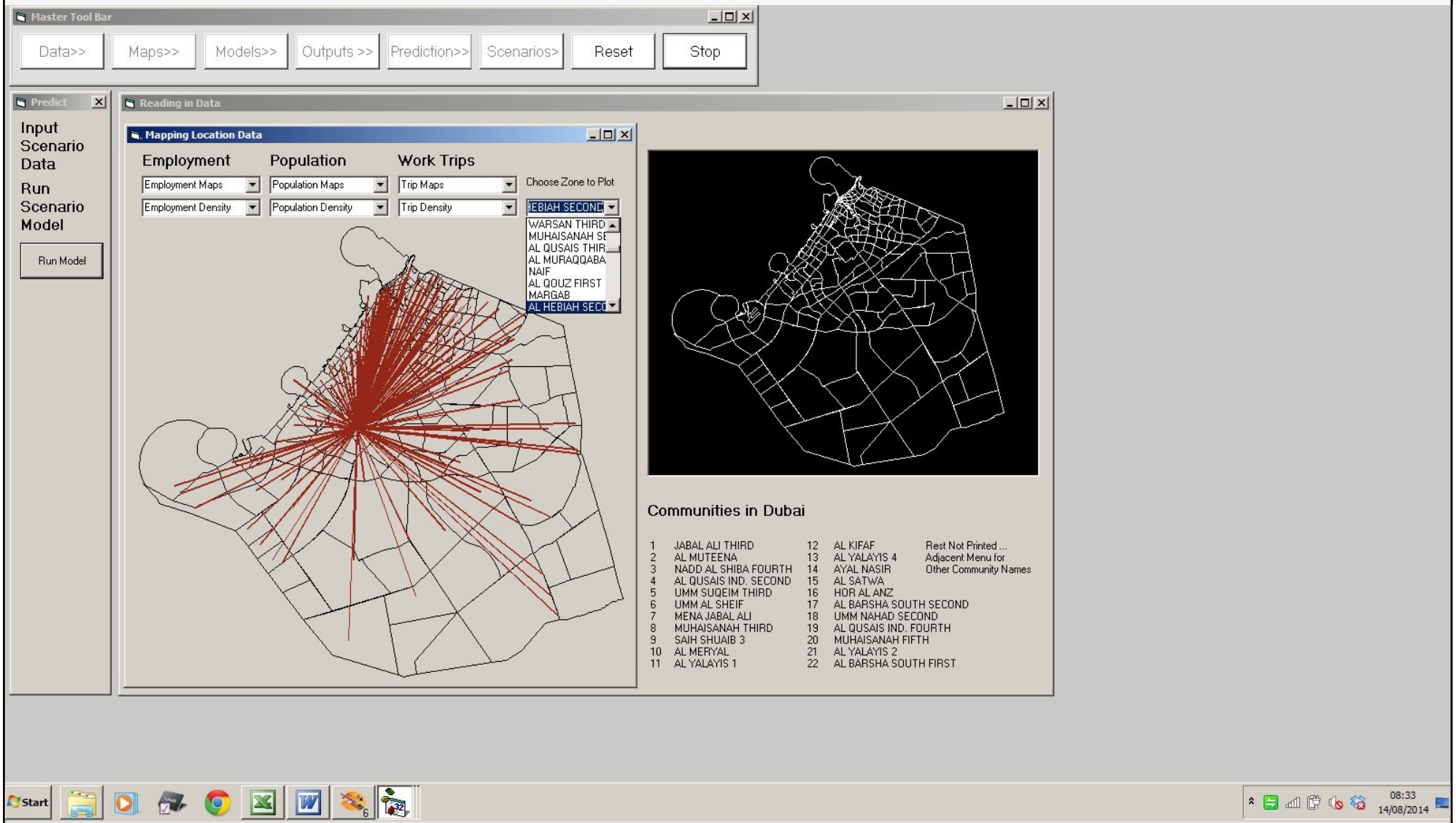

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Scenarios>
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Stop
Predict
Input Scenario Data
Run Scenario Model
Run Model
Reading in Data
The Impact of New Retail Activities
Employment
Population
Work Trips
w Employment Bar Map
Population Maps
Trip Maps
The Impact of a New City Centre in Area of AL HEBIAH SECOND
Communities in Dubai
1 JABAL ALI THIRD
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Maps>>
Models>>
Outputs >>
Prediction>>
Scenarios>
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Input Scenario Data
Run Scenario Model
Run Model
Reading in Data
The Impact of New Retail Activities
Employment
Population
Work Trips
New Employment Bar M
ew Population Bar Map
Trip Maps
The Impact of a New City Centre in Area of AL HEBIAH SECOND
Communities in Dubai
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19 AL QUSAIS IND. FOURTH
20 MUHAISANAH FIFTH
21 AL YALAYIS 2
22 AL BARSHA SOUTH FIRST
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Master Tool Bar
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Maps>>
Models>>
Outputs >>
Prediction>>
Scenarios>
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Stop
Predict
Input Scenario Data
Run Scenario Model
Run Model
Reading in Data
The Impact of New Retail Activities
Employment
Population
Work Trips
New Employment Bar M
w Population Area Map
Trip Maps
The Impact of a New City Centre in Area of AL HEBIAH SECOND
Communities in Dubai
1 JABAL ALI THIRD
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21 AL YALAYIS 2
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Master Tool Bar
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Maps>>
Models>>
Outputs >>
Prediction>>
Scenarios>
Reset
Stop
Predict
Input Scenario Data
Run Scenario Model
Run Model
Reading in Data
Mapping Location Data
Employment
Population
Work Trips
Employment Maps
Employment Density
Population Maps
Population Density
Trip Maps
Trip Density
Choose Zone to Plot
ABAL ALI THIRD
JABAL ALI THIRD
AL MUTEENA
NADD AL SHIBA
AL QUSAIS IND.
UMM SUQEIM TH
UMM AL SHEIF
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9 SAIH SHUAIB 3
10 AL MERYAL
11 AL YALAYIS 1
12 AL KIFAF
13 AL YALAYIS 4
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Urban Simulation 4

General Urban Models: QUANT

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Monday, February 2026